Early embryo differentiation in zebrafish

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1 Glossary and preliminaries

Because biology likes to use big, fancy words, we first have to build a glossary of terms that will appear in this report. Most are relatively "standard" in the literature, but many have slightly different meaning across subfields, so this section can be also viewed as term definitions.

- A **transcription factor (TF)** is a protein that directly influences the transcription process, either by participating in it, or by modifying the environment that influences the actual transcription machinery (e.g., opening or closing chromatin). Note that this is intentionally broader than the strict textbook definition, which only covers proteins that bind specific DNA sequences; here, we also include co-factors and chromatin remodelers.
- Time points in zebrafish embryo development are usually denoted in **hours post fertilization (hpf)**. We will shorten this to **hours (h)** when clear from context. We will be mostly interested in the development processes taking place until approx. **12hpf**.
- Since zebrafish are simple organisms, the embryo development is also relatively stable and easy to reproduce. The most important environmental factor is **temperature**, which is typically **28.5 °C**. Other temperatures within acceptable rates should have roughly linear influence on the development speed [*citation needed*]. Other factors like pH, medium density or oxygen levels also impact development, but the embryo is robust to small, reasonable variations in them [*citation needed*].
- An embryo cell can be **totipotent**, **pluripotent**, or **multipotent**. Totipotent means it can differentiate into internal embryo cells, or external cell layers enveloping the embryo (**extra-embryonic cells**). Pluripotent means it can differentiate into any embryo cell, but not into an extra-embryonic cell. Multipotent means the cell can differentiate within a specific lineage (e.g., endoderm), but not outside of that specific lineage (at least not without a significant intervention).
- We can also distinguish cells by how *robustly* they are differentiated so far. Note that these terms are originally defined based on what happens to the cell across various environments, not necessarily by the epigenetics and transcriptome, so we are "extrapolating" a bit here to what is causing these terms to be well defined in the first place.
  - A cell is **primed** if its epigenetics admit a particular differentiation event (relevant chromatin is open, important TF is expressed but not causing transcription yet, etc.). A cell can be primed towards many lineages at the same time. Importantly, a primed cell will not differentiate without some additional "signal" triggering the differentiation.
  - A cell is **specified** if it is actively transcribing RNA of a particular cell type, and will continue this process in a neutral environment. In other words, the transcription has been started and is somewhat self-sustaining, but can be easily disrupted by an unexpected "signal".

  - A cell is **determined** if it is specified, but it can maintain its own identity when encountering an opposing, but reasonable "signal" (i.e., a signal that would naturally occur somewhere in the organism). Determination can be driven by a stronger self-sustaining GRN feedback loop, as well as by epigenetic changes.
  - A cell is **differentiated** if it is determined, and it has undergone morphological changes associated with its cell type, including additional epigenetic changes to further increase robustness against signaling perturbations.
  - The expression profile of a specified / determined / differentiated cell can be different, despite being "the same cell type". The cell is actively producing TFs that will lead to epigenetic changes required for determination / differentiation, but this transcription can eventually turn off once the cell is "sufficiently committed".
  - The process is hierarchical (cell type A differentiates into B, which turns into C, ...), and for the intermediate cell types, the cell does not have to pass all stages. For example, pluripotent cells will never be truly differentiated, or even determined. Similarly, stages can interleave across cell types, so a cell can be determined type A, but specified type B, and primed type C, meaning it will maintain identity A on its own, is already developing expression profile of type B, and is able to eventually become type C if a signal is given.

- A zebrafish embryo goes through several "high-level" developmental stages: **zygote (0-0.75h)**, **cleavage (0.75-2.25h)**, **blastula (2.25-5.25h)**, and **gastrula (5.25-10h)**.
  - In the zygote and cleavage stage, the embryo looks mostly uniform. However, it is not fully symmetric: some maternal factors (dorsal determinants, germ plasm) are already localized asymmetrically (see MBT below).
  - During blastula, the surface layer of extra-embryonic cells is starting to split off. These are the **enveloping layer** (EVL) and **yolk syncytial layer** (YSL); one guards the embryo from the environment, and one is an interface with the yolk on which the embryo rests. Remaining cells are also called **deep cells**.
  - A **midblastula transition (MBT)** occurs around **2.75hpf**. Until that point, the cells are predominantly driven by the maternal transcriptome present in the zygote (a small first wave of zygotic transcription already starts around the 64-cell stage). Until MBT, the whole cell population is essentially synchronous. Nevertheless, several important decisions are already made in this period, driven by maternal factors:
    * formation of the YSL and EVL (YSL forms at 512-1024 cells; Figure 1);
    * specification of a small group of **primordial germ cells**, which inherit maternal **germ plasm** during cleavage;
    * the **dorsal-ventral** axis (see below), determined by maternal dorsal determinants that are transported to one side of the embryo.

    The causes of these decisions mostly can't be observed through RNA expression (only the consequences), since they rely on localization of maternal proteins. Also, *the overall mass of the embryo does not change much* during blastula and gastrula (the cells mostly thin out and spread); actual growth comes later. After MBT, the cells divide asynchronously, hence there is no longer a clear cell doubling pattern.
  - During gastrula, the embryo splits along the *radial axis* (the deep–superficial direction, perpendicular to the embryo surface; not to be confused with the radial symmetry of the blastula around the animal-vegetal axis) into **hypoblast** (develops into **mesendoderm**) and **epiblast** (develops into **ectoderm**). Note that hypoblast and epiblast are purely positional terms: Both are part of the **germ ring**, with hypoblast being the inner layer (adjacent to the yolk), while epiblast is the outer layer (see also

Figure 1). In other words, a cell can be a part of hypoblast and of mesendoderm; one is the location, the other is the expression profile. The germ ring is the **margin** of the embryo (you can see cells called "marginal", meaning they belong to this region). The process of separation into hypoblast/epiblast along the germ ring is called **internalization**. During gastrula, the epiblast still contains mesendoderm precursors that internalize later; only once internalization finishes does epiblast fully correspond to ectoderm.

  - Within the gastrula phase, the embryo also visibly splits along the **dorsal-ventral** axis (dorsal is back / spine, ventral is front / belly). This split is marked by a **shield** formation on the dorsal side of the embryo. Note that the dorsal side itself is determined much earlier (maternally, see MBT above); the shield is where it first becomes morphologically visible. This is again mostly a positional term (i.e., shield is not an expression profile). However, cells in the shield do produce various signaling molecules that reinforce the dorsal-ventral split, and later also the anterior-posterior separation (see below). Sometimes, they are called **organizer cells**.
  - The term **lateral** (and its opposite, **medial**) is used in two related senses. During gastrula, it refers to position along the margin: viewed from the animal pole, the margin is a circle with the dorsal side (shield) on one end and the ventral side on the opposite end; the *lateral* margin are the two sides in between (left and right). In the body, it refers to the distance from the midline (the future spine): *medial* is close to the midline, *lateral* is far from it. The two are connected, because during gastrula the cells converge towards the dorsal side, which becomes the midline of the body. A related term is **paraxial**, meaning directly alongside the midline (i.e., the body axis).
  - After gastrula, the embryo undergoes **early segmentation**, where the split along the **anterior-posterior** (A-P) axis becomes morphologically visible (as with the other axes, A-P patterning, e.g., of the neural plate, already starts during gastrula). The A-P axis is often confused with the **animal-vegetal** axis, but these are different axes: animal-vegetal is the axis of the egg (animal pole = top, where the blastoderm sits; vegetal pole = bottom of the yolk), while A-P is the body axis, which only forms during gastrula. They are only roughly aligned (the animal region gives rise to anterior head structures, while the tail bud forms near the vegetal pole). In Figure 1, AP/VP mark the animal/vegetal pole, while Ant./Pos. mark anterior/posterior. Segmentation is beyond the scope of this document.

- During blastula and gastrula, the embryo also undergoes **epiboly**, during which it envelops the connected yolk. This is again largely used as a timing / position marker, with Figure 1 showing the individual stages. Note that the formation of a hypoblast and epiblast, as well as of a germ ring and shield are happening relatively quickly between 50% and 60% epiboly, but this is a morphological change; transcriptomic and epigenetic changes leading up to the formation are already happening sooner.

- Another note is that in terms of morphology and transcriptome, epiblast is essentially just the continuation of the existing deep cells, without major changes at the moment of separation. It is the hypoblast that represents a new lineage that is splitting off of epiblast in the marginal region. However, later during gastrula, the ectoderm itself splits into **neural** and **non-neural** ectoderm (the latter becoming the **epidermis**), driven by a signaling gradient along the dorsal-ventral axis, and this split is visible in the transcriptome.

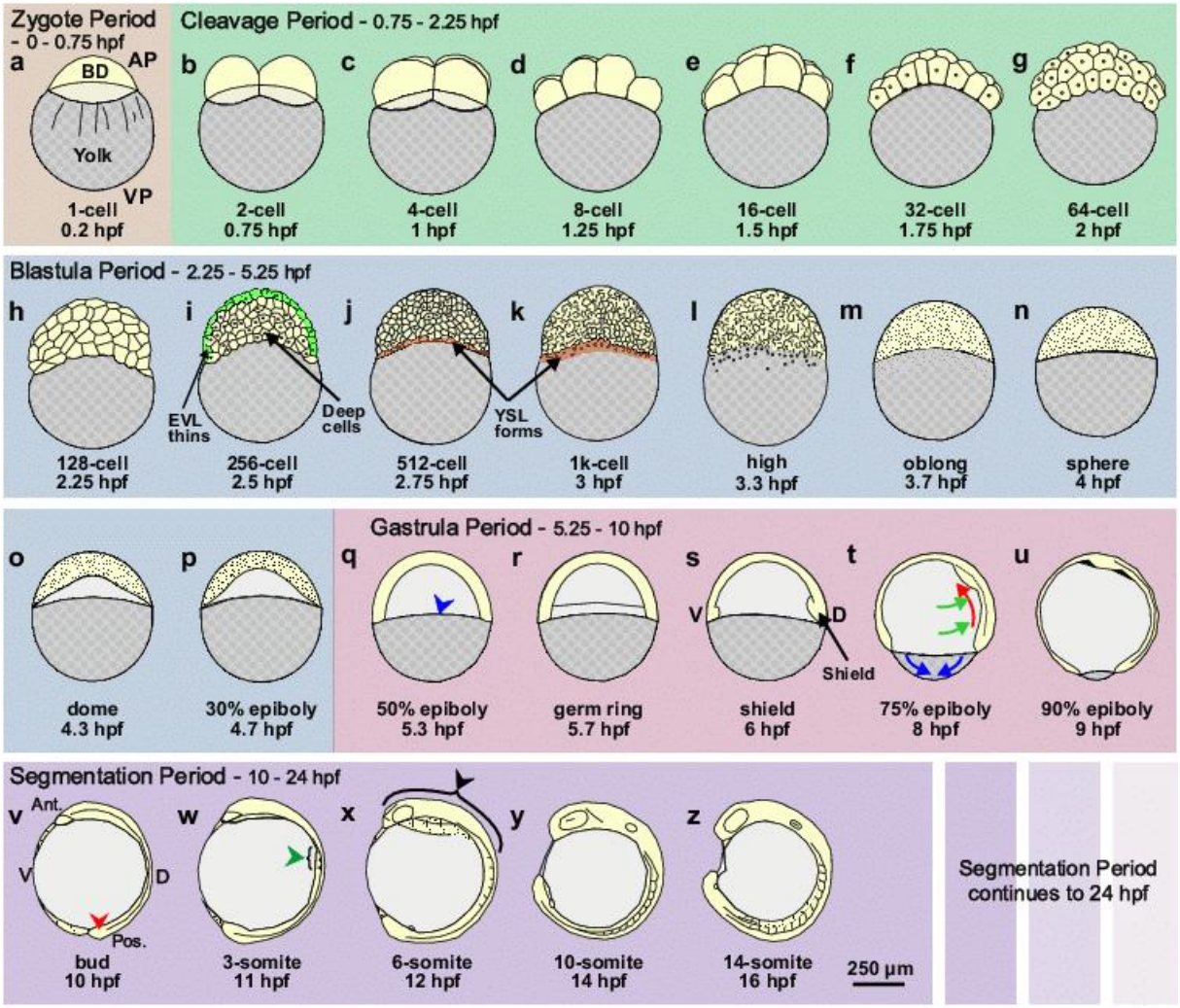


Figure 1: `https://www.aups.org.au/Proceedings/38/43-51/`

2 Transcriptomic and epigenetic timeline

In the previous section, we described the important terms and formations which appear during embryo development. However, these are mostly *morphological* terms dictating what the embryo looks like or what is happening externally. They do not capture how the cells are evolving internally, which typically precedes morphological changes quite a bit. Also, as we discussed, these do not map directly to any meaningful definition of "cell types" (i.e., a "shield cell" is not a well defined cell type). In this section, we thus summarize the *expression programs* that we expect to observe in the transcriptome during the first 12 hours, roughly when they emerge or diminish, what they do, and which earlier program they develop from. A few general remarks:

- All time ranges are approximate. They denote when a program should become *distinguishable in the transcriptome*, which usually precedes the corresponding morphological change by an hour or more. Also, the transitions are gradual: a new program typically first appears as a weak, graded bias within its parent population, and only later becomes a clearly separated cluster.
- Programs rarely "end"; most of them keep splitting into more specific sub-programs. When we say a program diminishes, we mean it is no longer a meaningful description of the cells, because they have been replaced by its more specific descendants.
- Not everything is equally visible in single-cell data. Some programs are not captured at all (e.g., the YSL, which is a syncytium, not a set of cells), or involve only a handful of cells (e.g., germ cells), and the GSE106587 atlas only starts at 3.3hpf, i.e., after MBT.

Table 1 gives an overview, and Figure 2 shows the same programs as a timeline, highlighting the branching points and the spatial axis along which each split happens. The individual programs are described in more detail below.

2.1 Cleavage and blastula (0-5.25hpf)

- **Maternal program** (0hpf, cleared by approx. 4-6hpf). Transcripts deposited in the egg by the mother. They drive the cleavage divisions and activate the zygotic genome. After MBT, a large fraction of maternal transcripts is actively degraded; by the start of gastrula, the transcriptome is dominated by zygotic expression [*citation needed*]. In the atlas, the earliest stages (high, oblong) should show a strong maternal component shared by all cells.
- **Blastomeres** (approx. 2-3hpf, diminishing 5-8hpf). The "default" program of deep cells after zygotic genome activation (ZGA). Cells are pluripotent and largely uniform; the observable differences between them are initially small and graded (distance from the margin, i.e., position along the animal-vegetal axis, and position along the dorsal-ventral axis). This program gradually disappears as the cells commit to germ layers; it persists longest in the epiblast, which is its direct continuation.
- **Yolk syncytial layer (YSL)** (2.75-3hpf, persists). Forms when marginal blastomeres collapse into the yolk cell. It drives epiboly and, importantly, is a source of signals that induce the marginal mesendoderm and contribute to the dorsal organizer [*citation needed*]. Because it is a syncytium inside the yolk, *it is not captured in single-cell data*; we only see its effect on neighbouring cells.
- **Enveloping layer (EVL)** (approx. 3-4hpf, persists as **periderm**). Develops from the outermost blastomeres, which become lineage-restricted during the blastula [*citation needed*]. Forms a protective epithelium around the embryo. In single-cell data, it is expected to be the first clearly separate cluster, already at the earliest atlas stages.

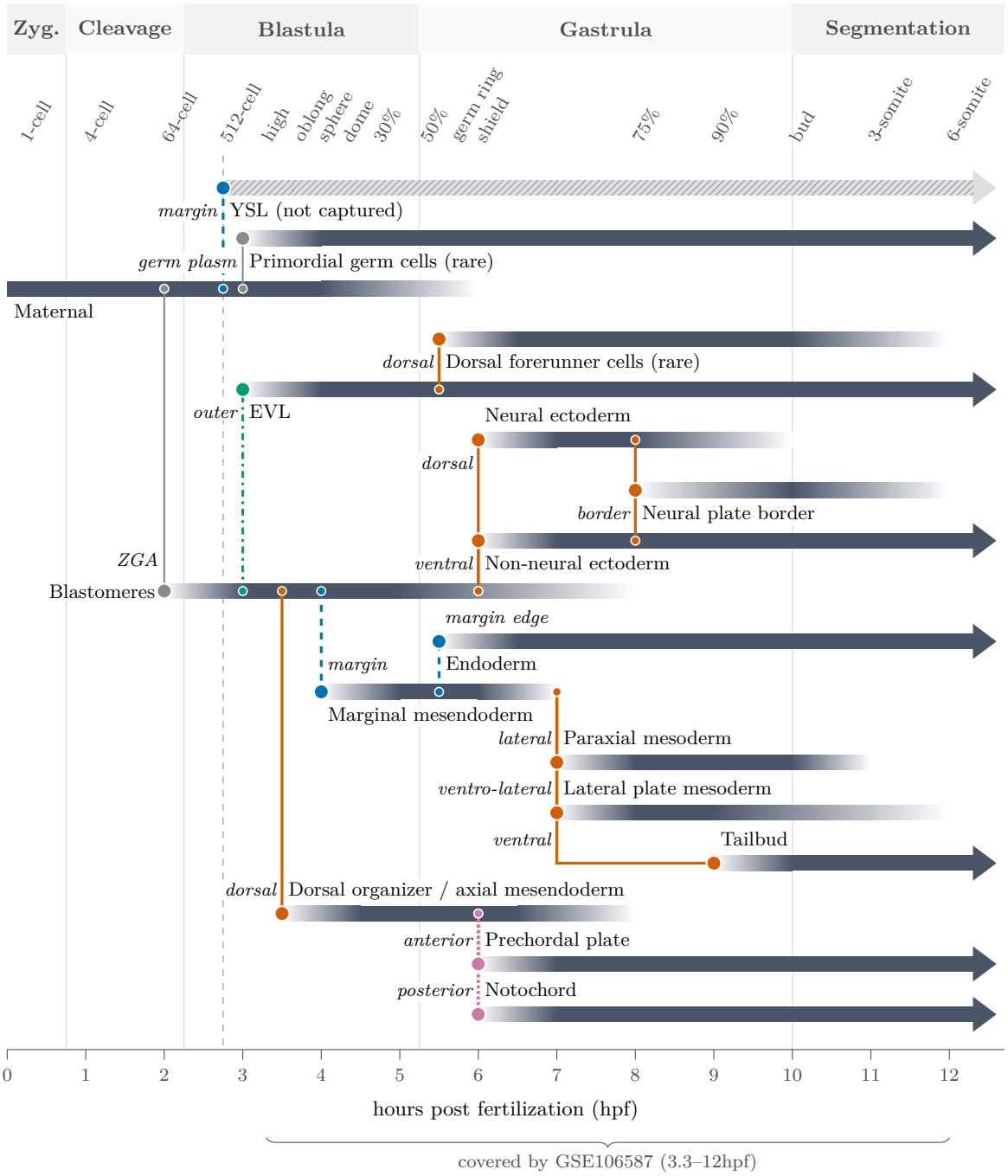


Figure 2: Timeline of the expected expression programs (Table 1). Each bar is one program; shaded ends mark the (approximate) intervals in which it emerges or diminishes. The dashed vertical line marks the MBT.

Program	Emerges (hpf)	Diminishes (hpf)	Develops from	Conditions	Note
Maternal	0	4-6	—	deposited in the egg	actively degraded after MBT
Blastomeres	2-3	5-8	maternal	zygotic genome activation (MBT)	replaced by germ layer programs; persists longest in the epiblast
YSL	2.75-3	—	maternal	marginal blastomeres fusing with the yolk	syncytium; not captured in single-cell data
EVL	3-4	—	blastomeres	outermost cell layer	becomes periderm
Primordial germ cells	3-4	—	maternal	inherited germ plasm	rare (a handful of cells)
Dorsal organizer	3.5-4.5	6.5-8	blastomeres	dorsal margin; maternal dorsal determinants	morphologically the shield (6hpf); splits into prechordal plate and notochord
Marginal mesendoderm	4-5	6-7	blastomeres	margin; signaling from YSL	forms the hypoblast; splits into endoderm and mesoderm programs
Endoderm	5.5-6.5	—	marginal mesendoderm	closest to the margin; strongest signaling	few cells at this stage; gut, liver, pancreas, pharynx
Dorsal forerunner cells	5.5-6.5	10-12	EVL	dorsal margin, ahead of the shield	rare; form the Kupffer's vesicle
Prechordal plate	6-7	—	dorsal organizer	first internalized cells	anterior part becomes the hatching gland (10-12hpf)
Notochord	6-7	—	dorsal organizer	later internalized cells	
Paraxial mesoderm	7-8	10	marginal mesendoderm	lateral margin; further from the margin edge	splits into adaxial cells and presomitic mesoderm; segments into somites
Lateral plate mesoderm	7-8	9-12	marginal mesendoderm	ventral and lateral margin	splits into heart, blood, endothelial and kidney precursors
Non-neural ectoderm	6-7	—	blastomeres	ventral epiblast; ventralizing signaling	becomes epidermis
Neural ectoderm	6-7	8-10	blastomeres	dorsal epiblast; protected by the dorsal organizer	subdivides along the A-P axis (forebrain, midbrain, hindbrain, spinal cord)
Neural plate border	8-10	10-12	neural and non-neural ectoderm	boundary between the two; intermediate signaling	splits into neural crest and placodes
Tailbud	9-10	—	marginal mesendoderm	ventral margin converging at the vegetal pole	includes overlying epiblast; neuromesodermal progenitors

Table 1: Overview of expected expression programs. Times are approximate and refer to when the program becomes distinguishable in the transcriptome; — means the program persists beyond 12hpf. *Develops from* is always a preceding program in the table; *conditions* describe where (location) or how (signaling, inheritance) the new program arises within it.

- **Primordial germ cells (PGCs)** (specified during cleavage, own zygotic program from approx. 3-4hpf, persists). Blastomeres that inherited maternal germ plasm. They give rise to gametes and do not participate in somatic differentiation. Only a handful of cells per embryo, so they will form a very small cluster.

- **Dorsal organizer** (approx. 3.5-4.5hpf, splits at 6.5-8hpf). Dorsal marginal cells (together with the dorsal YSL) that respond to the maternally determined dorsal side of the embryo. They produce signals that dorsalize the surrounding tissue, protect the dorsal ectoderm from ventralizing signals (enabling neural fate), and later contribute to A-P patterning. Morphologically, this becomes the **shield** at 6hpf; transcriptionally, the program is present well before that. Its cells later give rise to the axial mesoderm (prechordal plate and notochord). From approx. 6hpf, the internalized organizer cells (the inner layer of the shield) are often called the **axial mesendoderm**, i.e., midline mesendoderm that has not yet split into prechordal plate and notochord. We do not treat it as a separate program, because it is transcriptionally a gradual continuation of the organizer program; the clear split only happens with the prechordal plate and notochord.

- **Marginal mesendoderm** (approx. 4-5hpf, splits at 6-7hpf). Cells within a few cell tiers of the margin, induced by signals from the YSL and neighbouring marginal cells. The program is graded: the closer to the margin (and the stronger the signal), the more the cells are biased towards endoderm, while cells further away are biased towards mesoderm [*citation needed*]. These are the cells that internalize to form the hypoblast.

2.2 Gastrula (5.25-10hpf)

- **Endoderm** (approx. 5.5-6.5hpf, persists). Develops from the marginal mesendoderm cells that received the strongest inducing signal (mostly close to the margin, more on the dorsal side). Will form the gut and associated organs (liver, pancreas) and the pharynx. Relatively few cells at this stage.

- **Dorsal forerunner cells** (approx. 5.5-6.5hpf, forming the **Kupffer's vesicle** at approx. 10-12hpf). A small group of cells at the dorsal margin, ahead of the shield, partially derived from the EVL [*citation needed*]. They do not internalize; instead, they migrate ahead of the margin and form the Kupffer's vesicle, which establishes the left-right asymmetry. Only a handful of cells.

- **Prechordal plate** (approx. 6-7hpf, persists). The anterior part of the axial mesoderm, derived from the leading (first internalized) cells of the dorsal organizer. It migrates towards the animal pole and contributes to head patterning. Its most anterior part becomes the **hatching gland** (approx. 10-12hpf).

- **Notochord** (approx. 6-7hpf, persists). The posterior part of the axial mesoderm, derived from the remaining dorsal organizer cells. Forms the midline rod that later supports the body axis, and is an important source of signals that pattern the neighbouring neural tube and paraxial mesoderm.

- **Paraxial mesoderm** (approx. 7-8hpf, starts to form somites at 10hpf). Mesoderm from the lateral margin that converges on both sides of the notochord. Splits into **adaxial cells** (directly adjacent to the notochord; slow muscle precursors) and **presomitic mesoderm**, which will be segmented into somites (muscle, skeleton).

- **Lateral plate mesoderm** (approx. 7-8hpf, splitting at 9-12hpf). Mesoderm from the ventral and lateral margin. Gives rise to the heart, blood and endothelial precursors, and the pronephros (kidney). Many of these sub-programs only become clearly distinct around the end of gastrula or early segmentation.

- **Non-neural ectoderm** (approx. 6-7hpf, persists as **epidermis**). The ventral part of the epiblast, where ventralizing signals are strong. It is the first ectodermal program to diverge from the blastomere program.
- **Neural ectoderm** (approx. 6-7hpf, subdivided at 8-10hpf). The dorsal part of the epiblast, protected from ventralizing signals by the dorsal organizer. Forms the **neural plate**. During late gastrula, it is patterned along the A-P axis into forebrain (including eye field), midbrain, hindbrain and spinal cord programs.
- **Neural plate border** (approx. 8-10hpf, splitting at 10-12hpf). Cells at the boundary between neural and non-neural ectoderm, receiving intermediate levels of signals. Gives rise to the **neural crest** (very diverse lineage: pigment cells, craniofacial skeleton, peripheral neurons, ...) and **placodes** (sensory organs).
- **Tailbud** (approx. 9-10hpf, persists). Forms from ventral marginal cells converging at the vegetal pole at the end of epiboly. Contains bipotent **neuromesodermal progenitors** that produce both spinal cord and paraxial mesoderm during body axis elongation. Note that this is one of the few programs which mix "neural" and "mesodermal" signatures.

3 Dataset overview

The analysis is based on the GEO SuperSeries GSE106587 [34]. It consists of three single-cell datasets:

- **Drop-seq atlas** (SubSeries GSE106474). Wild-type embryos at 12 stages from high (3.3hpf) to 6-somite (12hpf); 39,505 cells, 23,974 genes. This is the main dataset.
- **SMART-seq2** (SubSeries GSE106466). 416 cells at 50% epiboly: 52 wild-type cells and 364 cells of MZ*oep* mutants, which lack Nodal signalling and hence most of the mesendoderm. Fewer cells, but deeper coverage per cell.
- **10x Genomics** (GSE106587 itself). Wild-type (3,000 barcodes) and MZ*oep* (2,200 barcodes) embryos at 6-somite (12hpf). The barcode lists are not QC-filtered, so not every barcode is a cell.

All three are available only as processed count matrices (raw reads were not downloaded). In the rest of this section, we only describe the Drop-seq atlas. **The GEO matrix has already passed the basic quality filters of the original study, but not its later outlier removal** (Section 3.4); here, we use all 39,505 cells without any additional filtering. The experimental details below are taken from the supplementary materials of the paper. The numbers, figures and table are produced by `scripts/01_dataset_structure.py`.

3.1 Stages, batches and libraries

The Drop-seq atlas has three nested levels (Figure 3 and Table 2):

- **Stage**: the developmental time point (12 stages, each with a single hpf value).
- **Sequencing batch** (DS2, DS3, DS4 and DS5): the "sequencing batch" field of the GEO samples. It is the main expected source of technical variation.
- **Library**: one sample of pooled embryos, processed in one Drop-seq run; equivalently, one GEO sample (GSM) and one SRA experiment. There are 28 libraries in total. Most (stage, batch) pairs have exactly one library. Six pairs have two, in which case the cell names of the second library carry a `b` suffix (e.g., `ZF50_WT_DS4b_...`).

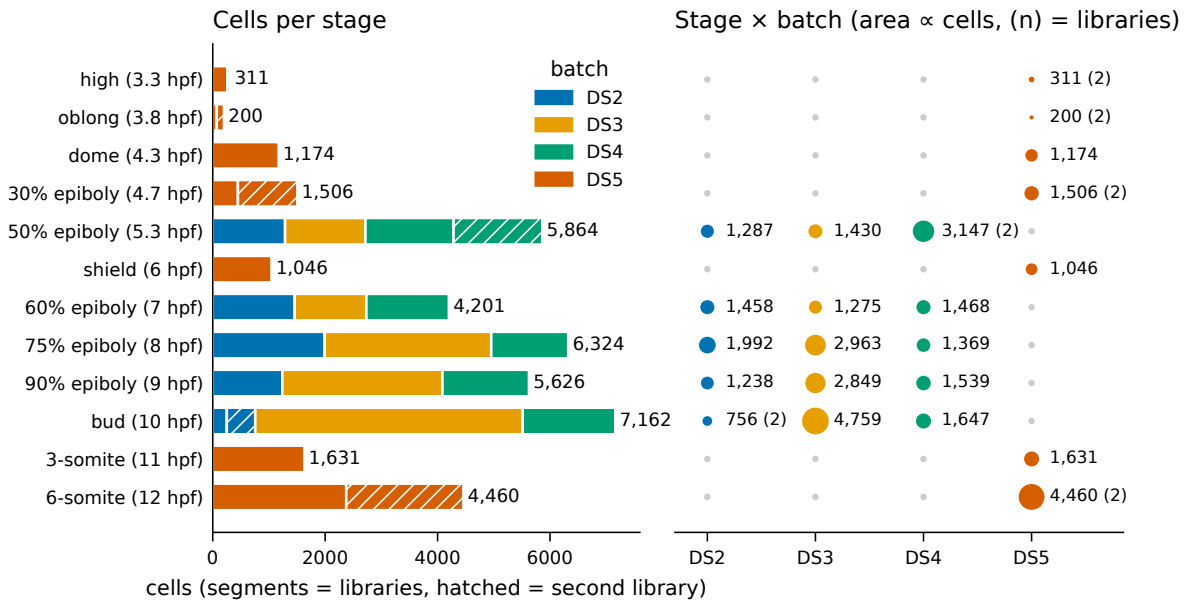


Figure 3: Structure of the Drop-seq atlas (GSE106474). Left: number of cells per stage; each segment is one library, coloured by sequencing batch, and second libraries of the same stage and batch are hatched. Right: the same counts per stage and batch. Stages of batch DS5 and stages of batches DS2–DS4 do not overlap.

Each library was prepared from a pool of 20–40 embryos of the same stage. The embryos were manually deyolked (**which also removes the YSL**), mechanically dissociated, and the resulting cell suspension was encapsulated in Drop-seq droplets. Consequently, cells carry no information about which embryo (or which position in the embryo) they came from. Some libraries were sequenced twice to increase depth, but the reads of such re-sequencing are merged into the same library; they do not form a second library.

Second libraries are separate embryo pools. GEO itself does not say whether the second library of a (stage, batch) pair comes from a separate embryo pool (biological replicate) or is a second run of the same pool (technical replicate). However, the supplementary Table S1 of the paper lists the total number of embryos used for each stage (Table S1). For high (56 embryos), oblong (76) and 30% epiboly (59), this is more than a single pool of at most 40 embryos, so the two libraries of these stages must come from different pools. For the remaining stages with a second library (50% epiboly, bud and 6-somite), the embryo counts are consistent with separate pools of approx. 20 embryos each, but they do not rule out a shared pool. The paper also treats each library as its own unit: Table S1 calls the libraries "batches" (two for high, four for 50% epiboly), and the two 6-somite libraries are described as "acquired in two batches" and batch-corrected before trajectory reconstruction. We thus treat second libraries as biological replicates sequenced in the same sequencing batch. Note that the paper's "batch" corresponds to our library, not to the GEO sequencing batch.

GEO replicate numbering. The "Rep N" in the GEO sample titles numbers the libraries of a stage *across batches*. It does not correspond to our a/b label. For example, "50% epiboly Rep 2" is the DS3 library, while "high Rep 2" is the second DS5 library of the high stage.

Table 2: Drop-seq libraries. Lib. "b" is the second library of the same stage and batch (a separate embryo pool). Medians are per cell, over all cells in the GEO matrix.

Stage	hpf	Batch	Lib.	GEO sample	Cells	Median UMIs	Median genes
high	3.3	DS5	a	GSM2838529	263	6,731	3,009
		DS5	b	GSM2838530	48	13,385	4,566
oblong	3.8	DS5	a	GSM2838531	72	11,866	4,171
		DS5	b	GSM2838532	128	3,603	2,048
dome	4.3	DS5	a	GSM2838528	1,174	4,398	1,931
30% epiboly	4.7	DS5	a	GSM2838506	448	3,858	1,717
		DS5	b	GSM2838507	1,058	3,260	1,564
50% epiboly	5.3	DS2	a	GSM2838509	1,287	5,600	2,106
		DS3	a	GSM2838510	1,430	4,562	1,730
		DS4	a	GSM2838511	1,567	5,179	1,988
		DS4	b	GSM2838512	1,580	5,630	1,962
shield	6	DS5	a	GSM2838533	1,046	2,904	1,291
60% epiboly	7	DS2	a	GSM2838513	1,458	6,173	1,898
		DS3	a	GSM2838514	1,275	6,074	1,826
		DS4	a	GSM2838515	1,468	4,970	1,640
75% epiboly	8	DS2	a	GSM2838518	1,992	5,404	1,768
		DS3	a	GSM2838519	2,963	3,352	1,258
		DS4	a	GSM2838520	1,369	5,582	1,712
90% epiboly	9	DS2	a	GSM2838521	1,238	5,814	1,793
		DS3	a	GSM2838522	2,849	3,377	1,282
		DS4	a	GSM2838523	1,539	3,603	1,343
bud	10	DS2	a	GSM2838524	250	6,642	1,915
		DS2	b	GSM2838525	506	6,446	1,864
		DS3	a	GSM2838526	4,759	2,556	1,063
		DS4	a	GSM2838527	1,647	3,683	1,381
3-somite	11	DS5	a	GSM2838508	1,631	2,609	1,047
6-somite	12	DS5	a	GSM2838516	2,375	2,000	884
		DS5	b	GSM2838517	2,085	3,180	1,237

3.2 Batch design

The most important property of the atlas is that the batches split the stages into two blocks that *share no stage*:

- **Block A** (batch DS5 only): high, oblong, dome, 30% epiboly, shield, 3-somite and 6-somite; 11 libraries, 10,328 cells.
- **Block B** (batches DS2, DS3 and DS4): 50% epiboly, 60%, 75% and 90% epiboly, and bud; 17 libraries, 29,177 cells. Every stage of this block was sequenced in all three batches.

Within block B, the design is balanced, so stage and batch effects can be separated. Within block A, there is only one batch, so stage differences are not confounded with batch, but there is also nothing to estimate the DS5 batch effect from. This is particularly unfortunate for our focus on the decisions around 6hpf. Shield (6hpf, DS5) lies between 50% epiboly (5.3hpf) and 60% epiboly (7hpf), which are both in block B. A batch effect will therefore appear as an abrupt developmental change at shield and back again at 60% epiboly.

3.3 Library size and sequencing depth

The number of cells per stage ranges from 200 (oblong) to 7,162 (bud) (Figure 3). The five block B stages (50% epiboly to bud) contain 74% of all cells. The two earliest stages are tiny: high and oblong have 511 cells together, and three of their four libraries have fewer than 150 cells (48, 72 and 128). Sequencing depth (Figure 4) is a median of 3,798 UMIs and 1,444 detected genes per cell over the whole atlas. We can make the following observations:

- **Depth decreases with developmental time**, from library medians of 4–13 thousand UMIs per cell at high and oblong to 2–3 thousand at 3- and 6-somite. Part of this is biological (cells get smaller and contain less mRNA with each division, which is also why the original study used stage-specific filtering thresholds).
- **Libraries of the same stage and batch are not interchangeable.** At oblong, the median depth of the first library is 3.3× that of the second; at high, the second library is 2× deeper than the first, and at 6-somite 1.6×.

3.4 Cell filtering

The original study filtered the cells in two steps; the dataset includes the first, but not the second filtering.

Basic quality filters (already applied in GEO). First, cells were filtered by the number of detected genes and UMIs, using a separate range for each stage (Table S1), because RNA content per cell decreases over early development. Cells below the range are likely low-quality (e.g., broken cells or empty droplets with ambient RNA), while cells above the range are likely doublets. Afterwards, cells with more than 45% mitochondrial RNA (a sign of stressed or dying cells) were removed. The GEO matrix is consistent with these filters having been applied:

- Only 51 of the 39,505 cells fall outside the gene ranges, all of them just slightly.
- The mean number of detected genes per stage agrees with Table S1 to within 1.5%.
- The GEO matrix contains no mitochondrial genes. This explains why the mean UMIs per cell are 20–25% lower than in Table S1 (where approx. 20–29% of UMIs are mitochondrial), and why some cells fall below the UMI ranges.

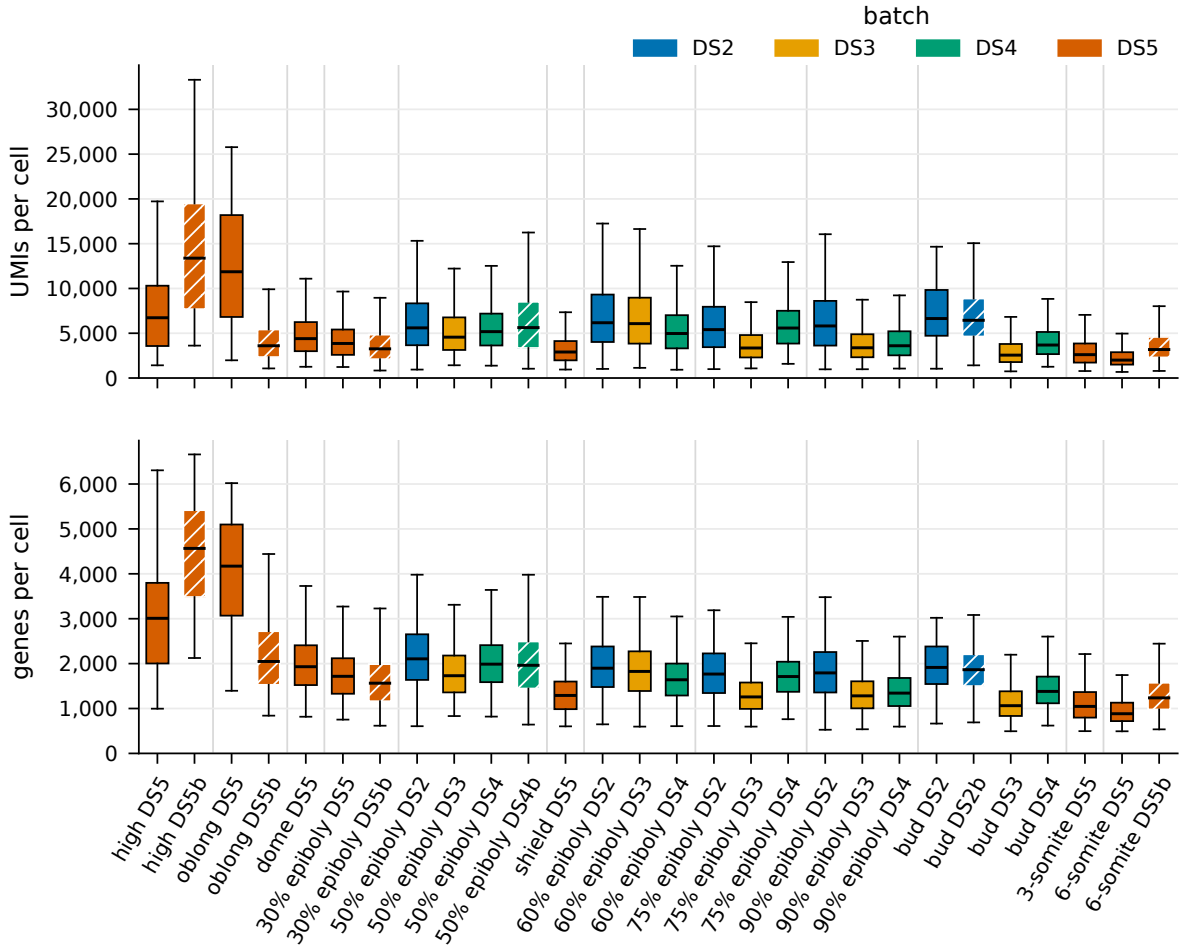


Figure 4: Per-library sequencing depth: UMIs (top) and detected genes (bottom) per cell. Boxes show the quartiles, whiskers extend to 1.5× the interquartile range (outliers not shown). Colour is the sequencing batch; second libraries are hatched. Thin vertical lines separate stages.

Outlier removal (not applied in GEO). Before the trajectory reconstruction, the paper removed 618 cells forming a cluster dominated by an "apoptotic-like" expression program, which appears in cells of many different types and would therefore create artificial connections between unrelated cell types. It also removed 139 cells that were poorly connected in the nearest-neighbour graph. The paper's full-data visualization uses 39,488 cells, and $39{,}488 - 618 - 139 = 38{,}731$, which is the number of cells reported in the paper. The GEO matrix contains 17 cells more than 39,488, which the paper does not explain. The identities of the removed cells are not available in GEO. Finally, the per-stage cell counts in Table S1 sum to 38,671, i.e., 60 fewer than 38,731, so the paper is not fully consistent here either.

3.5 Consequences for the analysis

- **Quantitative analysis within library, qualitative comparison across libraries.** To avoid batch and stage effects, quantitative analysis (normalization, binarization, etc.) should be calibrated and applied within individual libraries. Between libraries, we should primarily draw conclusions from qualitative results (e.g., gene A is higher than B in cells X across DS2/DS3/DS4; the quantity can be different, but the effect stays the same).

- **When to compare quantities across libraries?** We will most likely need to break the

first rule in a few places. First, comparing **time stages within batch** is reasonable when assessing long-term changes in transcription (but should probably carry greater burden of proof than within-library tests). Similarly, **batch-effect calibration within stage** is possible for DS2, DS3 and DS4. This is not possible with respect to DS5, but maybe we can at least extrapolate some useful information from it. Batch correction ideally stays within each stage to avoid "correcting for" biological development. But most algorithms can take this into account, so in theory, for DS2/DS3/DS4, a full batch correction run is viable; however, it is questionable whether it is needed to derive useful qualitative comparisons. Finally, **library-effect within batch and stage** can be also quantified using the a/b replicates, but we only have these for a small number of libraries, and they do come from separate embryo pools, so there can be genuine biological differences between them.

- **Some pseudo-libraries may be needed.** The early development stages (high, oblong) contain a very small number of cells, further subdivided into the a/b replicates. Assuming we don't observe significant expression profile differences between the a/b replicates, it may be reasonable to merge them and treat them as one library to obtain a bigger cell pool for the following experiments. In this context, **down-sampling or weighting the remaining, much larger libraries may be necessary** for some analysis methods to avoid over-representation of the 50%/60%/75%/90%/bud stages.

- **Outliers must be handled by us.** The GEO matrix still contains the apoptotic-like cells and the poorly connected outliers removed in the paper. We likely need our own (comparable) outlier filtering step. Furthermore, some cell types not relevant for the Boolean model may be also eliminated outright (e.g. the germ cells, or the EVL). However, for operations such as discretization, it may be useful to have the whole expression distribution, even if it is not fully assigned to cell types.

4 Result: markers of heterogeneity within libraries

Many of the following analyses need a set of "interesting" genes: genes that define a low-dimensional embedding of the cells, genes whose expression we try to predict, or genes that pass a filter because they mark a subpopulation. The standard choice is a fixed number of *highly variable genes* (HVGs). In this section, we replace it with a test that asks, for every gene and library, whether the gene varies between the cells of the library more than a gene expressed equally in every cell would. Intuitively, we expect genes that define specific subpopulations of the library to fall within such a subset. **We call such genes *heterogeneous*** (within a library).

The test relies on the assumption that the **UMIs of a uniformly expressed gene should follow the negative-binomial distribution**, which accounts for the variance of bursty gene expression (the telegraph model) and sampling noise [59, 146]. The test quantifies the distance of the real expression from such an ideal distribution. The cutoff is calibrated against simulated libraries of uniformly expressed genes, which leaves approx. 10 false positives per library, and combined with an effect size estimated from the library itself, which gives three tiers of genes with an estimated false discovery rate of 50%, 10% and 1% (weak, moderate, strong). **The test selects 421–1,527 genes per stage** (weak tier, dome onward; high and oblong are mostly uniform with 33–164 het. genes per library). We show that the test is more consistent across comparable libraries than HVGs and it aligns better with genes with a localized expression pattern according to ZFIN [10] records; the advantage is large over the original Seurat HVGs and moderate over newer HVG methods. The numbers, figures and tables are produced by `scripts/01b_heterogeneous_genes.py`, `scripts/01c_heterogeneity_zfin.py` and `scripts/01d_hvg_flavours.py`.

4.1 A model of a homogeneous gene

Consider one library and one gene. Cell i has a relative depth s_i (its number of UMIs divided by the library average); in this dataset, s_i varies by approx. 10× within a library. The variation in s_i is mostly technical (capture efficiency), but it also absorbs differences in cell size: we only assume that a homogeneous gene makes up the same fraction of every cell's transcripts. Therefore, if the gene were to be expressed at the same level in every cell, cell i would contain on average $s_i \cdot m$ of its UMIs, where m is the gene's mean per average cell. However, the observed count of such a homogeneous gene is still random for two reasons:

- **Sampling.** Only a fraction of the transcripts of a cell is captured and sequenced. Given the true expression, the captured count follows Poisson distribution, with variance $s_i \cdot m$.
- **Biological noise.** Even in identical cells, the true expression is not identical: transcription occurs in bursts [146], and cells can differ in size and cell-cycle stage. If the true expression varies between cells with a squared coefficient of variation φ (gamma-distributed), the count is negative binomial (NB), with variance $s_i \cdot m + \varphi \cdot (s_i \cdot m)^2$.

NB counts describe droplet-based UMI data well, without any need for extra "dropout" zeros [59, 180, 61]. Meanwhile, the Poisson model ($\varphi = 0$) is not a useful baseline: For a well-expressed gene, the extra variance $\varphi \cdot (s_i \cdot m)^2$ is small in each cell, but with a thousand cells it is easily detectable: at shield, 88% of the genes with 100–200 UMIs have $\hat{\varphi} > 0$. Under the Poisson model, nearly every well-expressed gene would have higher variance than expected. In other words, if we only accounted for sampling noise (Poisson model), but not for the biological expression noise (negative binomial model), we would wrongly conclude most genes are heterogeneous.

However, in the NB model, we do not know the true biological value of φ for each gene. Instead, we estimate its value based on *typical genes with the same mean*, and then ask whether the observed φ differs significantly from this "typical" φ_t (in other words, we assume most genes are homogeneous, which allows us to estimate their φ_t and use it to distinguish the heterogeneous ones). A gene restricted to a subpopulation has a much larger φ: if it is expressed in a fraction f of the cells (and uniformly within them), $\varphi \approx 1/f - 1$, e.g., 9 for $f = 10\%$ (this is only the impact of the subpopulation; true φ will also depend on the biological noise; consequently, φ is not zero even if $f = 100\%$).

Estimating the squared coefficient of variation. We estimate φ separately for every gene in every library. Let y_i be the UMI count of the gene in cell i, where i runs over all cells of the library (no cells are removed). Then we use the method of moments:

$$\hat{\varphi} = \frac{\sum_i (y_i - s_i\hat{m})^2 - \sum_i s_i\hat{m}}{\hat{m}^2 \sum_i s_i^2}, \qquad \hat{m} = \frac{\sum_i y_i}{\sum_i s_i}.$$

Here $\hat{m}$ is the gene's mean per average cell, the first sum is the observed variance around the expected counts $s_i\hat{m}$, and the second sum is the part of it that Poisson sampling explains. Thus $\hat{\varphi}$ is the variance left after subtracting the sampling noise, divided by the scale at which φ enters the variance. The estimate is 0 when the gene varies exactly as much as sampling alone predicts, and negative when it varies less, which happens by chance.

We only estimate φ for genes with at least 10 UMIs in the library (6,756–12,555 of the 23,974 genes, depending on the library). Below that, a gene is detected in a handful of cells and its variance says almost nothing about its distribution. Such a gene is considered *not measured* in that library: it has no estimate and is never considered heterogeneous there. In particular, it also counts as not heterogeneous when we combine multiple libraries of a single stage (see below).

The *typical* φ_t at a given mean is the value of a dispersion–mean trend fitted to the genes of the same library (see next paragraph). The *residual* of a gene is $\log_{10}\hat{\varphi} - \log_{10}\varphi_t$, i.e., $\log_{10}$

of the ratio of the two: 0 means typical, 1 means ten times more variable than typical. We use a ratio rather than a difference because φ_t itself changes by orders of magnitude with the mean. An absolute difference of 0.5 would thus be negligible for rare genes but extreme for well-expressed ones. Meanwhile, ratio remains comparable across genes with vastly different φ_t. The ratio requires a positive $\hat{\varphi}$, so negative and zero estimates are floored at 10^{-3}; such genes are far below any reasonable cutoff and are clearly not heterogeneous anyway.

Typical overdispersion: a dispersion–mean trend. In every library, genes of similar mean have similar $\hat{\varphi}$, and $\hat{\varphi}$ falls with the mean. We describe this by a trend with three parameters,

$$\varphi_t(m) = a_0 + a_1\, m^{-b}.$$

Its shape follows from the mechanism of the noise. Under the telegraph model of bursty transcription [146], a gene's NB shape parameter is its burst frequency and its mean is the burst frequency times the burst size, so $\varphi \approx$ burst size$/m$: rarer genes are noisier, approx. as $1/m$ if burst sizes are similar. The constant a_0 is the noise floor shared by all genes (e.g., cell-cycle stage and cell size beyond what s_i absorbs), which becomes comparable to the power law only for the most expressed genes; the same "a/mean + floor" pattern was found across the proteome of single *E. coli* cells [184]. Comparing every gene with this trend thus absorbs the variation that all genes share, without having to model each source.

The trend is fitted only to genes with at least 100 UMIs in the library. Below that, the estimate is dominated by chance: a gene with 10–30 UMIs in a library of 1,000 cells is detected in a few dozen cells, typically with one UMI each. For example, at shield, only 33% of the genes with 10–15 UMIs have a positive estimate, compared with 88% of the genes with 100–200 UMIs. For genes with fewer than 100 UMIs, $\varphi_t(m)$ is therefore a pure extrapolation. To check that the extrapolation does not depend on this choice, we refitted the trend to genes with at least 50 and 30 UMIs instead of 100: the typical value of the rarer genes changes by a median of at most 0.01 in $\log_{10}$ (at most 0.11, in the smallest library from dome on). At 20 UMIs, the fit breaks in some libraries, because the bins of such genes have medians at or below 0, i.e., the threshold must stay well above the point where most estimates are dominated by sampling.

The genes with at least 100 UMIs are sorted by mean and divided into bins of 200 genes, and $\log_{10}\varphi_t$ is fitted by least squares to the medians of $\hat{\varphi}$ in the bins (6–35 bins per library). The medians make the fit insensitive to heterogeneous genes, which are a minority of every bin. In the 24 libraries from dome on, the trend describes the bin medians within 0.03 in $\log_{10}$ (root mean square), with very similar exponents $b = 0.72$–0.85 (median 0.77), i.e., slightly less steep than $1/m$, as expected if burst sizes grow somewhat with expression. The floor is small ($a_0 = 0$–0.026, median 0.008): over the observed range, the trend is mostly the power law, and it levels off only for the most expressed genes. There, it is slightly too high in the late libraries (median residual of the 100 most expressed, non-heterogeneous genes -0.15 to -0.22 at 3- and 6-somite, -0.11 to $+0.07$ elsewhere), which makes the test more conservative for these genes. In the four libraries of high and oblong (48–263 cells), only 6–20 bins are available and their medians are close to the noise of the estimate; the fitted curves are flat or erratic ($b = 0$–0.66), the typical value of rare genes is unreliable (Figure S1), and which of their genes are selected depends mostly on the chance level and on high effect-size floors (see below).

Relation to DESeq2 and other methods. The model is the one used for bulk RNA-seq by DESeq2 [118]: NB counts with variance $\mu + \alpha\mu^2$, size factors that play the role of s_i, and a dispersion α (our φ) that is assumed to depend smoothly on the mean, with the parametric trend $\alpha(\mu) = a_0 + a_1/\mu$ (ours has a free exponent). The two methods use the trend in opposite ways. DESeq2 has few samples per condition, so each gene's dispersion is very uncertain; the trend serves as a prior towards which the gene-wise estimates are shrunk, so that the *means* of the conditions can be compared reliably. Deviations from the trend are treated as noise,

except for "dispersion outliers" (genes far above the trend), which are not shrunk because they plausibly have extra variability. We have hundreds to thousands of cells per library, so the gene-wise estimates are precise for well-expressed genes, and the deviations from the trend are exactly what we look for: our heterogeneous genes are the dispersion outliers of a single library. In single-cell data, sctransform [61] also regularizes NB parameters as a smooth function of the mean, but uses them mainly to normalize cells; the variance of its residuals is used to rank genes, without a calibrated cutoff; and Brennecke et al. [11] select genes whose CV^2 lies above a trend of the same form, $a_0 + a_1/\mu$, in which, unlike in our φ, the sampling noise is included.

Selecting heterogeneous genes. A residual above 0 does not yet make a gene heterogeneous. A homogeneous gene can lie above the trend for two reasons:

1. Its estimate is noisy (*sampling noise*), which matters most for rare genes; this is handled by a **chance level** obtained from simulated libraries.

2. It is intrinsically noisier than the median gene of its expression level (*intrinsic noise*, e.g., larger bursts), which matters most for well-expressed genes, where sampling noise is small; this is handled by an **effect-size floor** obtained from the spread of the real genes.

A gene is heterogeneous in a library if its residual reaches the larger of the two.

Sampling noise: chance level. Even relative to a sensible typical value, the residual of a rare gene is noisy. A fixed cutoff therefore does not work: a residual of at least 0.5 (3.2× typical) selects 953–1,977 genes in the libraries of more than 1,000 cells, but also 126–615 genes in a simulated library with the same depths s_i and gene means $\hat{m}$, in which every gene is drawn from an NB with the typical φ of its mean, i.e., in which no gene is heterogeneous; in the small high and oblong libraries, the simulation even reaches 54–78% of the genes selected in the data (Figure S2a). We therefore use such simulations to set the cutoff. Eight simulated copies of each library give, for every expression level, the 99.9% quantile of the residual of a homogeneous gene (running quantile over the approx. 1,000 genes of most similar mean, i.e., 8,000 simulated values; the residual is taken against the typical φ the gene was simulated with, so that the trend of φ with the mean does not enter the quantile). This chance level rises smoothly towards rare genes: at shield, from 2.4× typical above 1,000 UMIs to 13× at 10–15 UMIs. The simulations are random; each library has its own fixed seed (derived from its name), so the results are reproducible, and with other seeds, 2–4% of the genes selected in a library change (up to 15% in the smallest libraries; tested on four libraries).

Intrinsic noise: effect-size floor. The simulation gives every homogeneous gene exactly the typical φ, but real genes differ in their intrinsic noise even when they are expressed uniformly. Among well-expressed genes (at least 1,000 UMIs), the residuals of the bulk of the genes have a robust standard deviation of 0.26–0.38 (from dome on), approx. 3–4× more than in the simulation, while the chance level there is only 1.4–2.7× typical. The chance level alone would thus select every gene that is somewhat burstier than the median gene. We therefore require a minimum effect size, derived from an *empirical null* [27]: the residuals of the well-expressed genes (or of the 300 most expressed genes, if fewer are well expressed) are modelled as normal, with their median and robust standard deviation. For any floor, the estimated false discovery rate (FDR) is the number of well-expressed genes this null distribution would place above the cutoff, divided by the number of well-expressed genes that actually are above it. Each library gets three floors, the smallest ones with an estimated FDR of at most 50% (**weak**), 10% (**moderate**) and 1% (**strong**); a gene belongs to a tier if its residual reaches the larger of the tier's floor and the chance level. In the four libraries of high and oblong, there are hardly any well-expressed genes, and the 300 most expressed ones still carry much sampling noise (standard deviation 0.35–0.64), so their floors are high (0.83–1.49 for weak), and at oblong, the three tiers coincide.

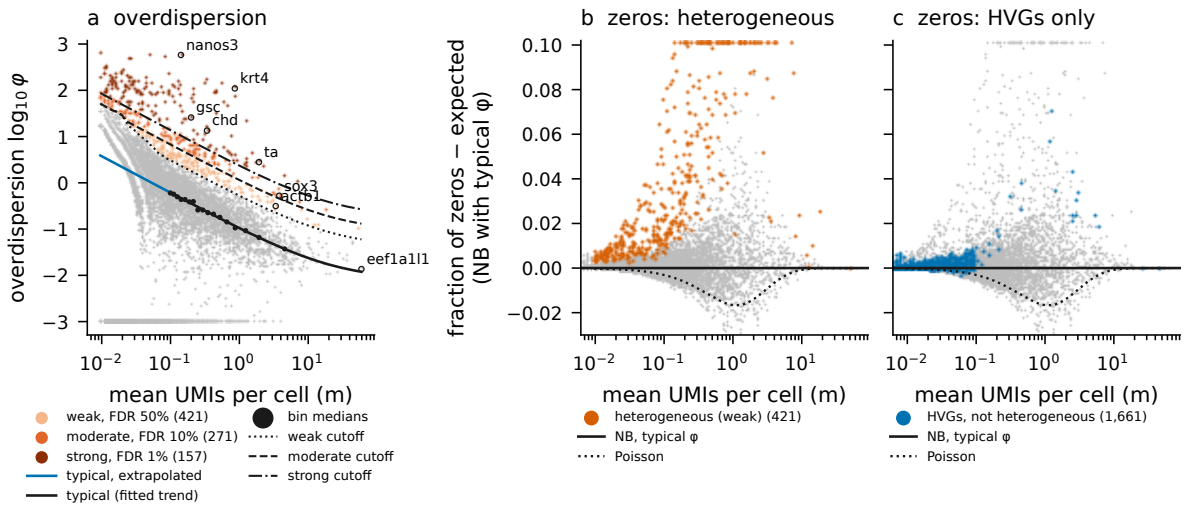


Figure 5: Heterogeneous genes in the shield library. (a) Estimated overdispersion against the mean (both logarithmic). Solid: typical value (fitted trend $a_0+a_1m^{-b}$), fitted to the bin medians (dots); blue where it is extrapolated to rarer genes. Dotted, dashed and dash-dotted: the cutoffs of the weak, moderate and strong tiers; light to dark orange: genes of these tiers. Estimates ≤ 0 are shown at -3. (b, c) Fraction of cells without a UMI of the gene, minus the fraction expected for a typical gene of the same mean (NB with typical φ, the library's depths), against the mean (logarithmic); values above 0.1 are drawn at the top of the axis. Dotted: the same for a Poisson gene. (b) heterogeneous genes (weak tier); (c) the HVGs of the library (Seurat, top 2,000) that are not heterogeneous. Grey: all other genes.

What the tiers mean. Table S2 summarizes the tiers and the two kinds of false positives. The sampling-noise false positives are counted in an independent, ninth simulation; the simulation cannot produce intrinsically noisy genes, because all its genes have the typical φ. The FDR, in turn, refers only to well-expressed genes, where the empirical null can be measured: only 17% (median) of the weakly heterogeneous genes are well expressed, so the intrinsically noisy ones are at most approx. 9% of the weak tier. Since the bulk of well-expressed genes also contains genes with mild, real differences between cells (e.g., gradients), its spread, and hence the FDR, is if anything overestimated. For rarer genes, intrinsic noise cannot be measured under the sampling noise; there, the chance level is already 4–13× typical, at or above two standard deviations of the intrinsic spread among well-expressed genes (approx. 4×), so it also excludes most of them.

Defaults. **Heterogeneous** without qualification means the weak tier. We use it by default, i.e., we accept that up to half of the well-expressed heterogeneous genes may be merely noisier than typical, in exchange for missing few real markers; the moderate and strong tiers are available where a smaller, more reliable set is needed. Rare genes are selected if their variation is large: at shield, *sox17* (endoderm) has only 26 UMIs, but a residual of 1.20 (16× typical), and is moderately heterogeneous. A gene is in a tier **at a stage** if it is in that tier in at least half of the stage's libraries (Table 3). The overdispersion, typical value and cutoffs of every library are shown in Figure S1.

Zeros. The same model also predicts how many cells should contain no UMI of a gene. For a cell with expected count $\mu_i = s_i m$, this probability is $(1+\varphi\mu_i)^{-1/\varphi}$ under the NB, and $e^{-\mu_i}$ under Poisson. A gene expressed in only some cells has *more* zeros than a typical gene of the same mean, because its UMIs are concentrated in fewer cells. Figure 5b shows these excess zeros at shield. Typical genes lie on the NB curve (0), and slightly above the Poisson curve,

which predicts too few zeros. The heterogeneous genes have up to 40% more cells without a UMI than expected (the figure is cut at 10%), i.e., they are absent from a substantial part of the library. The excess grows with the mean, as an expressed gene becomes increasingly unlikely to be missed by chance; at the lowest means, zeros are expected anyway, so the zero fraction carries little information, and the variance (panel a) is the more sensitive measure. Excess zeros are a consequence of heterogeneity, not a criterion: a gene can be heterogeneous with few extra zeros, if its variance comes from graded expression levels or from a few cells with many UMIs (36 of the 421 heterogeneous genes at shield have fewer extra zeros than an NB gene exactly at the weak cutoff would).

4.2 Why not highly variable genes?

The original Seurat HVG method [160] (Seurat v1/v2, still the default of scanpy [209]) normalizes every cell to 10,000 UMIs, takes $\log(1 + x)$, and ranks the genes by their dispersion (variance over mean) relative to genes in the same bin of mean expression, keeping the top 2,000. This has three problems:

- **Rare genes look variable.** After depth normalization and the logarithm, a gene with a single UMI in a few cells has a large variance relative to its tiny mean, and its bin contains only similarly rare genes. As a consequence, many HVGs are genes we cannot measure at all: per stage, 1,083–1,924 of the 2,000 HVGs are not heterogeneous, with a median of 5–22 UMIs in the *whole stage*, and 761–1,074 of them (38–54% of all HVGs) have fewer than 10 UMIs in every library (Table 3). Accordingly, the HVGs that are not heterogeneous have no more zeros than a typical gene: at shield, only 13 of these 1,661 genes have more than 2% extra zeros (median 0%; Figure 5c), against a median of 2.2% for the heterogeneous genes. Most of them are barely detected (median 0.005 UMIs per cell).
- **Depth is only removed from the mean.** Normalizing to a common total changes the scale, but not the sampling noise, which is larger in shallow cells; the log transform distorts it further [190]. The NB model accounts for depth explicitly.
- **Pooled libraries.** Selected on the pooled cells of a stage, a gene that differs between libraries (e.g., the heat shock genes of 50% epiboly DS4*a*, Section 5) is "variable" even if it is uniform within each library. Our test runs within each library.

Conversely, 82–610 heterogeneous genes per stage (from dome on) are not among the HVGs, although they are well expressed (median 120–1,578 UMIs in the stage). Newer HVG methods fix the first two problems in part; we compare with them in Section 4.4.

Replication. If a gene ranking reflects the cell types of a stage, it should be similar in two libraries of the same stage. Over all pairs of libraries of the same stage, the 1,000 genes with the highest residual overlap approx. twice as much (Jaccard index 0.27–0.50 from 30% epiboly on) as the 1,000 genes with the highest Seurat dispersion (0.14–0.29; Figure S2c). The same holds for the sets we actually use: the weakly heterogeneous genes of two libraries overlap 0.38–0.61, against 0.16–0.28 for as many top Seurat genes. At high and oblong, neither replicates (0.06–0.21).

4.3 Heterogeneous genes reflect known expression patterns

A gene selection that replicates better across libraries than HVGs could still be the wrong one, e.g., it could favour genes with some technical property. As an independent check, we ask whether genes known to be expressed in a restricted part of the embryo at a given stage are heterogeneous at that stage, and whether genes known to be expressed everywhere are not. The knowledge comes from the expression records of the Zebrafish Information Network (ZFIN) [10],

Table 3: Heterogeneous genes per stage. Weak, Mod., Strong: genes in the tier in at least half of the stage's libraries. The last three columns compare the heterogeneous (weak) genes with the 2,000 Seurat HVGs of the pooled stage (Both: in both sets); in brackets, the median number of UMIs of the genes in the whole stage.

Stage	Lib.	Cells	Weak	Mod.	Strong	Both	HVG only	Het. only
high	2	311	182	106	58	138	1,862 (22)	44 (62)
oblong	2	200	102	102	102	76	1,924 (14)	26 (37)
dome	1	1,174	429	258	175	347	1,653 (6)	82 (241)
30% epiboly	2	1,506	527	368	276	343	1,657 (7)	184 (340)
50% epiboly	4	5,864	823	602	453	570	1,430 (7)	253 (1329)
shield	1	1,046	421	271	157	339	1,661 (5)	82 (120)
60% epiboly	3	4,201	744	483	316	547	1,453 (9)	197 (1578)
75% epiboly	3	6,324	908	656	449	614	1,386 (8)	294 (1494)
90% epiboly	3	5,626	1,031	771	500	688	1,312 (8)	343 (949)
bud	4	7,162	953	676	422	617	1,383 (8)	336 (1405)
3-somite	1	1,631	899	608	301	612	1,388 (6)	287 (191)
6-somite	2	4,460	1,527	1,037	560	917	1,083 (5)	610 (264)

compiled from the literature and from large in situ screens. The numbers and table are produced by `scripts/01c_heterogeneity_zfin.py`.

ZFIN records. We use the wild-type expression records of ZFIN (downloaded on 2026-10-26; `scripts/download_zfin.sh`). Each record states that a gene was detected in an anatomical structure (e.g., "shield", "neural plate", "whole organism") during a range of developmental stages, by a given assay, in a given publication. We keep:

- **mRNA in situ hybridization** only. This is the assay that shows *where* in the embryo a gene is expressed; other assays (e.g., RT-PCR) record the whole embryo by necessity.
- **Publications before 2018**, so that the records are independent of the atlas [34]. The year is taken from ZFIN's publication list or, for direct data submissions such as the in situ screens of Thisse et al., from the date in the publication identifier.
- **Genes of the atlas**: ZFIN genes are matched to the gene symbols of the count matrix via their current symbol or an unambiguous former symbol (the matrix often uses older names).

This leaves 76,143 records of 8,350 genes (of 84,705 in situ records of 10,247 genes from 4,263 publications).

Stages. ZFIN stages are time intervals (e.g., "Shield" is 6–8hpf), and a record covers the time from the start of its first stage to the start of the stage after its last one. A record is assigned to an atlas stage if it covers the time of the atlas stage (the start of the ZFIN stage of the same name; 60% epiboly, which ZFIN does not distinguish, at 7hpf, i.e., within "Shield"). At each atlas stage, a gene is

- **restricted** if at least one of its records at that stage names a localised structure, i.e., any anatomy term except 15 broad terms ("whole organism", "unspecified", "ectoderm", "head", "trunk", ...; `data/curated/zfin_anatomy_programs.tsv`) and the yolk and YSL, which were removed with the yolk before dissociation;
- **whole organism** if all of its records at that stage name broad terms only, i.e., it was seen everywhere (or no more specific location was recorded).

Table 4: Heterogeneous genes and ZFIN expression records (in situ hybridization, before 2018) per stage. All measured: genes with at least 10 UMIs in one library of the stage, and the percentage of them that are heterogeneous. ZFIN: measurable genes that are restricted (recorded in a localised structure at that stage) or recorded in the whole organism only, and the percentage of them selected as heterogeneous (Het., weak tier), among the same number of top Seurat HVGs (HVG_k), or among the 2,000 top HVGs. High and oblong have too few restricted genes to be informative.

	All measured		ZFIN: restricted				ZFIN: whole organism only			
Stage	Genes	Het. %	Genes	Het. %	HVG_k %	HVG_{2000} %	Genes	Het. %	HVG_k %	HVG_{2000} %
high	9,515	2	15	47	27	53	3,105	1	1	9
oblong	8,062	1	10	40	10	50	2,887	0	1	10
dome	10,730	4	52	44	33	48	3,244	2	1	5
30% epiboly	9,978	5	131	61	40	57	3,204	2	1	5
50% epiboly	12,489	7	533	47	26	38	3,785	3	2	3
shield	8,370	5	552	35	18	36	3,295	1	1	6
60% epiboly	11,394	7	662	45	27	42	3,759	3	1	4
75% epiboly	11,847	8	652	51	29	44	3,729	4	1	4
90% epiboly	11,832	9	608	57	34	45	3,681	4	2	4
bud	12,623	8	813	48	27	40	3,840	4	1	3
3-somite	10,425	9	1,491	32	16	28	3,328	3	2	4
6-somite	11,992	13	1,763	40	23	27	3,567	4	2	3

We expect restricted genes to be heterogeneous, and whole-organism genes not. Neither expectation is strict: a restricted gene may be too rare to reach our cutoff, or expressed in a large domain (e.g., the whole neural plate), and a record may cover several stages of which only some show the restricted pattern. We compare three selections of the genes measurable at the stage (at least 10 UMIs in one of its libraries): the heterogeneous genes, the same number of top Seurat HVGs (HVG_k), and the usual 2,000 HVGs.

Results. Table 4 lists all stages. High and oblong are not informative: ZFIN records the blastula almost only as "whole organism", so there are only 15 and 10 measurable restricted genes, and these stages have few heterogeneous genes (Section 4); we exclude them from the summary. From dome to 6-somite:

- **Restricted genes are heterogeneous 3–12× more often than an average gene** (32–61% of them, median 46%, while the heterogeneous genes are 4–13% of the measured genes). In the moderate and strong tiers, 22–54% (median 35%) and 11–39% (median 26%) of them are selected.
- **The HVGs recover fewer restricted genes:** 16–40% (median 27%) for the same number of HVGs, and 28–57% (median 41%) for 2,000 HVGs, i.e., with 1.3–4.8× as many genes.
- **Whole-organism genes are depleted:** 1.4–4.4% of them are heterogeneous (weak), 0.3–0.5× the rate of an average gene, and 0.8–2.4% and 0.4–1.3% in the moderate and strong tiers. The method therefore does not simply select expressed or well-studied genes.

The recall is highest during gastrulation (30% epiboly: 61%; 90% epiboly: 57%) and lowest at 3- and 6-somite (32–40%), where ZFIN records many genes with organ-specific patterns that begin during segmentation and are recorded over long stage ranges (i.e., partially out of scope for the studied atlas).

4.4 Comparison with newer HVG methods

The test is a variant of an established idea: compare the variance of a gene with that of genes of similar mean under a count model, as do the technical-noise model of Brennecke et al. [11], the current Seurat default ("vst": the variance of the counts standardized by a fitted mean–variance trend [178]) and the analytic Pearson residuals [107] (the variance of NB Pearson residuals with one fixed dispersion). The comparisons of the previous sections use the oldest method, so we repeat them for the two newer ones, each on the pooled cells of a stage, and also with the genes ranked within each library and then combined (the recommended use with several batches). As usual, these methods first drop genes detected in fewer than 3 cells; we also show the old method with this filter. Figure 6 summarizes the stages from dome on (ranges in Table S3) (`scripts/01d_hvg_flavours.py`):

- **Most of the gap is the old method.** Of the ZFIN genes with a restricted pattern at a stage (Section 4.3), the top k genes of the old method contain 16–40%, those of both newer methods 25–56% and our ranking by the NB residual 29–59% (k = the number of heterogeneous genes of the stage). In replication, the top 1,000 genes of two libraries overlap 0.14–0.29 (old), 0.29–0.49 (vst), 0.24–0.46 (Pearson residuals) and 0.29–0.50 (NB residual; median pair per stage). The newer methods are close to our ranking: ours is ahead at most stages by at most 0.05, and behind at a few (e.g., Pearson residuals replicate better at bud). The filter of rare genes explains only a small part of the difference.

- **Rare genes remain.** Even the newer methods put 380–790 genes with fewer than 10 UMIs in every library of the stage among their top 2,000 when selected on the pooled stage (ranked per library, fewer at some stages, e.g., 1 at bud for Pearson residuals); in the top k, still 90–330 (pooled) or 0–256 (per library). Our test does not rank such genes at all.

- **The calibrated set is better than any ranking of the same size.** The heterogeneous genes contain 32–61% of the restricted genes, 2–9 percentage points more than the best ranking at every stage, also at single-library stages (shield 35% vs. 27–32%, 3-somite 32% vs. 25–29%), where it cannot profit from replication. The cutoff that depends on the mean (the chance level of Section 4) admits genes that a single ranking puts too low. And the calibration gives the number of genes (421–1,527 per stage), where an HVG method needs a fixed number such as 2,000.

We therefore keep the heterogeneous genes, but the advantage over modern HVG methods is moderate; most of what Table 3 shows is a weakness of the old method.

4.5 Consequences for the analysis

- **The heterogeneous genes of a stage replace the HVGs** wherever we need "interesting" genes: for embeddings and neighbourhoods, as evaluation targets, and as gene filters. By default, these are the weakly heterogeneous genes; where a smaller, more reliable set is needed (e.g., as markers of a population), the moderate or strong tier should be used. They are listed per stage and library, with their tier, in `results/tables/heterogeneous_genes.tsv.gz` and `..._calls.tsv.gz` (0: none, 1: weak, 2: moderate, 3: strong). The function `heterogeneity_calls` in the script applies the same calibration to any subset of cells (e.g., half of a library).

- **The two earliest stages have little detectable structure.** High and oblong have 182 and 102 heterogeneous genes, with 3–8 false positives per library: with 48–263 cells per library, only strong differences can be told from chance, and the rankings of their two libraries hardly overlap. This agrees with the expectation that cells before dome are largely equivalent (apart from small populations such as the germ plasm-carrying cells:

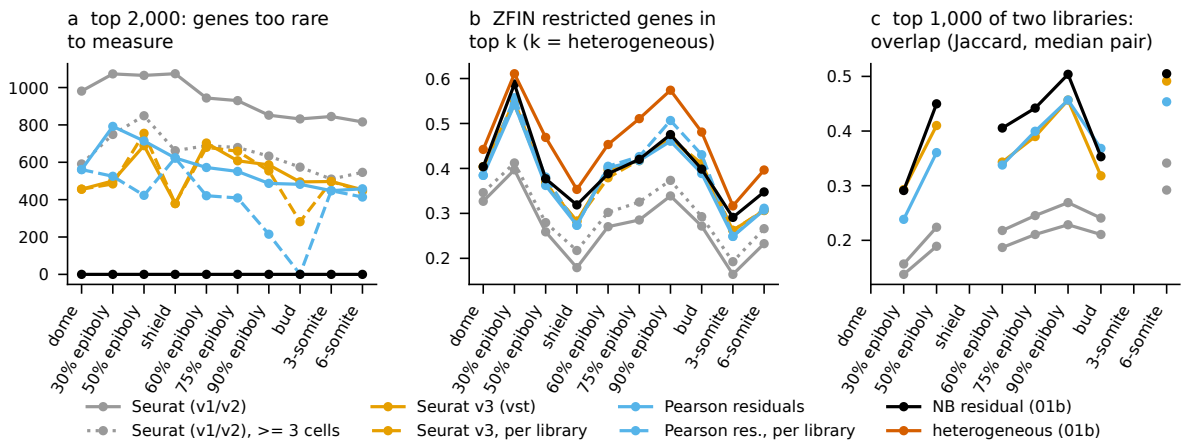


Figure 6: Heterogeneous genes vs. newer HVG methods, from dome on. (a) Genes with fewer than 10 UMIs in every library of the stage among the top 2,000 genes of each method (selected on the pooled stage, or per library and combined; dashed). (b) Share of the ZFIN genes with a restricted pattern at the stage among the top k genes, k = the number of heterogeneous genes. (c) Overlap of the top 1,000 genes of two libraries of the same stage (median over pairs; stages with two or more libraries).

nanos3 is strongly heterogeneous at every stage where it is measured), but it also means that we have little power there.

- **The number of heterogeneous genes grows over development,** from 421–1,031 between dome and bud to 1,527 at 6-somite, as cell types diversify. Shield (421) and dome (429) are single libraries; at stages with several libraries, a gene must be heterogeneous in at least half of them.

5 Result: a/b libraries have distinct expression profiles

Six (stage, batch) pairs have two libraries, a and b (Table 2). If the two libraries of a pair were interchangeable, we could merge them, which would remove one level of the library hierarchy and, in particular, give larger cell pools for the tiny high and oblong stages. In this section, we test whether the two libraries of a pair contain the same population of expression profiles. **They do not: in every pair, the libraries can be told apart, in some pairs as clearly as two neighbouring stages, and neither depth normalization nor depth matching changes this.** We therefore keep the a/b libraries separate. The numbers and figures are produced by `scripts/02_replicate_mixing.py` (mixing) and `scripts/03_replicate_differences.py` (gene-level evidence).

5.1 Experiment

For each pair, we embed the cells of both libraries together using a standard pipeline: each cell is normalized to 10,000 UMIs and log-transformed, the genes are restricted to those heterogeneous in either of the two libraries (Section 4; 74–1,527 genes per comparison), and the cells are projected onto the first 30 principal components using PCA. We do not use the usual highly variable genes: selected on two libraries jointly, they include genes that differ between the libraries as a whole rather than between cells, which would bias the test towards separation. In this space, we find the 20 nearest neighbours of every cell. If the libraries contain the same population of cells, the neighbourhood of a cell is a random sample of both libraries, i.e., the number of neighbours from library b follows a binomial distribution around the overall share of

b cells. If one library contains cells that the other lacks, or if all cells of one library are shifted in expression, the neighbourhoods are dominated by one library. We summarize this with three measures; the first two are standard metrics of batch mixing in single-cell data [119]:

- **Rejection rate** (in the style of kBET [13]): the fraction of cells whose neighbourhood composition differs from the overall composition (binomial test, $p < 0.05$).
- **Local inverse Simpson's index** (LISI [95]): the effective number of libraries in a neighbourhood (between 1 and 2). It is computed as $1/(p_a^2 + p_b^2)$, where p_a and p_b are the fractions of neighbours from each library, so a 50/50 neighbourhood gives 2 and a single-library neighbourhood gives 1. For simplicity, we weight the 20 neighbours uniformly instead of using the original distance-based weights.
- **Classifier AUC**: a logistic regression predicts the library of a cell from its principal components (5-fold cross-validation). This is a classifier two-sample test [117], a general statistical method rather than an established single-cell metric. A related idea was used by the authors of the atlas, who identified batch-specific gene modules by how well they predict the batch (ROC AUC, via Seurat's `BatchGene` function) [34]. An AUC of 0.5 means the libraries are indistinguishable, 1.0 means they are fully separable.

None of these has a natural threshold for "mixed well", and with thousands of cells, even a negligible difference becomes statistically significant. We therefore calibrate every pair against reference comparisons of the same size:

- **Permutation null**: the a/b labels are randomly shuffled (200 distinct repetitions) on the same neighbourhood graph. This is what perfect mixing looks like.
- **Negative control**: a random split of one library into two parts of the same sizes as the pair, analysed with the whole pipeline. This is what two samples of the same embryo pool look like.
- **Positive control**: library a against an adjacent stage of the same batch (e.g., 50% epiboly DS4 against 60% epiboly DS4). This is what a real developmental difference looks like.
- **Other batch** (where available): library a against the same stage from another batch (50% epiboly DS3, bud DS3).

Finally, we know that the libraries of a pair differ in sequencing depth (up to 3.3× in the median, Figure 4). Normalizing each cell to the same total removes differences in the average expression level, but not in sparsity: a shallow cell has more zeros, which can by itself move it in the embedding. We thus run every comparison twice:

- **Normalized**: all UMIs, only the per-cell normalization described above.
- **Thinned**: before normalization, each UMI of the deeper library is kept with probability equal to the ratio of the two median depths (binomial thinning [49]). Afterwards, both libraries have the same median depth, as if the deeper one had been sequenced less.

5.2 Results

The results are summarized in Figure 7. The full per-cell distributions of neighbourhood composition for all comparisons are in Figure S3, the UMAPs of all six pairs in Figure S4 (after thinning: `results/figures/replicate_mixing_umap_thinned.pdf`), and all numbers in `results/tables/replicate_mixing.tsv`.

- **The pipeline itself does not separate the libraries.** The random splits are indistinguishable from the permutation null (AUC 0.46–0.54, rejection rate at most 4%). AUCs slightly below 0.5 are not a systematic effect, but the noise of cross-validation, which goes in either direction.
- **All six pairs are clearly distinguishable.** The classifier AUC is 0.87–1.00, while the highest AUC in the permutation null is 0.52–0.65. Between 12% and 82% of cells have a neighbourhood dominated by one library, compared to at most 4% under the null.
- **The difference is sometimes comparable to a stage step.** At high, a against b is as separable as a against the adjacent stage (AUC 0.97 vs. 0.96). At 50% epiboly, DS4a is as different from DS4b as from DS3 (82% vs. 79% rejected neighbourhoods). The bud DS2 pair is the most similar (AUC 0.94, 12% rejected), but there, the libraries of different batches (DS2 against DS3) are also unusually similar (AUC 0.90, 24% rejected; after thinning even less separable than the pair, AUC 0.77 vs. 0.94).
- **Depth is not the cause.** In the normalized experiment, depth alone predicts the library with AUC up to 0.87 (oblong, where library a is 3.3× deeper; 0.76 at high). After thinning, this drops to 0.48–0.55 in all pairs, but the libraries remain separable (AUC 0.83–1.00).
- **The difference is spread over many genes.** Removing the 1,000 genes with the largest a/b fold change and repeating the analysis leaves five of the six pairs clearly separable (AUC 0.68–0.78, against at most 0.59 in the permutation null). Only the high pair becomes nearly indistinguishable (AUC 0.60, null up to 0.56).
- **The libraries share their structure, but are shifted within it.** In the UMAPs (Figure 7, bottom), both libraries cover the same branches, but within each branch, the cells of a and b occupy neighbouring, mostly non-overlapping regions. This suggests that most of the difference is a shift of the whole library rather than a cell population unique to one of them.

Consequently, merging the a/b libraries would mix two distinct expression profiles. Normalization or thinning does not make them compatible. If we need larger cell pools for some analysis, the two libraries have to be treated like two batches (i.e., with the same care as any other between-library comparison), not as one population. This agrees with the original study, which treated libraries as batches: at 50% epiboly (four libraries), 7 of the 20 gene modules it identified reflected batch effects and were removed, and the two 6-somite libraries were batch-corrected before trajectory reconstruction [34].

5.3 Hypotheses: why do the libraries differ?

The analysis above shows *that* the libraries differ, not *why*. Here, we collect the evidence for three plausible explanations. This is not meant to be exhaustive or conclusive. It is based on a pseudobulk comparison of each pair (mean expression per 10,000 UMIs, and the fraction of cells in which a gene is detected, computed after thinning the deeper library so that detection rates are comparable). With only one library per side, there is no replicate-level variance, so we do not report p-values. Table S4 lists the genes supporting each explanation; all genes with at least a 2-fold difference are in `results/tables/replicate_de_genes.tsv`.

Different parents (clutch effect). Each library is a separate pool of embryos, quite possibly from different parents. The supplement of [34] only states that the embryos come from wild-type crosses of the Tupfel longfin/AB background, a mix of two laboratory strains, and does not say whether the pools of one stage come from different crosses or days. Wild-type zebrafish lines are genetically heterogeneous, and before and shortly after zygotic genome activation, nearly

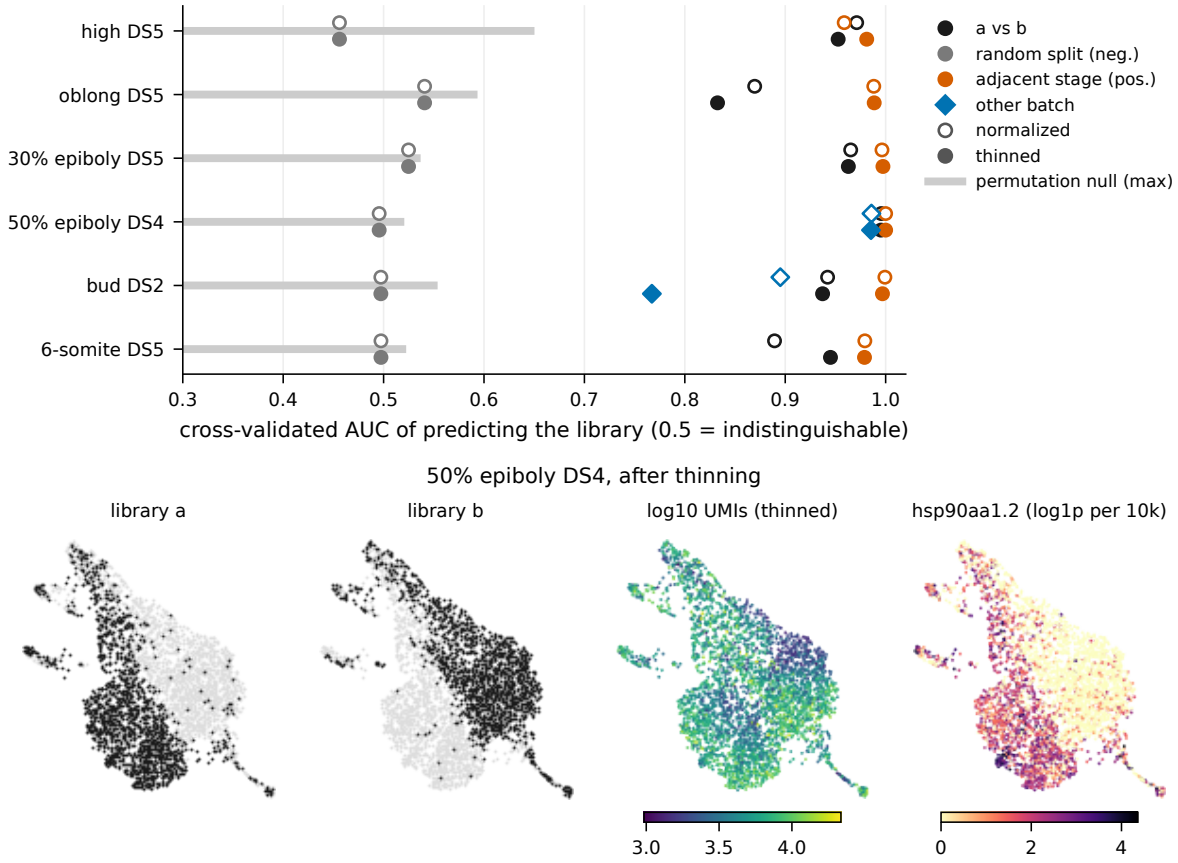


Figure 7: Mixing of the a/b libraries. Top: cross-validated AUC of a classifier predicting the library of a cell from its principal components, for each pair (a vs. b, black) and its reference comparisons; open markers without, filled markers with binomial thinning of the deeper library. The grey bar extends to the highest AUC obtained with randomly permuted labels. Bottom: UMAP of the thinned 50% epiboly DS4 pair, with separate panels for the cells of each library, the depth after thinning, and the expression of the heat shock gene *hsp90aa1.2* (Section 5.3).

all mRNA in a cell is inherited from the mother. A genetic difference between parents should show up as *on/off genes*, detected in many cells of one library and almost none of the other, regardless of cell type. Such genes exist in high (e.g., *gbgt1l4*: 6% vs. 71% of cells) and 50% epiboly DS4 (e.g., *si:dkey-108k21.10*: 49% vs. 5%), see Table 5. At high, library b also contains approx. 2–4× more of the maternal germ plasm transcripts *nanos3* and *dnd1*, which is consistent with a different maternal contribution. Checking whether the on/off genes lie in genomic regions known for copy-number variation would strengthen this explanation.

Handling stress. In the 50% epiboly DS4 pair, library a strongly expresses heat shock genes (*hsp90aa1.2* 16×, *hsp70.3* 9×, *serpinh1b* 7×). *hsp90aa1.2* is detected in 81% of the cells of library a, but only in 14% of library b, and it is high throughout library a in the UMAP (Figure 7), i.e., this is a shift of the whole library, not a subpopulation. Heat shock genes respond within minutes to hours, so this most likely reflects the handling of the embryos or cells (temperature, dissociation) rather than development. The protocol leaves room for such differences: dissecting one sample took up to 15 minutes, cells were dissociated mechanically, and droplets were held on ice for up to 1.5 hours to allow several sequential collections before reverse transcription [34], so two libraries of the same stage and batch may have been handled for different lengths of time. Note that this also affects any comparison involving DS4a, e.g.,

heat shock genes appear among the top differences between 50% and 60% epiboly in DS4.

Developmental timing. Embryos of one stage are collected by morphology: each dish was inspected visually, and 20–40 embryos were taken once they reached the desired stage [34]. Two pools of the same nominal stage can therefore differ by some minutes of development. We use two independent indicators of developmental time, each expressed in units of one "stage step" (the difference between the pair and an adjacent stage of the same batch, so that, e.g., +0.25 means that b is a quarter of a stage step older than a):

- **Ribosomal fraction**: the share of UMIs from the 79 ribosomal protein genes. Across the atlas, it increases monotonically with developmental time, from approx. 2% at high to 21–24% at bud, and within a library, it does not correlate with the depth of a cell (median Spearman correlation −0.05; `results/tables/library_ribosomal.tsv`). However, it no longer increases consistently between 3- and 6-somite.
- **Stage-marker score**: the ratio of UMIs in genes that are more than 2× higher in the older stage to UMIs in genes that are more than 2× higher in the younger stage (in both libraries of the pair; ribosomal genes excluded). For high and oblong, the reference stage is dome, so this score measures the progress of zygotic genome activation.

We consider timing supported only if both indicators agree (Table 5). This is the case for **bud DS2, where library a is older** (by 0.5 and 0.2 stage steps). This is consistent with library a containing more cells expressing the muscle genes of the adaxial cells (*myod1*, *myl10*, *actc1a*), which start to express muscle genes during late gastrulation, and fewer cells expressing *sox19b*, an early ectodermal SoxB1 gene. The low detection rates of these genes (at most 21%) suggest that this is a difference in the number of such cells, as expected if library a is slightly more advanced. Removing the 1,000 most different genes does not eliminate the difference (AUC 0.78), so timing explains at most part of it. In 50% epiboly DS4 (library b older) and oblong (library a older), the indicators agree in direction, but the marker score shift is small. At high and 30% epiboly, they disagree. In particular, at high, the zygotic score suggests that library b is *younger*, while its ribosomal fraction is higher. At the earliest stages, development changes the ribosomal fraction by less than one percentage point, and in both the high and oblong pairs, the library with the higher ribosomal fraction is the much deeper one, so we suspect a technical component and do not trust this indicator before dome. For 6-somite, neither indicator is usable, because neither changes consistently between 3- and 6-somite.

Unexplained differences. For 30% epiboly and 6-somite, we found no specific explanation: both are separable (AUC 0.97 and 0.89), but none of the three explanations is clearly supported. Furthermore, this analysis cannot separate a difference in cell type composition from a shift in expression within cell types. For example, the higher expression of EVL keratins (*krt8*, *krt18*, *cldne*) in oblong b comes with higher detection rates (4–7% vs. 12–21%), so library b probably simply contains more EVL-like cells. We will revisit this distinction once the cell types are identified.

6 Method: aligning libraries

Section 5 showed that libraries cannot be pooled: even two libraries of the same stage and batch are separable. Still, we need to compare cells across libraries, e.g., to ask whether a population found in one library is also present in another. We therefore *align* the libraries: we remove what differs between them on average, and keep how their cells differ from each other. This section defines the alignment, tests whether it does what we need, and states when it may be used.

Table 5: Summary of the a/b differences. On/off genes are detected in at least 40% of the cells of one library and at most 10% of the other (after depth matching). Ribosomal % is the median share of UMIs from ribosomal protein genes. Shifts are in stage steps, positive if library b is older; n/a if the indicator does not change between the pair and the reference stage. Timing is "a/b older" if both shifts agree and are at least 0.1, "weak" if they agree but one is smaller, and "conflicting" if they disagree. Reference stages: dome (high, oblong), shield (30% epiboly), 60% epiboly DS4 (50% epiboly), 90% epiboly DS2 (bud) and 3-somite (6-somite).

Pair	Cells (a/b)	On/off	Ribo. % (a/b)	Ribo. shift	Marker shift	Timing
high DS5	263/48	2	1.8/2.7	+0.58	−0.14	conflicting
oblong DS5	72/128	2	2.9/2.0	−0.70	−0.04	weak, a older
30% epiboly DS5	448/1,058	0	6.6/6.5	−0.10	+0.07	conflicting
50% epiboly DS4	1,567/1,580	3	7.8/9.1	+0.33	+0.10	weak, b older
bud DS2	250/506	0	24.3/21.3	−0.49	−0.20	a older
6-somite DS5	2,375/2,085	0	29.4/25.9	n/a	n/a	n/a

Alignment. Every library is transformed separately. A UMI count y of a gene in a cell is replaced with its Pearson residual:

$$r = \frac{y - s_i m}{\sqrt{s_i m + \varphi_t(m)\,(s_i m)^2}},$$

i.e., the deviation of the count from what the negative binomial model of the library (Section 4) expects for a homogeneous gene, in units of the expected standard deviation. Here m is the mean of the gene in *this* library, s_i is the sequencing depth of cell i relative to the library average, and $\varphi_t(m)$ is the typical dispersion of a gene of mean m in the library. This is the transformation of sctransform [61], with our own dispersion trend. Residuals are clipped at ± 10, the same limit for every library (a limit that depends on the library size makes two unequal halves of one library look different).

The residuals do most of the alignment. The model is fitted to each library separately, so the expected count $s_i m$ is built from the library's own gene mean. A gene that is, on average, twice as abundant in library A as in library B has twice the m in A, and its residuals measure each cell's deviation from its own library's mean. Since the s_i average to 1, the numerators $y - s_i m$ sum to exactly zero over the library's cells, so every gene has a mean residual close to zero in every library before anything else is done. The difference between library averages, whatever its cause (batch, a/b, timing, but also the composition of the library), is thereby removed. What remains are deviations from the average, scaled by the noise expected for the gene's mean, so they are comparable between libraries of different depth and dispersion. Fitting one model to the pooled libraries would instead keep the differences between library averages in the residuals.

Centring. The mean residual is close to zero, but not exactly zero: the denominator gives cells of different depth different weights, and the clip cuts off the largest values of genes expressed in only a few cells. We therefore subtract every gene's mean residual within its library (*centring*). The correction is small: before centring, the mean residual of a heterogeneous gene is a median of 0.02 of its standard deviation (90% of the genes below 0.11; checked at 50% epiboly and bud), and only a few sharply restricted genes reach 0.6–0.85. After centring, the average of every gene is exactly zero in every library. Centring shifts all cells of a library by the same amount, so it does not change the distances between them; the residual transformation does change them, as does any normalization.

By *alignment* we mean both steps together: per-library residuals, then centring. We use the genes heterogeneous at the stage; one principal component analysis (30 components) of all libraries of the stage gives shared axes.

What we test. We compare the aligned data with the unaligned embedding of Section 5 (joint log-normalization and PCA of the compared libraries, on the same genes), in four ways (Figure 8):

1. **Libraries of a stage mix.** For every pair of libraries of a stage, the share of cells whose 20 nearest neighbours are significantly unbalanced between the two libraries (as in Section 5), compared with a random split of one library. Aligned, the random splits give 1–6% (as unaligned), the a/b pairs 1–28% (median 18%; unaligned 12–82%, median 39%), and pairs from different batches 1–36% (median 18%; unaligned 4–95%, median 63%). The alignment removes most, but not all, of the difference: in about half of the pairs, 20–36% of the neighbourhoods remain unbalanced (e.g., 50% epiboly DS4 a/b, 6-somite a/b, 50% epiboly DS2 vs. other batches).

2. **Stages stay apart.** Pairs of libraries of different stages from the same batch (so that the batch does not contribute), 800 cells each. Aligned, stages approx. 1 h apart are blurred (11–44% unbalanced, median 29%; unaligned 35–97%), stages 1.3–2 h apart mostly separate (57–86%, median 73%), and stages more than 2 h apart separate nearly as well as without alignment (78–96%, median 92%). The alignment removes the stage averages too, but cells of distant stages still have no neighbours in common: their populations and gene combinations differ.

3. **Structure within a library is kept.** For every cell, the share of its 20 nearest neighbours in the shared PCA that are also among its 20 nearest neighbours in a PCA of its own library alone: 58–100% (median 79%). The rest is due to the shared choice of axes. The unaligned joint embedding keeps much less (32–73%, median 47%), because its axes also describe the differences between libraries.

4. **Cell types find each other.** Six populations with unambiguous markers, at the stages where they are unambiguous (EVL up to shield, primordial germ cells, endoderm, notochord from 75% epiboly, hatching gland from bud, and the p53-like cells), called in every library from a marker score (positive if above the 99.9% quantile of the scores of random gene sets with the same expression levels); the marker genes are removed from the features. For every pair of libraries, each cell of one library takes its 20 nearest cells of the other, and we ask how well the share of marker-positive neighbours predicts the cell's own call (AUROC). Aligned, the median AUROC is 0.96 (EVL), 0.99 (endoderm, notochord), 0.98 (hatching gland), 0.88 (p53-like) and 0.71 (PGCs, only approx. 10 cells per library), the same as unaligned. Populations that are distinct in every library are found by both; the alignment does not lose them.

Pictures. Figure S5 shows the aligned libraries of every stage; the libraries overlap. Figure S6 shows, on the same coordinates, the marker-called populations of the cell type check in every library (the marker genes are not among the features, so the islands are not drawn by the markers themselves). Where a population is distinct, it forms the same island in every library of the stage: the EVL from dome to 50% epiboly, the p53-like cells at 50% epiboly, the endoderm from 50% epiboly to bud, the PGCs at 75% epiboly, the notochord from 75% epiboly and the hatching gland from bud. Elsewhere, the called cells are partly scattered: part of the EVL at 30% epiboly and shield, the p53-like cells after 60% epiboly, and the PGCs at most stages (approx. 10 cells per library), which matches their lower transfer AUROC (Figure 8d). Figure 9

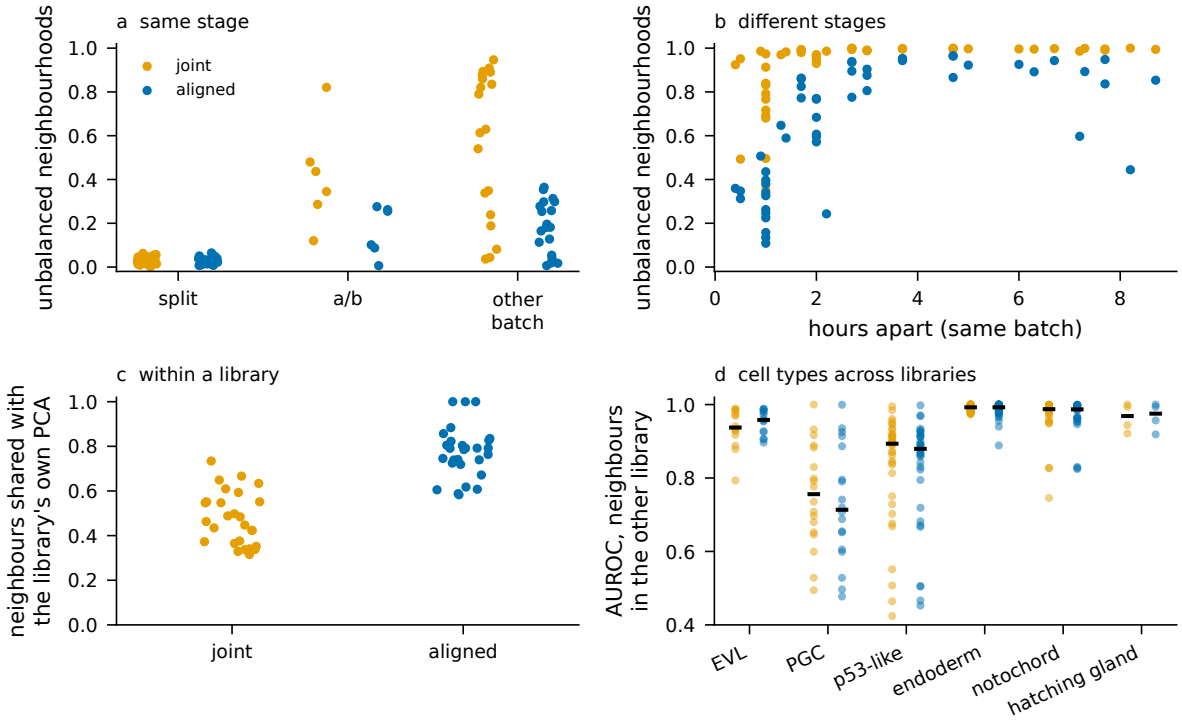


Figure 8: Checks of the alignment (blue) vs. the unaligned joint embedding (orange). (a) Share of cells with unbalanced neighbourhoods for pairs of libraries of one stage, and for random splits of one library. (b) The same for libraries of different stages of one batch, by the time between the stages. (c) Share of a cell's 20 nearest neighbours that are also its neighbours in a PCA of its own library. (d) AUROC of the share of marker-positive neighbours in another library of the stage, per population (black: median).

shows the whole atlas, aligned and unaligned. Unaligned, the atlas is ordered by stage (the DS5 libraries, which hold the earliest and the latest stages, sit at both ends). Aligned, the batches overlap (panel b), and the stages form a gradient rather than separate regions: centring removes the stage averages, so stages separate only where their cells differ, and adjacent stages partially overlap. The aligned atlas is therefore not a developmental time line, and its picture should not be read as one. The other side of the same effect is shown by the marker populations (panels c and f): aligned, each population forms one island across all the stages where it is called (e.g., the endoderm from 50% epiboly to bud, the notochord from 75% epiboly on, and a compact island of p53-like cells), while unaligned, the endoderm is split between early and late stages and the p53-like cells are scattered over the stage gradient. The aligned atlas groups cells by what they are rather than by when they were sampled.

When to use which.

- **Aligned data** for everything that compares *cells* across libraries: finding a population of one library in another, joint neighbourhoods and pictures of a stage, and links between populations of nearby stages. Even here, two caveats: 20–36% of the neighbourhoods of some pairs remain unbalanced, so conclusions should not require libraries to be interchangeable, and stages approx. 1 h apart are partly blurred, so a match between nearby stages is weaker evidence than a match within a stage.
- **Original libraries** for everything about library *averages*: library-level programs, pseudobulk expression, a/b differences, timing, and the composition of a library (how many

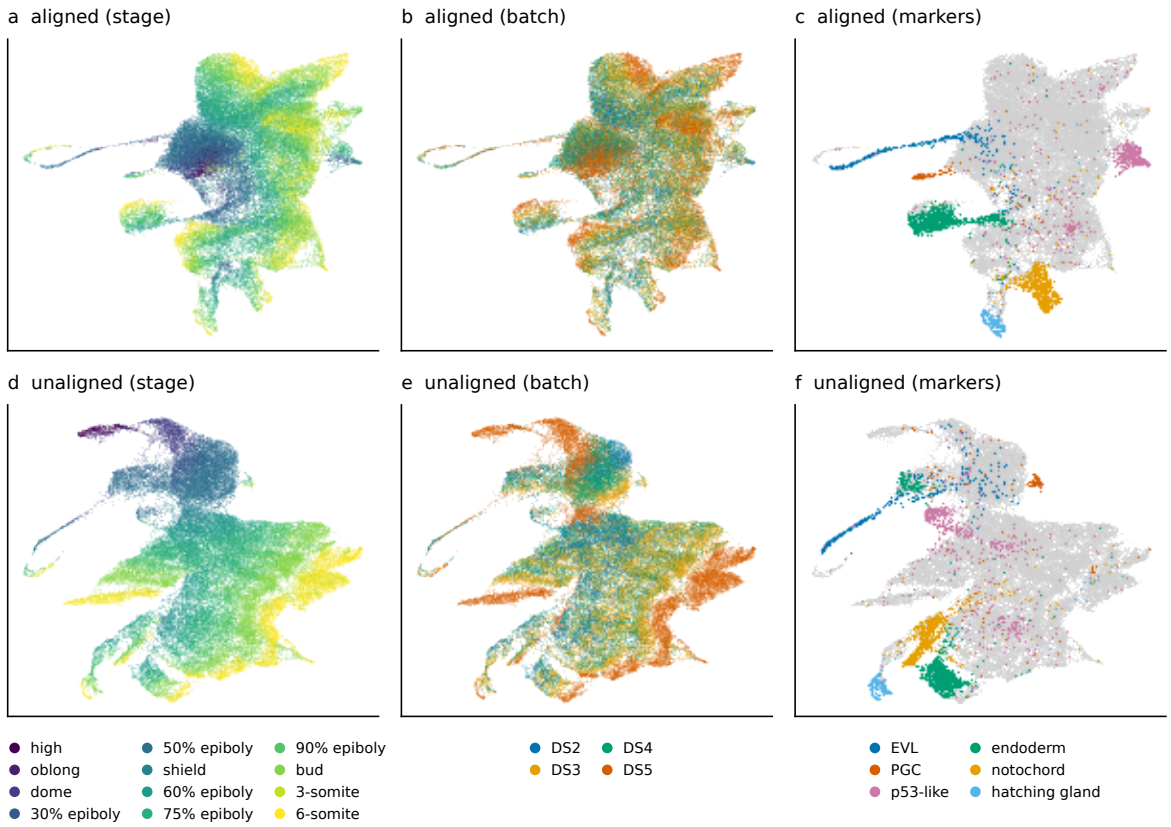


Figure 9: UMAPs of the whole atlas on the genes heterogeneous at any stage (marker genes excluded). Top (a–c): aligned (per-library residuals, centred); bottom (d–f): unaligned (log-normalized, scaled). Colour: stage (a, d), sequencing batch (b, e), and the marker-called populations of Figure S6 (c, f; grey: no call, or a stage where the population's markers are not used).

cells a population has). The alignment removes all of these by construction.

7 Result: expression programs

An *expression program* is a set of genes associated with one cell type, tissue, cell state or signalling pathway at one stage, e.g., the genes recorded in the notochord at bud, or the feedback targets of BMP. Programs may share genes, and a cell may have several active programs. A program is a hypothesis about co-expression, not a cell type: whether its genes vary together in the cells is tested in Section 8. In this section, we define several sets of programs, from the data and from prior knowledge, describe how the curated set is built, when its genes vary between cells, and how strongly each program is expressed in whole libraries. Section 8 asks whether each program is visible in the data and how much of the variation between cells each set explains, and Section 9 relates the programs to the signalling profiles.

7.1 Program sets

A program set is a collection of programs that together describe the cells of a stage (Table 6; `scripts/16_program_sets.py`). A program contains only genes measured at the stage (at least 10 UMIs) and has at most 30 genes (the number listed per module by [34]); the gene lists of every set are in `results/tables/program_sets/`. The sets differ in how their programs relate across stages (column "Programs" of Table 6):

- **Per stage, unrelated**: the programs of a stage are unrelated to those of the next.
- **Per stage**: a program keeps its name (an anatomy term, a pathway), but its genes are chosen anew at every stage, and it exists only at the stages where it has enough genes.
- **Fixed**: a program has one gene list, used at every stage, so that its score means the same thing at every stage; only genes not measured at a stage are excluded.

Reference sets. These are compared with the curated set (and inform future curation decisions) but cannot be used as model phenotypes, because their genes vary per stage:

- **Farrell et al. modules** (per stage, unrelated): the gene modules that [34] found by non-negative matrix factorization in each stage of this dataset (their Table S2: the 30 listed genes of each module, with weights). They were found in these cells, so they are informed solely by data. Consequently, they are are expected to predict the data well, but may not have clear biological interpretations.
- **ZFIN terms** (per stage): one program per anatomy term of ZFIN [10], with the genes that have an in situ (i.e., localized) record in the term at the stage. Broad terms (whole organism, unspecified, ...) are left out, and a term needs at least 3 genes. Nested terms (e.g., periderm within the EVL, the rhombomeres within the neural plate) are kept as separate programs. **A second version that keeps only the genes heterogeneous at the stage is also considered**. As a sanity check, the heterogeneous version was also run with the records published before 2018 only, which are independent of the cell atlas; it behaves like the all-years version (Section 8.1) and is not reported further.

The curated set (fixed; used in the rest of the report) combines four kinds of programs, built as described in Section 7.2:

- **Anatomy**: ZFIN terms grouped into curated, named programs, e.g., shield, embryonic shield and organizer into *dorsal organizer*. The result is adjusted to minimize gene overlap between programs. This process is automated, with small but notable human interventions documented in Section 7.2.
- **Signalling profiles**: genes that indicate (a) direct targets of a pathway (transcription directly downstream of the receptor; typically first response); (b) feedback inhibitors (modulate signal strength locally; typically delayed); (c) ligands that carry the signal (a necessary condition for sending it); (d) secreted antagonists and degrading enzymes (which block the signal at short range). Some of these indicate that a cell *receives* the signal (a, b), others that it *sends* (c) or *blocks* (d) it, and some do both (e.g., Nodal ligands are also Nodal targets). Where these genes are present, the pathway has one *general* profile; where not, it has separate *receiving*, *sending* and *blocking* profiles (Section 7.2).
- **Cell states**: maternal transcripts, the pluripotent blastomere state, S and G2/M phases, heat shock and the p53 response (a literature-curated table, `data/curated/archive/`). These are absent from ZFIN's anatomy terms, but represent behaviors that should be visible in the dataset through heterogeneous genes. They do not necessarily need to form coherent subpopulations (e.g., cell cycle should activate across cell types).

The curated programs are disjoint, except along presumed derivations: a program may share genes with the programs it derives from (e.g., the heart with the lateral plate mesoderm, the hindbrain with the neural plate), but not with any other (Section 7.2). The signalling profiles and the cell states are also tested as sets of their own (**Signalling**, **Cell states**; fixed, as in the curated set). The literature behind every readout, ligand and antagonist is in Table S5 (autogenerated from the associated `.csv` file).

Set	Across stages	Prog./stage	Genes/prog.	Genes/stage	Genes
Farrell et al. modules	new per stage	10–45	30	249–514	887
ZFIN terms	per stage	1–110	14	7–780	1,268
ZFIN terms, het.	per stage	2–103	10	6–515	777
ZFIN, het., pre-2018	per stage	2–101	10	6–505	760
Signalling	fixed	9–14	3	20–49	50
Cell states	fixed	6–6	10	40–51	51
Curated	fixed	39–45	10	298–599	616
Controls	fixed	5–7	4	33–65	73

Table 6: Program sets: how programs relate across stages (Section 7.1), programs and distinct genes per stage (range over stages), median genes per program, and distinct genes over all stages; het.: restricted to heterogeneous genes.

Controls (fixed) are programs that should not describe cell identity. In other words, it is normal for these to be present, but unusual for them to form well defined subpopulations (or even be heterogeneous in some instances).

- **Whole organism**: genes that ZFIN records in the whole embryo from high to shield and never in a localized structure up to 6-somite (the 30 with the most publications); they should be active, but largely uniform.
- **Silent pathways**: readouts of pathways that should be inactive or uniform at these stages (type I interferon, aryl hydrocarbon receptor, glucocorticoid receptor, circadian clock; Table S5).
- **Technical: dissociation stress**: the immediate early genes induced by enzymatic dissociation [195]. These are expected to be active in surface cell populations, since they flag cells stressed during sample preparation, but should not form a distinct population. They differ from the heat shock and p53 states: the p53 response takes hours and reflects DNA damage or apoptosis in the embryo before collection (it is a subpopulation across multiple stages and libraries); meanwhile, heat shock is induced more slowly than the immediate early genes and should be mostly expressed uniformly within a library.
- **YSL**: ZFIN's yolk syncytial layer terms, grouped as the curated programs, without the genes of any curated program or other control. The embryos were deyolked, so YSL cells should be absent, or present only in small numbers.

A control for differences in age between the embryos of a library (genes that change between adjacent stages of one batch) was tried and dropped: few genes change between nearby stages, none at all from 60% epiboly to bud, and the only coherent list was an uncharacterized gene family that marks a population (*si:ch211-113a14*; Section 7.4), not an age difference.

The number of programs per stage differs between the sources (Figure S7): ZFIN terms are few up to 30% epiboly (1–13 per stage, against 10–17 Farrell et al. modules), about as many as the modules from 50% to 90% epiboly, and more from bud on (68–110 vs. 31–42). Of the 45 curated programs, 39 have a measured gene at high and all from 75% epiboly on; a heterogeneous gene, 31 at high and 26 at oblong, and 37–45 from dome on.

7.2 Curated program structure

The curated set is the one later used to identify subpopulations and candidate phenotypes. Every curated program is defined at every stage, using a shared gene list; The gene list is based on ZFIN and other literature, but the program activity is entirely tied to stages expressing the listed genes (i.e., a program based on ZFIN can be tested also at stages not reported via ZFIN).

Curation choices. The curated set is built by `scripts/16_program_sets.py` from the term mapping `data/curated/zfin_anatomy_programs.tsv`:

- **Grouping.** ZFIN anatomy terms are grouped into programs that keep their name across stages, e.g., shield, embryonic shield and organizer into *dorsal organizer*. A gene belongs to a group if it has an in situ record in one of the group's terms and, at some stage, the group holds at least a third of the gene's records, so that a gene recorded mostly elsewhere does not join a group through a single record. Genes satisfying this criterion, but recorded in several groups over time are excluded. The groups of terms are defined manually.

- **One gene list per program.** In the grouped ZFIN programs, genes chosen anew at every stage change so much that only the name persists: e.g., an EVL program has 69 different genes over its 10 stages, none of them at all stages. This is not a suitable phenotype definition. Every program therefore has a single gene list, used at every stage: (a) the genes of the group's records from high to 6-somite, pooled; (b) to be included, a gene must be heterogeneous at some stage; (c) genes are ranked by their mean heterogeneity over all stages (0 = not heterogeneous, 1–3 = weak to strong; Section 4), then by their number of ZFIN publications. The list is truncated to 30 genes (median 30; the heart has 5).

- **Splits.** Groups whose identity changes the most over time are split into smaller programs (column `fixed_program` of the mapping table): marginal mesendoderm into *margin* (germ ring, margin) and *mesoderm* (mesoderm, hypoblast), the notochord into *axial mesoderm* and *notochord*, and the prechordal plate into *prechordal plate* and *hatching gland* (polster, hatching gland). This is mostly due to how ZFIN records are organized, where the same term between stages can genuinely represent a different phenotype (i.e., this cannot be addressed simply by more granular grouping).

- **Left out: ZFIN's maternal and blastomere terms.** The maternal records (egg, cleavage) end before the first atlas stage, so there is no ZFIN maternal program. The blastomere terms (blastomere, blastoderm, deep cells, epiblast) collect patterned genes that determine future cell identity, rather than the undifferentiated pluripotent state: even limited to the stages up to shield, the group is led by ventral, non-neural and EVL genes (*gata3*, *tfap2a*, *foxi1*, *vox*, *klf2a*), and later by ventral and marginal genes (*bmp2b*, *gata2a*). The literature sourced programs describe the maternal and blastomere states instead.

- **Signalling evidence.** Every readout, ligand and antagonist has a literature record (Table S5; `data/curated/signalling_evidence.tsv`). Of the 43 readouts, 26 rest on zebrafish evidence of a direct or loss/gain-of-function response (strong) and 17 on indirect zebrafish or other vertebrate evidence (moderate); the 21 ligands and 8 of the 11 antagonists have strong evidence.

- **Signalling profiles.** A profile's score is the mean of its genes (see Section 8.1), so a profile is useful only if its genes are expressed in the same cells; a general profile for a pathway whose receivers and blockers are different cells would average genes that are anti-correlated, and ignore both groups. We therefore chose the profiles of each pathway by where the literature places its genes (Table S5):

  - **General**: *Nodal* (the ligands are themselves Nodal targets, expressed at the margin with *lefty*), *BMP* (the ventral ligands *bmp2b*, *bmp4*, *bmp7a* are BMP targets, expressed where BMP is read), *Wnt* (*wnt8a* and *wnt3a* at the margin, with the readouts) and *FGF* (*fgf8a*, *fgf24*, *fgf3* and *fgf4*, mostly at the margin, with the readouts). Here, sending and receiving a signal cannot be distinguished, so the profile only marks activity. That readouts and ligands mark the same cells is tested by the coherence of the profile: all four are supported at every stage from 60% epiboly on (Section 8.1).

  - **Blocking**: the BMP antagonists (*chd*, *nog1/2*, *fsta*), the Wnt antagonists (*dkk1b*, *frzb*, *sfrp1a*) and RA degradation (*cyp26a1*). These cells block a signal for their neighbours and are expected to have low signalling themselves.
  - **Sending**: *admp*, the BMP ligand of the organizer (where BMP is low), RA synthesis (*aldh1a2*, in the paraxial mesoderm, away from the cells that degrade RA), the Delta ligands (in lateral inhibition, senders and receivers are neighbouring but different cells), and two pathways without usable readouts: Hedgehog (its readouts are never heterogeneous) and Wnt/PCP (no transcriptional readout).
  - **Receiving**: Notch (the *her* and *hey1* targets) and RA (*dhrs3a*, a weak readout).

  Two rules handle genes that do not fit a single profile. **A gene with two roles in one pathway** goes to the profile where it is expressed (i.e., observable through RNA): *szl* (BMP feedback target and antagonist), *lefty* (Nodal feedback target and antagonist) and the Nodal ligands (also Nodal targets) are expressed where their pathway is active, so they belong to the general BMP and Nodal profiles; *cyp26a1* is induced by RA but expressed where RA is degraded, following anterior identity [23, 100], so it marks RA blocking. **A readout of two pathways** is not used, because its expression cannot be attributed to either: *ved*, *vox* and *vent* (BMP and Wnt8 [148]), *nkd1* (Wnt and BMP), *nrarpa* (Notch and Wnt) and *nrarpb* (Notch and BMP [53]). These genes are not reserved for the signalling profiles, so the anatomy programs can still use them (*ved*, *vox* and *vent* are in the margin, mesoderm and tailbud programs).

  As the anatomy programs, a profile is defined at every stage, with the genes of the profile that are heterogeneous at some stage. The expected relations between the separate profiles (e.g., low BMP where BMP is blocked or *admp* is sent, Notch receivers next to Delta senders) are tested in Section 9.2 (Table 7).

- **Disjoint programs.** A gene shared by two programs makes each score respond to the other's cells, so the curated programs share no genes: the genes of the signalling profiles, cell states and pathway controls are not allowed in the anatomy programs, and a gene in two anatomy programs stays only in the one with more ZFIN publications for it (in case of a tie, in the program where it ranks higher). The exception are **derived programs**: a program may share genes with the programs it derives from, directly or not (column `derives_from` of `data/curated/curated_programs.tsv`; the arrows of Figure 10). The derivations are prior knowledge, an assumption of the model, but none of them is controversial: e.g., the organizer from the margin, the axial mesoderm and the prechordal plate from the organizer, the notochord from the axial mesoderm, the hatching gland from the prechordal plate, the presomitic, lateral plate and intermediate mesoderm from the mesoderm, the heart from the lateral plate, and the neural regions from the neural plate. Without this exception, a program would lose exactly the genes that its precursor passes on. However, this also allows the more differentiated cells to be labeled by their precursor program (i.e., margin program is still active in the dorsal organizer). This underscores the fact that cell identity is rarely a single program, but a *combination* of programs.

  Of the genes within the original limit, 91 are excluded from the anatomy programs due to a signalling, state or control program, and 188 were shared with an unrelated anatomy program, 65 of them with equal evidence (Table S6). The programs that shrank most are the primary neurons (30 to 22 genes), the intermediate mesoderm (23 to 15), the organizer, the endoderm (both 30 to 23) and the neural crest (27 to 20).

- **Programs left without specific genes.** Three groups lost most of their markers to the signalling profiles or to other programs, so that what remained was not specific enough. ZFIN has a *ventral mesoderm* term, the ventral side of the dorsal–ventral decision, but its markers are BMP readouts and ligands, so it overlaps BMP signalling too much: what

remained were mostly genes with a single publication. The ventral side is described by the general BMP profile instead, and the program is dropped. The *floor plate* lost *shhb* and *lft2* to the Hedgehog and Nodal profiles and its other genes to programs with equal evidence, leaving 2 genes; it is dropped (Hedgehog sending marks the midline). The *presumptive neural* program (presumptive neural plate, neuroectoderm, presumptive brain) mixed anterior and posterior markers (*otx2*, *hoxb1b*, *hoxc10a*) and started after the neural plate; its terms are merged into the neural plate. Two more groups have only three genes, none of them a known marker, and are left out: the head mesoderm (*tpm4a*, *cdc42ep4a*, *prrx1a*) and the hemato-vascular group (*has2*, *ets2*, *ef1*; recorded from 3-somite).

- **Manual assignments.** A gene can be assigned to a program against its ZFIN records (`data/curated/gene_overrides.tsv`, with reason and references). This is used once: *dharma* (*bozozok*), the earliest organizer gene [213], which ZFIN records mostly in the YSL (it is expressed in the dorsal YSL beneath the shield). It is assigned to the dorsal organizer and so leaves the YSL control; it is heterogeneous from high to dome, before the organizer's ZFIN window.

What each program is. Table S7 (supplementary) describes every curated program: its source, the program of Table 1 it corresponds to (the curated set extends Table 1 beyond the early decisions, e.g., to the neural regions, somites and neural crest), the programs it is presumed to derive from, where and under which signals it arises, why it is part of the model and how to interpret it. The annotation is kept in `data/curated/curated_programs.tsv`, and the table is generated from it (`scripts/24_program_overview.py`).

7.3 Curated programs over time

Expression overview. Figure 10 shows the curated programs over the stages: how many of each program's genes are heterogeneous at each stage and how strongly (their mean tier, per Section 4), the stages at which ZFIN records the program (its *ZFIN window*: from the first to the last stage with at least 3 of the group's genes recorded), and, as indentation and arrows, the presumed derivations between programs. It compares the literature (the windows) with the data (the marker heterogeneity). ZFIN records no anatomy program at high and oblong, only the EVL and the margin at dome, 7 programs at 30% epiboly, 11 from 50% to 60% epiboly, 15–16 at 75–90% epiboly and 22–23 from bud on. In the data, the programs vary much earlier: 12 of the 25 anatomy programs have at least 3 heterogeneous genes at high (8 at oblong, with fewer cells), and 21–25 from dome on. Inside its ZFIN window, a median of 81% of a program's genes are heterogeneous (62% at the worst stage, 50% epiboly); outside it, 43%. In other words, gene heterogeneity is associated with ZFIN documented activity windows, but can appear earlier and last beyond the reported window. The reasons are discussed with the observations below.

Importantly, Figure 10 shows where a program's genes vary between the cells of a library. A program expressed evenly in every cell of a library does not vary, and so is invisible there (and in the program scores of Section 8.1, which are centred per library), and so is any change of a program between libraries. To see both, we compute the pseudobulk strength of every program in every original library (Figure 11; `scripts/25_program_pseudobulk.py`): the UMIs of each gene summed over the cells of the library, per 10,000 UMIs of the library (which removes library size, i.e., the number of cells and the depth), relative to the gene's peak (its highest stage, mean over the stage's libraries), and averaged over the program's genes (genes with a peak below 0.1 per 10,000 are excluded). Every program uses its gene list at every stage. The strength does not separate how many cells express a program from how strongly they express it. Libraries of the same stage agree closely (median difference 0.07), except for a few controls and states (e.g., heat shock in one library at 50% epiboly, interferon at 30% epiboly).

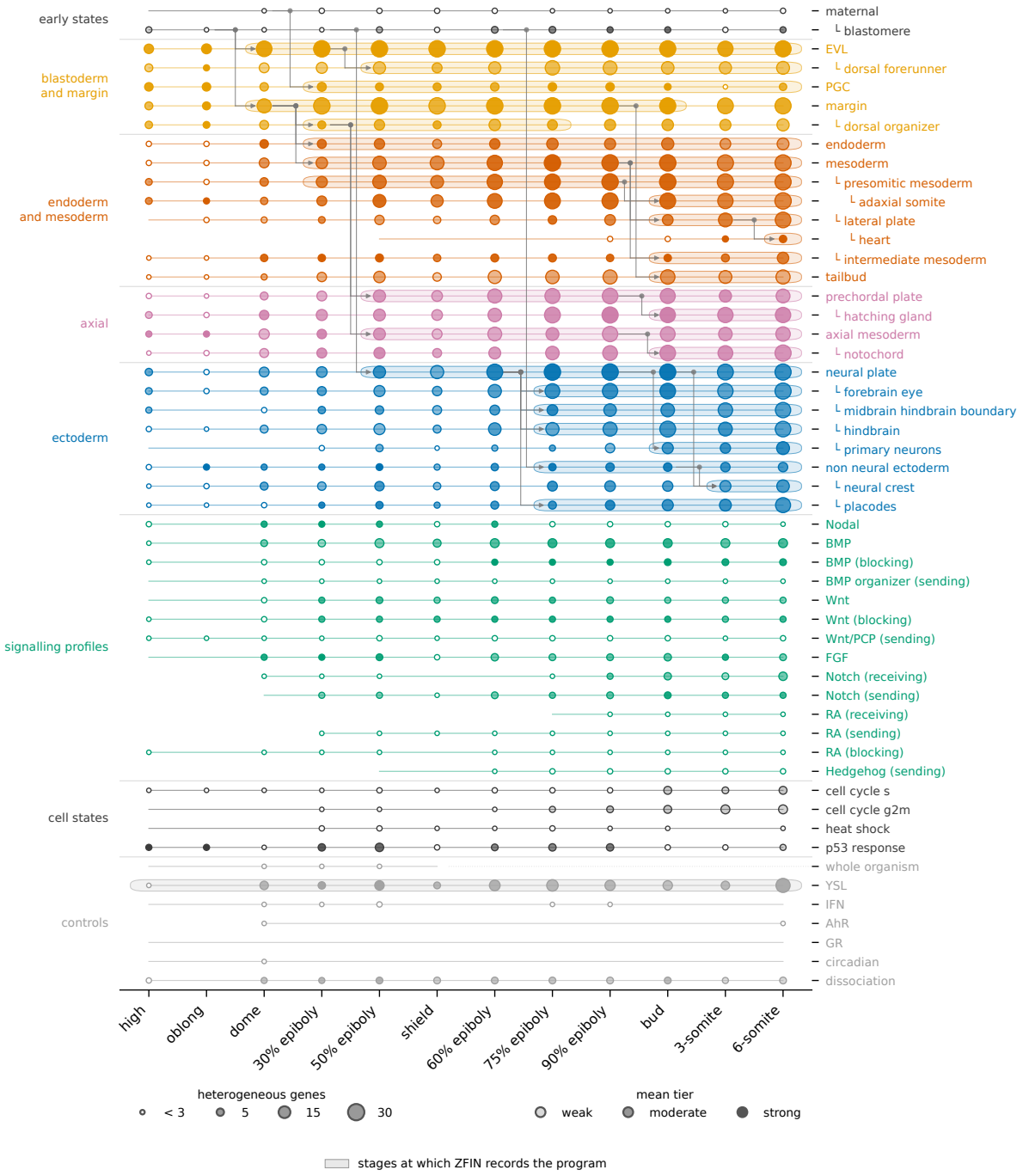


Figure 10: Curated programs and controls over the stages. Marker area: the number of the program's genes heterogeneous at the stage (none: no marker); fill: the mean tier of these genes (light: weak, dark: strong; per Section 4; below 3 genes, circle is always open as tier is not informative). Box: the stages at which ZFIN records the program (where ZFIN records are available). Names on the right. Indented: a program derived from the one above it in the same group. Groups of programs on the left, also by colour. Arrows: presumed derivation (Table S7), shown at the start of the ZFIN window; none if the source does not start earlier than target.

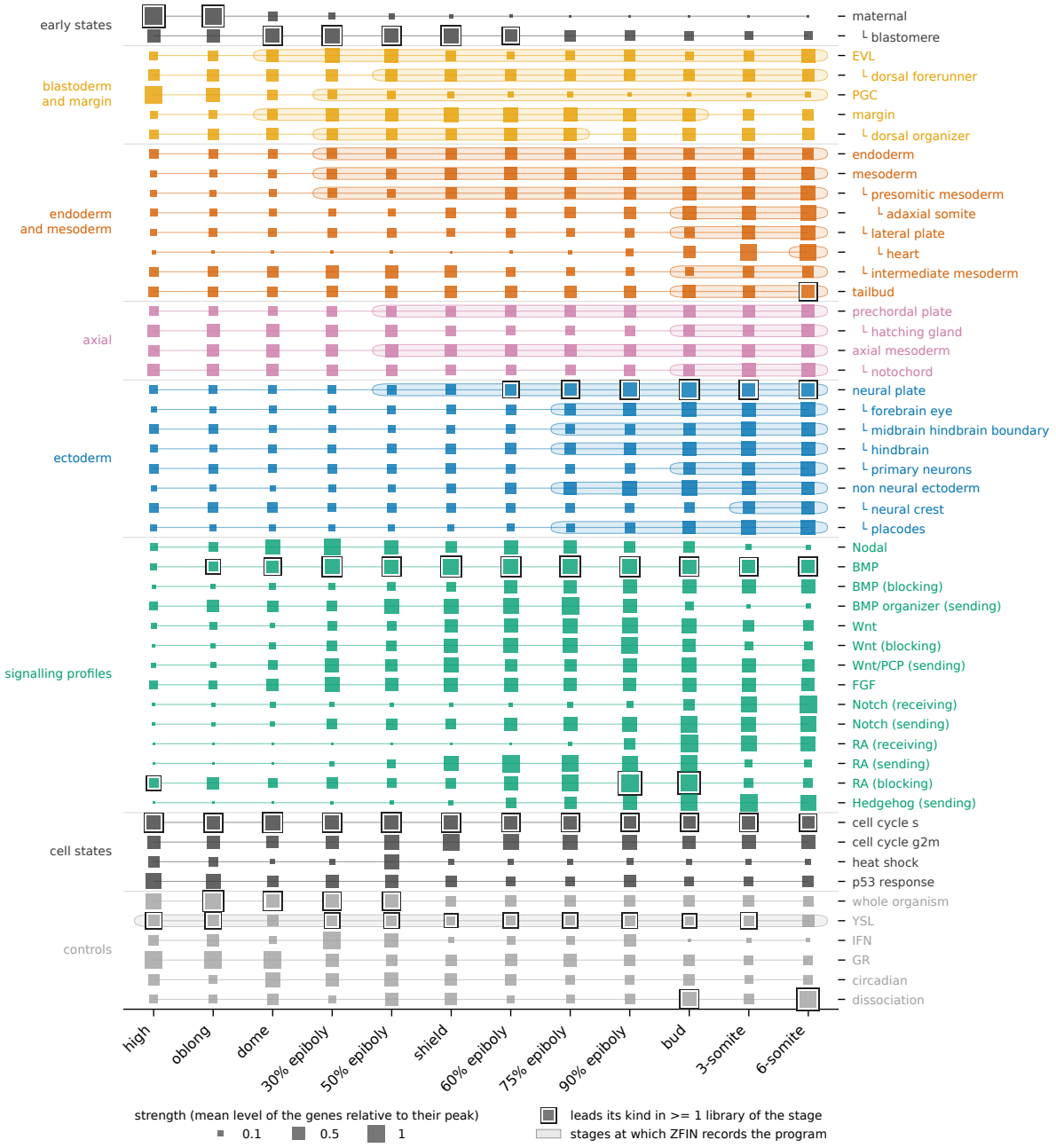


Figure 11: Pseudobulk strength of the curated programs and controls (squares, to distinguish the figure from Figure 10, whose rows it shares). Area: the mean, over the program's genes, of the gene's pseudobulk expression (UMIs per 10,000 of the library) relative to the gene's peak stage, averaged over the stage's libraries; 1 means every gene at its peak. Box: the stages at which ZFIN records the program, as in Figure 10. The strength is relative to each gene's own peak, so it hides how abundant a program is. Ringed: the leaders of a stage, i.e., the programs whose genes have the highest absolute mean expression (UMIs per 10,000 of the library) among the programs of their kind (anatomy programs and early states together; rest per category) in at least one library of the stage. The aryl hydrocarbon receptor control is left out (no gene expressed enough).

Observations. Several features of the two figures are worth noting:

- **Programs vary before ZFIN records them.** Most anatomy programs have heterogeneous genes before their ZFIN window. Some of this is expected: the PGCs are specified by maternal germ plasm (*dnd1*, *nanos3*, *ddx4*) from the start, and the organizer genes (*gsc*, *dharma*) are maternal β-catenin targets on the dorsal side; a structure is usually recorded only once it is morphologically visible, after its genes start to vary. Some of it is shared genes: a program can only be scored on genes it shares with its precursors, e.g., the hatching gland and the prechordal plate. And some of it may be genes that vary for unrelated reasons; whether the genes of a program vary *together* is tested later in Section 8.1. Over whole libraries, many anatomy programs keep a strength of 0.2–0.4 before their ZFIN window, because some of their genes are also expressed elsewhere at lower levels; they are stronger inside the window than outside it (median 0.45 against 0.27), and most rise towards the end of the time course. The strength compares a program with itself over time, not different programs.

- **Programs vary after ZFIN stops recording them.** The organizer (ZFIN: up to 75% epiboly) and the margin (up to bud) keep many heterogeneous genes up to 6-somite. Programs are not expected to switch off when a new one appears: the organizer and the margin give rise to the axial mesoderm and the tailbud, which keep using some of their genes, which explains why we see them continuous active.

- **Maternal and blastomere states** have almost no heterogeneous genes at the early stages: they are expressed in nearly every cell, so they do not vary between cells. The blastomere state has heterogeneous genes from 60% epiboly on, through *sox3*, *sox19a* and *pou5f3*, which are then expressed in the neural plate. Over whole libraries (Figure 11), the maternal state falls from 0.96 (high) to 0.25 (dome) and below 0.1 from 50% epiboly: the clearance of maternal transcripts. The blastomere state stays at 0.5–0.65 up to shield and falls to 0.22 by 6-somite. Within these early libraries, maternal and blastomere programs are the most expressed ones, supporting the assumption that they are uniformly active.

- **Leaders.** The strength compares a program only with itself; which program dominates a stage is shown by the leaders (ringed in Figure 11): in each library, the program of each kind whose genes have the highest mean expression; a program leads a stage if it leads in at least one of its libraries. The mean over genes, not their sum, is used, so that long gene lists are not favoured. Leaders are counted per library rather than per stage, because the mean expression of a program's genes differs by a median of 22% between the libraries of a stage. The maternal and blastomere states are compared with the anatomy programs, so that the shift between them shows: the maternal state leads at high and oblong, the blastomere state from dome to shield (in every library), the blastomere state and the neural plate share 60% epiboly (one and two libraries), and the neural plate leads from 75% epiboly on, with the tailbud in one of the two 6-somite libraries. Among the other cell states, the S phase leads at every stage; among the signalling profiles, BMP, with RA blocking at high and in some libraries at 90% epiboly and bud. A leader has abundant genes, which is not the same as many cells: small programs, such as the heart or the PGCs, never lead, but may matter as much.

- **PGCs and the p53 response.** The PGC program (0.91 at high, below 0.1 from 60% epiboly) contains maternal germ plasm transcripts. It is high early on and falls as other cell types take over. The p53 response is strongest at high and oblong (0.70–0.74), where it is also strongly heterogeneous (*cdkn1a*, *phlda3*, *gadd45aa*).

- **Early dorsal genes.** Some signalling profiles are heterogeneous from high on through a single strongly heterogeneous gene: *chd*, *dkk1b*, *ndr1*, *lft1* and *wnt11*. These are targets of

maternal β-catenin on the dorsal side and among the first zygotic genes.

- **Signalling profiles** rise and fall in the expected order: Nodal peaks at 30% epiboly (0.83) and is about 0.1 from 3-somite; RA synthesis peaks at 60% epiboly and RA receiving at bud; Notch receiving and Hedgehog sending rise towards the segmentation stages.
- **Cell cycle.** The S and G2/M states become heterogeneous only from 75% epiboly (G2/M) and bud (S) on, consistent with the cell cycle lengthening and becoming asynchronous during gastrulation.
- **The YSL control** has many strongly heterogeneous genes from 50% epiboly on, although the embryos were deyolked. Whether these genes mark YSL-like cells is tested in Section 8.1.
- **Single-library stages** (shield, 3-somite) have fewer heterogeneous genes in several programs (e.g., the axial mesoderm at shield, the PGCs at 3-somite): with one library and fewer cells, the heterogeneity test has less power and by extension fewer strong genes.

7.4 Farrell et al. modules and curated programs

The Farrell et al. modules are unnamed, so we name them by their overlap with the curated programs (their genes measured at the stage; Figure 12; `scripts/24_program_overview.py`). At shield and 75% epiboly, 15 of 25 and 17 of 24 modules share at least 20% of their genes (of the smaller of the two sets, and at least 3 genes) with a curated program: 8 and 12 with an anatomy program, the others with a cell state (cell cycle, blastomere, p53) or a signalling profile (BMP, Wnt). The modules of the margin are split as the curated set splits it: at shield into the mesoderm, the endoderm, the organizer and BMP, at 75% epiboly into the margin, the presomitic mesoderm and Wnt. Named by all genes of the stage's ZFIN terms instead (grouped as the curated programs, without the cap of 30 genes and the restriction to heterogeneous genes), 16 and 18 modules are named, so the curated gene lists lose almost nothing; the two modules they miss are a weak EVL module at shield (*epcam*, *oclna*) and a dorsal module at 75% epiboly whose genes the curated set divides between the blastomere state, the signalling profiles and the neural programs (*pou5f3*, *sox19a*, *admp*, *chd*, *foxb1a*). The unnamed modules are led by ribosomal, histone, DNA replication and RNA-processing genes, immediate early genes (*fosab*, *junba*, *atf3*), and the uncharacterized *si:ch211-113a14* gene family. This is variation that no curated program describes, and a part of the in-sample advantage of the Farrell et al. modules in the tests of Section 8.

7.5 Conclusions

- **The curated set** has 45 programs (25 anatomy programs from grouped ZFIN terms, 14 signalling profiles and 6 cell states), each with one gene list that is used at every stage. The programs share no genes, except along presumed derivations (prior knowledge, e.g., the hindbrain from the neural plate).
- **Literature and data differ in timing.** The genes of most anatomy programs vary between cells stages before ZFIN records the program (12 of 25 have at least 3 heterogeneous genes at high), and some keep varying after ZFIN stops recording them. Inside its ZFIN window, a median of 81% of a program's genes are heterogeneous; outside it, 43%.
- **Whole libraries** are dominated first by maternal transcripts (high, oblong), then by the blastomere state (dome to shield), and then by the neural plate, joined by the tailbud at the somite stages. Programs that are expressed evenly in all cells, such as the maternal and early blastomere states, are visible only at this level.

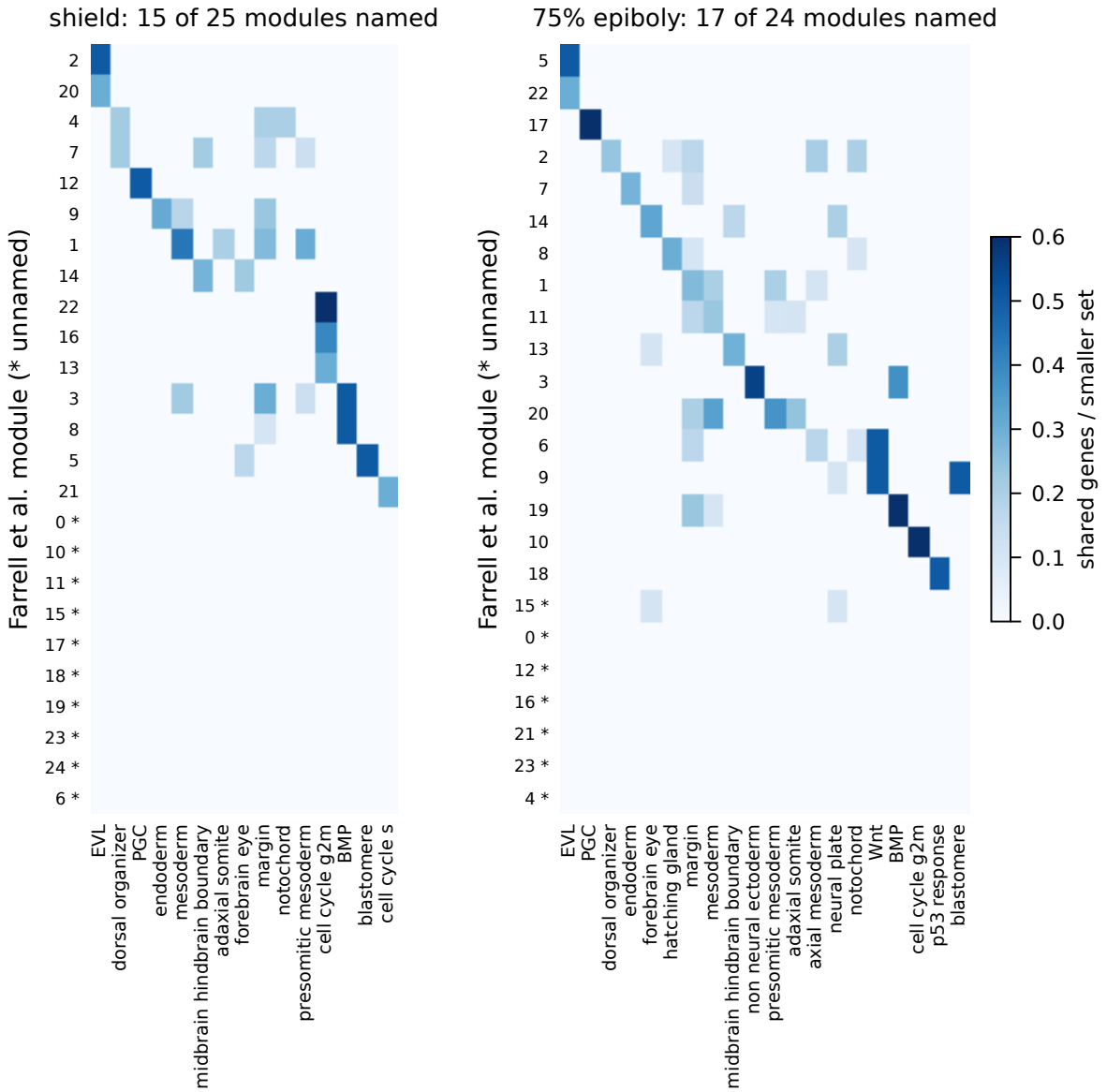


Figure 12: Farrell et al. modules (rows; their numbers in Table S2) against the curated programs (columns; their genes measured at the stage), at shield and 75% epiboly: shared genes as a share of the smaller of the two sets, if at least 3. A module is named by its best program if the share is at least 0.2; *: unnamed. Columns: programs that share at least 0.2 with some module.

8 Result: expression programs describe the cells

Here we test the programs against the cells: whether the genes of each program vary together within the libraries (Section 8.1), how much of the variation between cells each set explains (Section 8.2), and how many activity levels a program needs to keep that information (Section 8.3).

8.1 Is a program visible in the data?

Scores. The score of a program in a cell is the mean of the Pearson residuals of its genes under the model of the cell's library (Section 6), centred per library (weighted for the Farrell et al. modules). Centring (subtracting each gene's mean residual in the library, so that every gene has mean zero in every library) is required to make residuals comparable across libraries. The residuals already account for sequencing depth, so, unlike the usual gene-set score [209], no control genes are needed, and the score has the same meaning in aligned and per-library data. Since the residuals are centred per library, the score describes differences between the cells of a library; a program that is active at the same level in every cell of a library is constant (zero) within it, even if its level differs between libraries.

If the genes of a program vary together in the cells, the program is visible in the data. We measure this by the coherence

$$R = \frac{n \operatorname{Var}(\text{score})}{\text{mean variance of a gene's residuals}} \approx 1 + (n-1)\,\bar{\rho},$$

where n is the number of genes and $\bar{\rho}$ their mean correlation. Var(score) is the variance of the program's score across the cells of a library (the score being the mean of the residuals of the program's genes). The approximation is exact if all genes have the same variance. $R = 1$ if the genes vary independently: averaging n independent genes divides the variance by n, so $n \operatorname{Var}(\text{score})$ equals the mean gene variance. If the genes rise and fall together, averaging does not cancel their variation, and R grows up to n.

Supported programs To determine which programs vary together significantly, we could use an obvious null: shuffling every gene independently across the cells. However, this is too weak: random sets of heterogeneous genes pass it in 74% (50% epiboly) and 27% (bud) of cases, because heterogeneous genes vary along a few shared axes of the stage (e.g., margin vs. animal pole). In other words, the shuffle removes all co-variation, but almost any set of heterogeneous genes has some, so passing it does not show that the genes of a program belong together. We therefore require a program to be more coherent than 95% of 500 random sets of heterogeneous genes of the same size, in the same library, and call it **supported** at a stage if this holds in at least half of the stage's libraries (`scripts/17_program_tests.py`).

Relation to standard tests. R is (approximately) a rescaled mean pairwise correlation of the program's genes ($\bar{\rho} = (R-1)/(n-1)$), so it orders programs as the measures usually used for a co-expression module do, e.g., the mean correlation or Cronbach's $\alpha = n\bar{\rho}/R$ [18]; the variance form needs one variance per program instead of all pairwise correlations. The two nulls are the two kinds of gene-set test [51]: the shuffle is a *self-contained* test (do the genes co-vary at all?), the random sets a *competitive* one (do they co-vary more than other genes?), as in gene set enrichment analysis [179]. Our competitive null draws only heterogeneous genes, so a program has to beat genes that vary anyway. Unlike co-expression measured on log-normalised counts, the residuals remove sequencing depth first, which otherwise makes all genes correlate positively [61], and the test runs within each library, so differences between libraries do not appear as co-expression.

Relative coherence score The support decision is a test; to see how coherence changes over time, also below the threshold, we use the **relative coherence** $(R - \tilde{R}_{\text{random}})/(R^{95\%}_{\text{random}} - \tilde{R}_{\text{random}})$, with $\tilde{R}_{\text{random}}$ the median and $R^{95\%}_{\text{random}}$ the 95% quantile of the random sets of the same size in the same library, and its median over the stage's libraries (Figure 13). It is 0 for a program as coherent as a typical random set of heterogeneous genes, 1 at the threshold of support, and negative for a program less coherent than random heterogeneous genes (negative values are common in stages where the program's genes are not heterogeneous). It allows us to assess coherence relatively across stages and between programs.

Every curated program is tested at every stage, also outside its ZFIN window. From 60% epiboly on, 56–73% of the curated programs are supported (Farrell et al.: 53–81%, ZFIN terms: 54–68%, restricted to heterogeneous genes: 62–75%); up to shield, only 7–19% (Farrell et al. modules: 18–36% at dome to shield). Here, the inability to detect coherent programs in early stages largely lies in the strictness of the test: before gastrulation, the heterogeneous genes vary along the same few axes (few genes at all up to dome), so the 500 random sets are nearly as coherent as the actual programs. In fact, the typical anatomy program is less coherent than random heterogeneous genes up to 50% epiboly (median relative coherence -0.26 to -0.50 per stage) and about as coherent at shield (0.01); its genes vary, but not more together than any genes that follow the main axes of the stage. From 60% epiboly on, the median rises to 1.1–3.6 (Figure 13). The signalling profiles are the exception: up to shield, they are weakly above random (median 0.30), mostly below the threshold. Removing the unsupported programs at these stages therefore loses part of the predictive power (at dome and 50% epiboly, more than later; Section 8.2): they describe real variation that the strict supported program test cannot conclusively distinguish from that of random heterogeneous genes.

Literature and data. For the anatomy programs, support follows the ZFIN window, but not exactly. Inside the window, 74% of a program's stages are supported (73–91% from 60% epiboly on), outside it 16% (6% up to shield, 44% from 60% epiboly on). Programs that are coherent outside their window are those anticipated in Section 7.3:

- **Before ZFIN records them**: the EVL and the PGCs at high and oblong (the PGCs also at dome; maternal germ plasm), the non-neural ectoderm at oblong and 60% epiboly, and the precursors of programs that ZFIN records only from bud: the notochord, the hatching gland and the adaxial somites from 60% to 90% epiboly, the heart at bud and 3-somite. The relative coherence shows these as trends: the notochord rises from -0.3 at shield to 1.1 at 60% epiboly, 3.7 at 90% epiboly and 8.3 at bud, and the heart from about 0 at 90% epiboly to 1.2 at bud.
- **After ZFIN stops recording them**: the organizer from 90% epiboly to 6-somite and the margin at 3- and 6-somite, consistent with programs that persist in the tissues they give rise to (the axial mesoderm, the tailbud).

Inside the window, the unsupported programs are mostly broad or early but with rising coherence (e.g., early endoderm and mesoderm). Notably, neural plate is never supported. We hypothesize that this is because the program is very general (covers a wide range of precursors) and broadly expressed, so it suffers from similar problems as the maternal and blastomere programs early on (by Figure 11, it is the most expressed program from 60% epiboly onward). The other curated programs unsupported from 60% epiboly on are expected: maternal transcripts (gone), heat shock (uniform), the p53 response from bud on, the RA profiles and most sending and blocking profiles (see Section 9.2).

Controls. The control programs that should be invisible are: the readouts of the interferon, aryl hydrocarbon receptor, glucocorticoid and circadian pathways, and the whole-organism genes

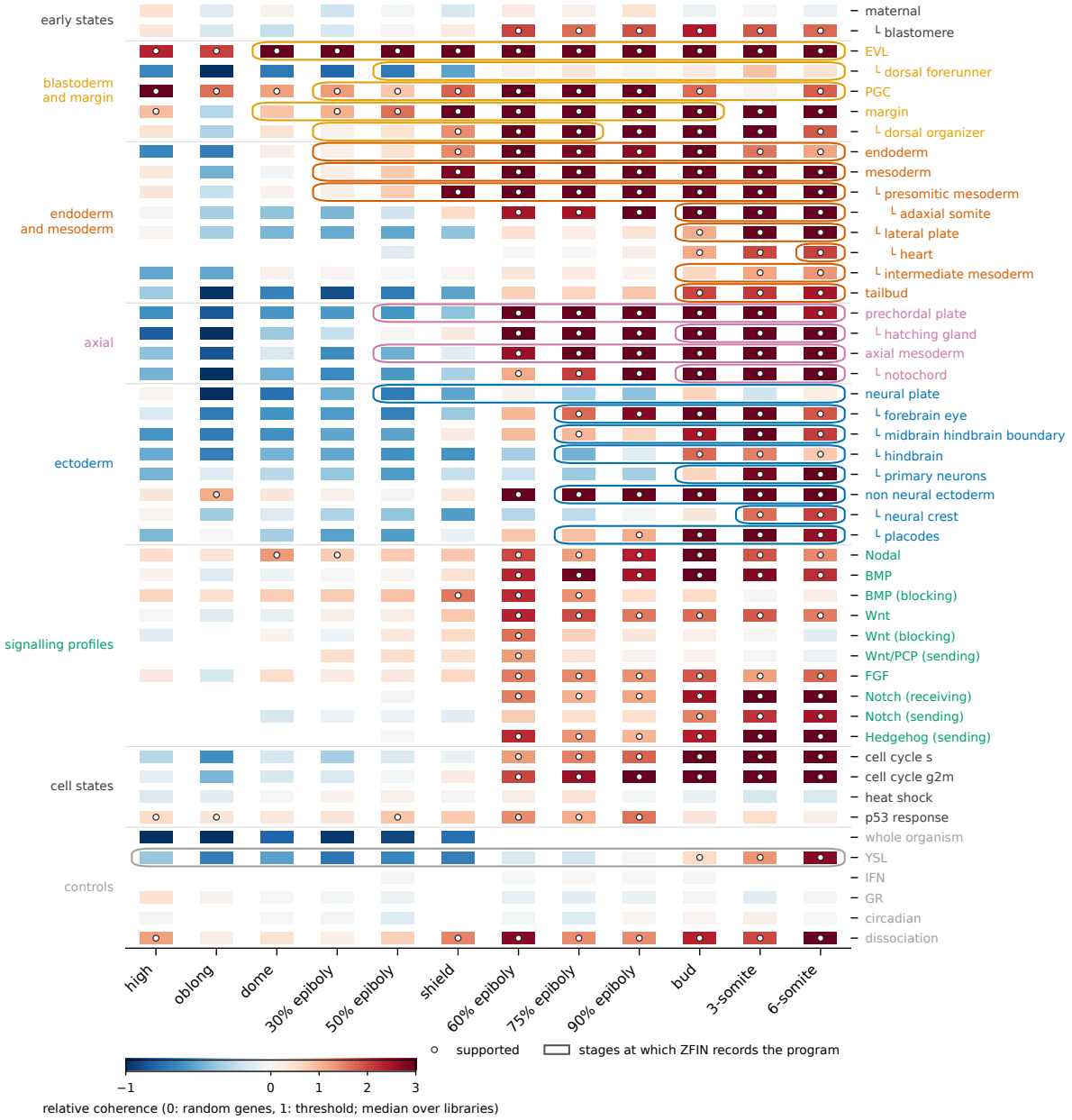


Figure 13: Relative coherence of the curated programs and controls per stage, in the rows following Figure 10 (`scripts/17_program_tests.py`, `scripts/24_program_overview.py`): 0 (white): as coherent as a typical random set, 1: the threshold of support, red: more coherent, blue: less coherent (colours clipped at −1 and 3). Circle: supported (coherent in at least half of the stage's libraries). No square: fewer than two of the program's genes measured at the stage. Box: the stages at which ZFIN records the program, as in Figure 10.

are not coherent in any library at any stage. Two controls are clearly visible, and both matter for future analysis (the YSL is discussed below):

- **Dissociation stress.** The immediate early genes (*fosab*, *junba*, *atf3*, *egr1*) are coherent in every library from shield on (and in one of the two high libraries): part of the cells responded to the dissociation, as in other tissues [195]. This is a technical axis of variation, and also one of the unnamed Farrell et al. modules at shield (Figure 12); it should be discarded as technical outlier.
- **The YSL.** The embryos were deyolked, which removes the YSL, so its genes should not form a visible program; the YSL is therefore a control. ZFIN's YSL term is coherent in no library up to 3-somite and in one of the two at 6-somite; restricted to heterogeneous genes, it is supported at 60% epiboly, bud and the somite stages, through secretory and ER genes such as *hspa5* and *sec61a1l*, which are expressed in many tissues. The YSL control leaves out every gene of the curated programs and other controls (12–29 genes per stage, led by *apoeb*, *tnfaip8l2b*, *tfa* and *sesn3*); it is coherent in no library up to 90% epiboly, in two of four at bud and in one library at 3- and at 6-somite. This late coherence is not a YSL signal: the cells in which the control is active are mostly EVL/periderm, non-neural and hatching-gland cells, and the yolk-specific transcripts are nearly absent (Section 10.2).

8.2 How much of the variation does each set explain?

The question. If a program set describes the states of the cells, then knowing a cell's program scores should tell us about the expression of genes that are in none of the programs. We test this at every stage by predicting held-out genes from the program scores, and compare each set with a model that knows only sequencing depth and library (the *floor*) and with the best model we could find (the *ceiling*). The test runs on all cells of the stage; its steps (`scripts/17_program_tests.py`):

1. **Evaluation genes** are genes detected in 2–98% of the stage's cells that belong to no program of any set at the stage (Farrell et al., ZFIN terms, signalling, cell states, curated programs, controls), so that no set predicts its own genes. They are taken in a fixed random order, in two groups: up to 400 heterogeneous genes (116–233 per stage from dome on, except shield: 49, since most heterogeneous genes are in some program) and 400 other measured genes (*all*).
2. **Target.** For each cell and evaluation gene: the share of the cell and its 10 nearest neighbours in its library that detect the gene (at least one UMI). A single cell detects a gene or not largely by chance; the neighbourhood frequency is a less noisy estimate of its expression. The neighbours are found among the evaluation genes themselves, so the target does not share neighbourhoods with the features.
3. **Features.** The score of every program of the set (Section 8.1), averaged over the cell and its 10 nearest neighbours in its library (neighbours in the principal components of the stage's heterogeneous genes that are not evaluation genes).
4. **Models.** For each evaluation gene, a logistic regression on 70% of the cells (training, stratified by library), with L2 regularization chosen by 3-fold cross-validation, so that uninformative features are shrunk towards the floor. The models differ only in their inputs: *floor* (smoothed log depth and library); each *program set*, i.e., the floor plus the set's scores (also the supported curated programs alone, and the random sets of step 6); and three models without programs: the 30 smoothed *principal components* of the heterogeneous genes that are not evaluation genes, *all* (principal components, curated and Farrell et al. scores together), and *nearest neighbours* (the gene's detection frequency among the cell's 20 nearest training cells).

5. **Score.** The log-loss on the other 30% of the cells (test), summed over the evaluation genes of a group. The *ceiling* is the model with the lowest loss at the stage, whichever it is. The **explained fraction** of a set is $(\text{loss}_{\text{floor}} - \text{loss}_{\text{set}})/(\text{loss}_{\text{floor}} - \text{loss}_{\text{ceiling}})$: 0 means no better than depth and library, 1 as good as the best model. The **explainable share**, $(\text{loss}_{\text{floor}} - \text{loss}_{\text{ceiling}})/\text{loss}_{\text{floor}}$, says how much of the variation there is to explain at all. The same losses are also summed over the test cells of each library (with the stage's ceiling model; libraries with fewer than 100 test cells get no explained fraction), which shows whether the libraries of a stage agree.

6. **Random reference.** For every program, a set of as many random genes with the same expression levels (50 bins of mean expression; never evaluation genes) is scored in the same way; a set should explain more than its random counterpart.

How much can be explained. Little of the variation of other measured genes can be explained by anything: the best model removes at most 6% of the floor loss (Figure 14a), and we report explained fractions only where at least 4% can be explained. Heterogeneous genes are much more predictable: 24–36% of the floor loss from dome on, except shield (6%); at high and oblong almost nothing (1% or less). The predictability of a gene follows its heterogeneity (Figure 14d): pooled over the stages from dome on, 4% of the floor loss of evaluation genes that are not heterogeneous can be explained, 22% of weak, 34% of moderate and 49% of strong heterogeneous genes. Most measured genes vary between cells only by chance, so there is little to predict about them; the heterogeneous ones carry the cell states. Shield is the exception: its heterogeneous evaluation genes are barely predictable in every tier (strong ones: 6%, against 27–70% at the other stages from dome on). The likely reason is the test itself: the neighbourhoods of the target are defined by the evaluation genes, and with 49 heterogeneous ones they carry little structure; the single library is not the reason (3-somite, also a single library, with fewer UMIs per cell: 24%). The ceiling is usually one of the feature sets (the Farrell et al. modules up to 30% epiboly and at 3-somite, the supported curated programs at shield, the principal components or all features together otherwise), never the nearest-neighbour model.

Results Figure 14b, heterogeneous genes, dome to 6-somite:

- **Every set explains far more than random genes.** Farrell et al. explains 0.92–1.00, ZFIN terms 0.86–0.99, curated 0.84–0.98, vs. 0.33–0.90 for random sets matched to the curated programs. Relative to random ((set − random)/(1 − random)): curated 0.60–0.91, ZFIN terms 0.54–0.97, Farrell et al. 0.73–1.00. The same holds for the other measured genes where defined (curated 0.83–0.97 vs. random 0.27–0.64, from 60% epiboly on).

- **Restricting to heterogeneous genes costs nothing.** ZFIN terms restricted to heterogeneous genes explain as much (0.92–0.99) with a third fewer genes.

- **The curated set is as good as the ZFIN terms with fewer programs.** With 41–45 programs per stage from dome on (ZFIN terms: 8 at dome, up to 110 at 6-somite), it explains 0.96–0.97 up to 50% epiboly and 0.94–0.98 from 60% epiboly on (ZFIN terms: 0.94–0.99 and 0.91–0.97); at shield, both explain less (0.84 and 0.86). The Farrell et al. modules remain the best, as expected from their in-sample advantage.

- **Unsupported programs matter little.** From 60% epiboly on, the supported curated programs alone (25–33 per stage) explain 0.93–0.97 (all curated: 0.94–0.98). Up to 50% epiboly, with only 3–4 supported programs, they lose more at dome and 50% epiboly (0.94 vs. 0.96; 0.91 vs. 0.96), but not at 30% epiboly (0.99 vs. 0.97).

- **Signalling and states alone** explain 0.41–0.94 and 0.32–0.88: they are a few programs each, which describe a part of the variation that overlaps the identity programs.

- **The controls explain much early, little late**: 0.61–0.83 from dome to 50% epiboly, at most 0.42 from 60% epiboly on. Before gastrulation, even genes that should not mark identity (whole-organism genes, silent pathways) follow the few shared axes of variation, which is another sign that these stages lack distinct programs.
- **The libraries of a stage agree** (bars in Figure 14a, b): the explainable share of the heterogeneous genes differs by at most 0.09 between the libraries of a stage, and the curated programs explain 0.94–0.98 in every library from dome on except shield (0.84), within 0.02 of each other at each stage; for every set, the libraries of a stage differ by at most 0.06 (random sets: 0.09), so the bars of Figure 14b are mostly hidden by the markers. The explainable share does not follow the size or depth of a library (Spearman correlation over the 24 libraries from dome on: -0.14 with the number of cells, 0.20 with the median UMIs per cell): the smallest bud library with a test (DS2*b*, 506 cells) reaches 0.37, the largest (DS3, 4,759 cells, with the fewest UMIs per cell) 0.33. For other measured genes, the explainable share follows the number of heterogeneous genes of the library ($\rho = 0.66$): little of a random gene is predictable while the cells are alike, and more as distinct cell states appear.

8.3 How many activity levels does a program need?

A Boolean or multi-valued model describes every program in a cell by one of a few levels. We ask how much of what the continuous scores say about a cell survives when every curated program is reduced to k levels (`scripts/22_program_levels.py`). We reuse the prediction test of Section 8.2 with the same cells, split, smoothing and target, but with evaluation genes of its own. Only the curated programs are tested here, so an evaluation gene needs to be in no curated program, but may be in a program of the other sets; this gives more and stronger heterogeneous evaluation genes (187–388 per stage from dome on, shield: 111, against 49 in Section 8.2). The floor and the ceiling (the best of the nearest-neighbour model, the principal components, the continuous scores and their levels) are fitted on these genes, so the explained fractions differ from those of Figure 14; what we compare is each discretisation with the continuous scores. The features:

- k **levels** (k = 2–6): the smoothed scores of each program are cut into k levels by 1-D k-means of the training cells, pooled over the stage's (aligned) libraries, i.e., one set of thresholds per program and stage, as a model would use; every cell gets the level of its nearest centre.
- **Boolean (null)**: a program is *active* in a cell if its smoothed score is higher than chance allows. The chance level comes from a null in which the program's genes do not vary together: in each library, every gene's residuals are shuffled independently across the cells (each gene keeps its own values, but which cells are high in one gene no longer relates to the others), the shuffled residuals are scored and smoothed exactly like the real ones, and this is repeated 5 times. The threshold is the 99.9% quantile of these null scores, so by chance only 1 in 1,000 cells would be called active; it is computed per program and library, so a program of a few sparse genes needs a stronger signal than one of many well-expressed genes. **3 states**: in addition, *depleted* below the 0.1% quantile of the same null. This is the call of "active" used elsewhere in the report (e.g., Section 9.2; Section 10.1 refines it per cell); unlike the coherence test (Section 8.1), it does not ask whether a program is real, only whether it is clearly on in a given cell.

A program with k levels enters the regression as $k - 1$ indicators "level $\geq v$". This allows any effect of every level, but the regularization of the regression shrinks a level towards the one below it rather than towards the lowest; indicators "level $= v$" span the same models but are worse

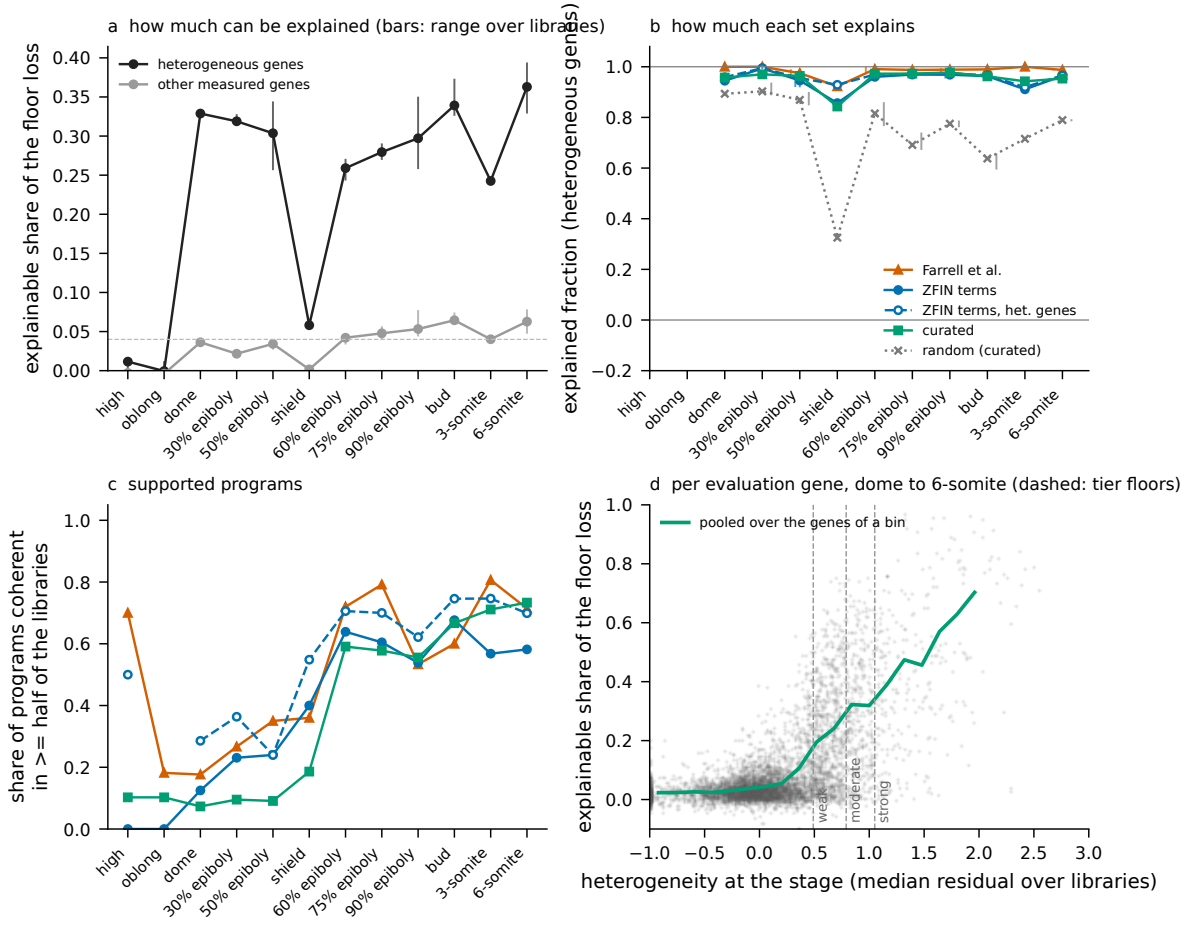


Figure 14: The prediction test (`scripts/17_program_tests.py`). (a) Explainable share of the floor loss: how much of the neighbourhood detection of the evaluation genes the best model predicts beyond depth and library, per stage; bars: range over the stage's libraries; dashed: 4%, below which no explained fraction is reported. (b) Explained fraction of the heterogeneous evaluation genes per set, 0 = depth and library only, 1 = best model; bars: range over the libraries (libraries with at least 100 test cells; none at single-library stages), mostly hidden by the markers, since the libraries differ by at most 0.06 (random sets: 0.09). Green: curated programs; grey: random gene sets matched to the curated programs. (c) Share of programs that are supported (more coherent than random heterogeneous genes in at least half of the stage's libraries; Section 8.1). (d) Explainable share of each evaluation gene against its heterogeneity at its stage (dome to 6-somite, both groups); line: pooled over the genes of a bin; dashed: the floors of the three tiers.

with many levels and few cells (e.g., 6 levels at shield: 0.56 vs. 0.75 of the continuous result). The levels of a program also allow effects that are not linear in its score, so a discretised set can sometimes explain slightly *more* than the continuous scores.

Result Figure 15; neighbourhood target, heterogeneous evaluation genes, dome to 6-somite; "kept" = explained fraction relative to the continuous scores, which explain 0.95–0.99:

- **Two levels keep over 85%** (median 0.94, range 0.86–0.97). **Three levels keep nearly everything** (median 0.97, 0.87–0.99), and four or five are as good (median 0.98); from 60% epiboly on, 3 levels keep 0.97–0.99 and 4 levels 0.98–1.01.
- **Where the threshold lies matters as much as the number of levels.** The Boolean call from the permutation null keeps only 0.90 (0.79–0.94), less than two k-means levels: its strict threshold marks only cells that are clearly active, and cells with an intermediate score fall into "off". Adding the depleted state recovers part of it (3 states: 0.94, 0.72–0.96), but less than three k-means levels.
- **Shield loses most.** At shield (one library, 1,046 cells), 2 levels keep 0.86, 3 levels 0.87, and more levels lose (4 levels: 0.84, 5 levels: 0.77), although the continuous scores explain as much as at the other stages (0.96). The test is weaker there: with more evaluation genes than in Section 8.2, 15% of the floor loss can be explained (6% there), but still much less than at the other stages from dome on (29–43%); with little to explain, extra indicators mostly fit the noise of the training cells. The single library is not the reason: the other single-library stages lose little (3 levels: dome 0.94, 3-somite 0.98).
- **The supported programs alone, with 3 levels**, keep 0.94–0.98 of the continuous curated set from 60% epiboly on, so restricting a model to them costs little; the broad unsupported programs (mesoderm, neural plate; Section 8.1) still carry information.
- **Random measured genes** show the same order with larger losses: from 50% epiboly on (where defined), 2 levels keep 0.80–0.92, 3 levels 0.82–0.96 and the Boolean call 0.70–0.83.

Which programs are switches? Separately from the prediction, we measure how much of the variance of a program's smoothed score its k-means levels keep (Figure 15c). Two levels keep a median of 69% of the variance (3 levels: 85%, 4 levels: 91%), much less than they keep of the prediction: what matters for the other genes is mostly which cells are at the top, and the variation within the bulk of the cells matters less. Per program (median over all stages), the programs that come closest to a switch (2 levels keep 74–87% of the variance, and the upper level holds 1–10% of the cells) are distinct populations and localized signalling centres: the EVL, hatching gland, endoderm and non-neural ectoderm, the Hedgehog senders, RA synthesis, the BMP antagonists, *admp* and Nodal. The most graded are the cell cycle (2 levels split the cells about in half), the uniform states (maternal, p53 response, heat shock), and the broad anatomy programs (dorsal forerunner, neural plate and its regions, lateral plate; 55–65%), as expected for gradients and continuous processes.

Caveats. The levels are computed from scores smoothed over each cell's 10 nearest neighbours, so they describe a cell's neighbourhood rather than the cell itself; levels of unsmoothed scores would be noisier, and we have not tested them. The thresholds are fitted per stage, so a program's levels need not mean the same at two stages, and they are fitted on aligned scores, so they say nothing about differences between whole libraries (e.g., the maternal program). Finally, the test measures what the levels say about other genes, not whether they are the right variables of a dynamical model.

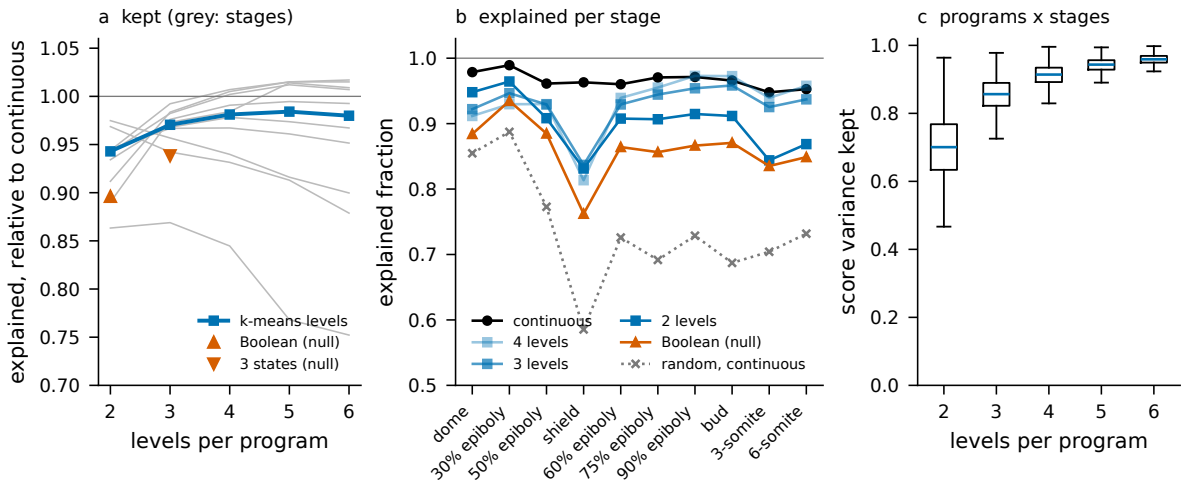


Figure 15: Discretised curated programs (neighbourhood target, heterogeneous evaluation genes in no curated program). (a) Explained fraction relative to the continuous scores, against the number of k-means levels per program (grey: stages, dome to 6-somite; blue: median), and for the Boolean and 3-state calls from the permutation null (orange; median). (b) Explained fraction per stage (0 = depth and library, 1 = best model). (c) Share of the variance of a program's smoothed score kept by its levels, over all programs and stages.

Conclusion for the model. Three levels per program (low / intermediate / high), with thresholds that are placed by the data rather than by a strict significance cut-off, keep practically all of what the continuous scores explain; a Boolean description keeps over 90% with data-placed thresholds and slightly less with the significance call. The exception is shield, at the focus of the model, where 3 levels keep only 0.87; most likely because the test has little to explain there (15% of the floor loss), not because of its single library. The two-level description is close to complete for programs of distinct populations, and loses most for graded programs (signalling, cell cycle), which are the natural candidates for more than two levels.

8.4 Conclusions

Programs from prior knowledge describe the cells nearly as well as the modules found in the data, and far better than random gene sets. Restricting them to heterogeneous genes loses nothing, and a first curated set of 39–45 named programs, each with one gene list for all stages, is as useful as the ZFIN terms. From 60% epiboly on, more than half of the curated programs are clearly visible in the data, three quarters of the anatomy programs within their ZFIN window; the rest are expected to be invisible (maternal, uniform states), too broad (neural plate), or not yet or no longer formed. Several programs are visible before ZFIN records them (precursors of the notochord, hatching gland, adaxial somites and heart) or after (organizer, margin). Before 60% epiboly, the variation between cells is dominated by a few shared axes, and our test cannot tell individual programs apart. The libraries of a stage agree: the curated programs explain 0.94–0.98 in every library from dome on except shield, independently of its size and depth. How much of a gene's variation can be explained at all follows its heterogeneity (4% for genes that are not heterogeneous, 49% for strongly heterogeneous ones), and grows with the number of distinct cell states, not with sequencing; shield, with few heterogeneous evaluation genes, is barely testable. For a discrete model, three data-placed levels per program keep nearly all of the information of the continuous scores, and two levels about 90% (Section 8.3).

9 Result: programs and signals

The signalling profiles of the curated set (Section 7.2) mark where a pathway is active, or where a signal is sent or blocked. Here we relate them to the identity and state programs: which tissues take part in which signal, and when (Section 9.1); whether the expected signalling centres are where the literature puts them (Section 9.2); and which signals accompany the emergence of a program from the program it derives from (Section 9.3). All three analyses follow the rules of Section 8: they are computed within each library, and every number is compared with random sets of heterogeneous genes (`scripts/21_signal_coupling.py`).

9.1 Which programs co-occur with which signals?

The general and receiving profiles measure where a pathway is active, so their correlation with the identity and state programs tells us which populations take part in which signal, and when. Within each library, we correlate the smoothed scores of each profile with those of every program over the cells. The curated programs are disjoint (Section 7.2), so no correlation comes from shared genes. A raw correlation is, however, hard to read: the smoothing raises all correlations, and before gastrulation almost any two sets of heterogeneous genes correlate, because they vary along the same few axes (Section 8.1). We therefore compare each correlation with those of 100 random sets of background genes (heterogeneous at the stage, in no curated program) of the profile's size, scored and smoothed in the same way. As for the coherence, the **relative coupling** is 0 for a profile as coupled to the program as random genes, and $+1$ or -1 at the 95% or 5% quantile of the random sets; per stage we report its median over the libraries, and whether every library is beyond the random sets on the same side (Figure 16). For a general profile (Nodal, BMP, Wnt, FGF), a coupled program sends or receives the signal, or both; a coupling says that the pathway's genes are expressed in the program's cells, not that the signal causes the program.

The baseline removes the couplings that any heterogeneous genes would show: the Notch readouts, for example, correlate at $r = 0.65$–0.96 with the EVL, non-neural ectoderm, neural crest and hatching gland programs from dome to 50% epiboly, but their relative coupling is at most 1.0. It does not remove the main axis itself: before shield, the profiles that mark the margin (Nodal, FGF, Wnt, and BMP at the ventral margin) are coupled, positively or negatively, with nearly every program that lies on one side of the animal–vegetal axis (Figure 16), so the early couplings mostly say on which side of that axis a program's genes are expressed. Most couplings agree with what is known about these pathways in the zebrafish gastrula:

- **Nodal** is coupled to the margin, the endoderm and the mesoderm at dome (relative coupling 2.4–3.1), fading after 50% epiboly: Nodal signals are required for endoderm and most mesoderm [36, 58], and the duration of signalling during the blastula and early gastrula decides the fate [62]. It stays longer in the dorsal organizer (1.5–2.6 up to 75% epiboly), the prechordal plate (2.4 at shield) and the axial mesoderm (1.2–1.5 from dome to 60% epiboly), as expected from the highest Nodal requirement of the prechordal plate [57] and Nodal ligands and *lefty* in axial tissues [150, 186]. The blastomere program is anti-coupled (-1.2 to -2.6 from dome to shield): cells that receive Nodal have lost the pluripotency genes.

- **BMP** is coupled to the non-neural ectoderm and the placodes (1.2–1.8 from 60% epiboly to bud) and the neural crest (1.0–1.4 from 75% epiboly on), and anti-coupled to the neural plate, the forebrain and the midbrain–hindbrain boundary (up to -6 at 30% epiboly, -1.1 to -2.4 from shield to 75% epiboly, weaker later): the BMP gradient of dorsoventral patterning [63, 92, 134]. Before shield, the BMP profile is coupled to the tailbud program (1.9–3.7 from dome to 50% epiboly), i.e., to the ventral margin, where the BMP ligands

are expressed, and against the whole ectoderm, including the non-neural ectoderm; the tail requires BMP [144].

- **Wnt** is coupled to the margin and the mesoderm (1.9–2.8 from dome to 50% epiboly), the presomitic mesoderm and the tailbud (up to 3.0; the tailbud from dome to 75% epiboly and at the somite stages), and anti-coupled to the forebrain and eye (-1.2 to -1.7 from 30% epiboly to bud): Wnt8 at the ventrolateral margin is required for ventrolateral mesoderm and posterior fates [111, 188], and the forebrain requires Wnt to be off [91, 67].

- **FGF** is coupled to the margin, the organizer, the presomitic mesoderm and the axial mesoderm (1.8–2.9 from dome to 50% epiboly), and to the tailbud at every stage from dome on except bud, and anti-coupled to the blastomere program (-1.9 to -2.7 from dome to 50% epiboly): FGF at the margin is required for trunk and tail mesoderm [54, 25], where its feedback genes are expressed [42, 145].

- **Notch** (receiving) is coupled to the tailbud, increasingly from 60% epiboly (0.5) to 6-somite (1.7).

- **RA** (receiving, a single gene, *dhrs3a*) is coupled to the hindbrain (1.1–1.8 at 75% and 90% epiboly), as expected from RA in the hindbrain, and to the blastomere program, through its neural genes. A single gene has no coherence to test (Section 8.1), but its coupling can be tested like any other.

Some expected couplings are missing, and some are unexplained:

- Notch is not coupled to the presomitic mesoderm (at most 0.4 from shield on), although the segmentation clock depends on Notch [73]; the clock genes oscillate, so at any moment only part of the presomitic cells express the readouts. Notch is also not coupled to the primary neurons (at most 0.6), although *her4* is a Notch target in the proneural clusters [181]. FGF is anti-coupled to the midbrain–hindbrain boundary (-0.9 to -1.0 from 60% to 90% epiboly), although Fgf8 is made there [151]: the general FGF profile is dominated by the margin and the tailbud (Section 9.2).

- Hedgehog readouts (*ptch1/2*, *gli1*, *hhip*) are never heterogeneous, so Hedgehog has only a sending profile, and its reception by the floor plate and adaxial cells cannot be tested.

- The maternal program is coupled to Nodal and FGF from 30% epiboly to shield (1.0–2.9 in every library), so maternal transcripts are higher where the margin profiles are; whether the marginal cells keep maternal transcripts longer is open. The coupling of Nodal with heat shock (1.1–1.2 at 60% and 75% epiboly) differs between libraries (-0.6 to 1.4 at 75% epiboly) and is likely technical.

9.2 Which tissues send and block which signals?

The general profiles above mark where a pathway is active; the sending and blocking profiles mark where a signal is sent or blocked by cells that need not receive it (Section 7.2). We test the expected senders, receivers and blockers explicitly (Table 7). A correlation over all cells misses small source populations, such as the organizer among all gastrula cells, so each hypothesis is tested on the cells in which the tissue's program is active (Section 8.3). The **AUROC** says whether the profile's score is higher in these cells than in the others: 0.5 means no difference, 1 that every active cell scores higher than every other cell, 0 the reverse. The other cells are, where the program has parent programs (Figure 10), the active cells of the parents: whether the organizer sends *admp* is asked against the rest of the margin, not against the whole embryo, so that a profile shared by the whole margin does not count for the organizer. As for the couplings,

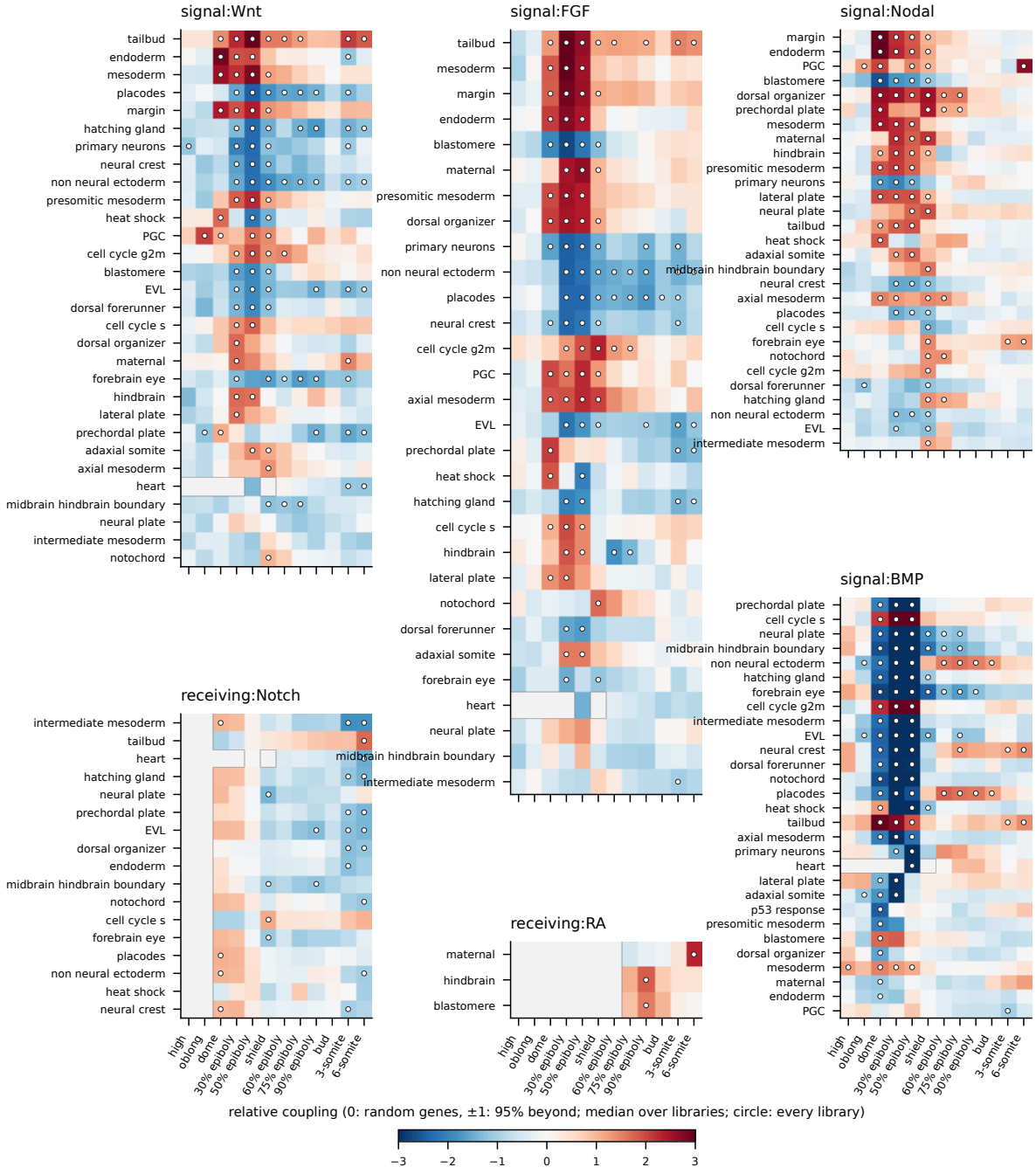


Figure 16: Relative coupling of the general and receiving signalling profiles with the identity and state programs, per stage: the correlation of the smoothed scores over the cells of a library, against random sets of background genes of the profile's size (0: as coupled as random genes; ±1: at the 95% / 5% quantile of the random sets), median over the stage's libraries (colours clipped at ±3). Circle: beyond the random sets on the same side in every library of the stage. Shown: programs beyond the random sets at some stage; grey: not defined.

every AUROC is compared with random sets of the profile's size (relative AUROC, median over the libraries), and a hypothesis is tested only at the stages at which the tissue's program is supported. A hypothesis is *consistent* if it is beyond the random genes on the expected side at one stage or more and never on the other side.

Of the 24 hypotheses, 19 are consistent, 3 weak (never beyond the random genes) and 2 opposite:

- **The expected tissues send, receive and block the expected signals**, also against their parent programs: Nodal, Wnt and FGF at the margin (best AUROC 0.97–0.99), BMP in the non-neural ectoderm (0.82–1.00 from 60% epiboly on), Wnt and FGF in the tailbud (0.88, 0.73), the RA-synthesizing *aldh1a2* in the presomitic mesoderm (0.83) and the RA-degrading *cyp26a1* in the tailbud and the forebrain (0.88, 0.79), Delta ligands in the presomitic mesoderm and primary neurons (0.88, 0.77), the organizer's *admp* (0.83), the BMP and Wnt antagonists in the organizer (0.95, 0.83) and the Wnt antagonists in the prechordal plate (0.90).
- **Delta senders are next to Notch receivers.** Cells with active Delta have high Notch readouts (AUROC up to 0.84, from bud on), although the two profiles are separate: in lateral inhibition, the cells that send and receive are neighbours, which the smoothing over neighbourhoods brings together.
- **Cells that block BMP have low BMP signalling** (AUROC 0.16 at dome and 30% epiboly, 0.22–0.36 from 75% epiboly to 3-somite), and so do the cells that make the organizer's BMP ligand *admp* (best 0.15, median 0.17): the organizer sends a BMP ligand but is itself shielded from BMP, consistent with ADMP acting at a distance [113, 206].
- **RA is made and degraded in different cells** at 90% epiboly and bud (AUROC down to 0.22).
- **Weak against the parent**: Nodal in the endoderm (median AUROC 0.62 against the margin, which also receives Nodal), Hedgehog in the notochord (0.68 against the axial mesoderm, which also makes *shha*) and FGF at the midbrain–hindbrain boundary (0.33 against the neural plate: the boundary has *lower* FGF scores than the rest of the neural plate).
- **Opposite**: Hedgehog in the axial mesoderm against the organizer (bud), since *shha* is already made in the organizer, and the Wnt antagonists against Wnt, which share the margin at 30% epiboly (AUROC 0.99 there) before they separate (down to 0.10 at 3-somite, not beyond the random genes).

The sending and blocking profiles are rarely coherent (Section 8.1; supported: Hedgehog from 60% epiboly, Delta from bud, the BMP antagonists from shield to 75% epiboly, the Wnt antagonists and Wnt/PCP at 60% epiboly): most pool genes of several tissues, e.g., *chd* of the organizer and *fsta* of the paraxial mesoderm, or have one or two genes. They are best read as markers of signalling centres within a tissue, together with the identity programs. Their couplings with every program are in Figure S8.

9.3 Which signals accompany a transition?

The derivations between programs (Figure 10) are presumed transitions between phenotypes, e.g., blastomere → margin. A Boolean model needs, for each transition, the signals under which it happens. For every derivation, we compare every signalling profile between the cells in which the derived program is active and the other active cells of its parent (relative AUROC, as above), at the first three stages at which the derived program is supported, i.e., at its emergence. A

Table 7: Expected signalling centres. Profile: a signalling profile (general, sending or blocking). Partner: an identity program (the AUROC of the profile's score for the cells in which the partner is active) or a signalling profile (the AUROC of the partner's score for the cells in which the profile is active). Exp.: expected direction. Against: the other cells, either the cells in which one of the partner's parent programs is active (parents) or all other cells (all). Stages: stages at which the identity partner is supported (signalling partners: all stages with enough active cells); best AUROC (stage) and median over these stages, and the median correlation of the two scores. Consistent: beyond random genes on the expected side at some stage and never on the other side (median over the libraries); weak: never beyond random genes; opposite: only on the other side; mixed: both.

Profile	Partner	Exp.	Against	Stages	Best AUROC	Median	Median r	Verdict
signal:Nodal	margin	+	parents	30% epiboly – 6-somite	0.97 (50% epiboly)	0.82	0.32	consistent
signal:Nodal	endoderm	+	parents	shield – 6-somite	0.73 (shield)	0.62	0.12	weak
signal:BMP	non neural ectoderm	+	parents	60% epiboly – 6-somite	0.98 (90% epiboly)	0.96	0.64	consistent
sending:BMP organizer	dorsal organizer	+	parents	shield – 6-somite	0.83 (75% epiboly)	0.73	0.50	consistent
sending:BMP organizer	signal:BMP	-	all	dome – 6-somite	0.15 (50% epiboly)	0.17	-0.29	consistent
blocking:BMP	dorsal organizer	+	parents	shield – 6-somite	0.95 (60% epiboly)	0.91	0.53	consistent
blocking:BMP	signal:BMP	-	all	dome – 6-somite	0.16 (dome)	0.33	-0.20	consistent
signal:Wnt	margin	+	parents	30% epiboly – 6-somite	0.98 (shield)	0.87	0.59	consistent
signal:Wnt	tailbud	+	parents	bud – 6-somite	0.88 (3-somite)	0.88	0.68	consistent
blocking:Wnt	dorsal organizer	+	parents	shield – 6-somite	0.83 (60% epiboly)	0.71	0.44	consistent
blocking:Wnt	prechordal plate	+	parents	60% epiboly – 6-somite	0.90 (60% epiboly)	0.53	0.63	consistent
blocking:Wnt	signal:Wnt	-	all	dome – 6-somite	0.10 (3-somite)	0.42	-0.09	opposite
signal:FGF	margin	+	parents	30% epiboly – 6-somite	0.99 (50% epiboly)	0.74	0.62	consistent
signal:FGF	tailbud	+	parents	bud – 6-somite	0.73 (6-somite)	0.68	0.48	consistent
signal:FGF	midbrain hindbrain boundary	+	parents	75% epiboly – 6-somite	0.56 (3-somite)	0.33	-0.22	weak
sending:RA	presomitic mesoderm	+	parents	shield – 6-somite	0.83 (60% epiboly)	0.78	0.83	consistent
sending:RA	blocking:RA	-	all	30% epiboly – 6-somite	0.22 (bud)	0.59	-0.03	consistent
blocking:RA	tailbud	+	parents	bud – 6-somite	0.89 (6-somite)	0.89	0.74	consistent
blocking:RA	forebrain eye	+	parents	75% epiboly – 6-somite	0.79 (75% epiboly)	0.73	0.42	consistent
sending: Hedgehog	axial mesoderm	+	parents	60% epiboly – 6-somite	0.53 (60% epiboly)	0.43	0.52	opposite
sending: Hedgehog	notochord	+	parents	60% epiboly – 6-somite	0.69 (75% epiboly)	0.68	0.49	weak
sending:Notch	presomitic mesoderm	+	parents	shield – 6-somite	0.88 (6-somite)	0.79	0.43	consistent
sending:Notch	primary neurons	+	parents	3-somite – 6-somite	0.77 (6-somite)	0.76	0.73	consistent
sending:Notch	receiving:Notch	+	all	30% epiboly – 6-somite	0.84 (3-somite)	0.72	0.32	consistent

profile that is higher in the derived cells at two of these stages or more, and never lower, is a candidate input of the transition; a lower one a candidate inhibitor. Table S8 lists them next to the signals expected from the annotation of the programs (Table S7). Of the 28 transitions, 26 can be tested (the neural plate and the dorsal forerunner cells are never supported):

- **Confirmed**: blastomere → margin with higher Nodal and FGF; margin → dorsal organizer with BMP and Wnt antagonists, *admp* and Hedgehog ligands, and lower Wnt; margin → tailbud with higher Wnt, FGF and BMP (and RA degradation); blastomere → non-neural ectoderm with higher BMP and lower BMP antagonists, Wnt and FGF; neural plate → forebrain with lower Wnt and higher RA degradation; neural plate → neural crest with higher BMP and Wnt (non-neural ectoderm → neural crest: Wnt); dorsal organizer → prechordal plate with lower Wnt and FGF.
- **Partly confirmed**: mesoderm → presomitic mesoderm with higher Nodal and RA synthesis and lower BMP, but not the expected Wnt, FGF and Notch; neural plate → placodes with higher BMP, but lower FGF, which the preplacodal ectoderm needs [101]; neural plate → hindbrain with lower RA degradation only.
- **Not seen**: BMP for the lateral plate, FGF for the midbrain–hindbrain boundary (which has higher Wnt), Notch for the primary neurons, Nodal for the endoderm and the prechordal plate against their parents (which also receive Nodal), and Hedgehog for the adaxial cells (no readouts).
- **No signal expected, none found**: maternal → PGC (germ plasm) and blastomere → EVL (position).

These are candidate rules, not proven causes: a signal can differ between two populations because one of them makes it, or because the two lie in different places. Where expected inputs are missing, the profile may lack readouts in the relevant tissue (Hedgehog, the Notch oscillator), or the parent receives the same signal (Nodal).

9.4 Conclusions

Compared with random genes, most couplings between the signalling profiles and the identity programs agree with the zebrafish literature: Nodal at the margin and in the endoderm, then along the axis; BMP first at the ventral margin, then in ventral and non-neural fates and against the neural plate; Wnt and FGF at the margin and in the posterior mesoderm. Of 24 expected signalling centres, 19 are consistent when tested against the neighbouring (parent) tissue at the stages at which the tissue is visible. For the transitions between programs, the data confirm the expected signals of the earliest decisions (margin, organizer, tailbud, non-neural ectoderm, forebrain, neural crest) and miss some later ones (Notch, FGF at the midbrain–hindbrain boundary, BMP in the lateral plate). Correlations show co-expression, not causation, and the sending and blocking profiles are best read as markers of signalling centres within a tissue; the RA and Hedgehog pathways have few or no usable readouts.

10 Result: cell populations

Before cells are grouped into populations, every single cell needs two answers: which program does it have, and should it take part at all? This section answers both (`scripts/26_cell_programs.py`); the populations themselves follow.

10.1 Which program does a cell have?

Calls. Within each library, a program's score in a cell is the mean Pearson residual of its genes, smoothed over the cell and its 10 nearest neighbours. The neighbours are found on the stage's heterogeneous genes *without* the genes of the scored programs, so that a program's own genes do not decide over which cells it is averaged, and without the heat shock, dissociation and cell-cycle genes (Section 10.2). The chance level is the gene-wise permutation null of Section 8.3 (20 permutations): a program is *active* above the null's 99.9% quantile, and its *significance* is the standardised excess $z = (\text{score} - \text{null mean})/\text{null SD}$, which, unlike the score, is comparable between programs with different numbers of genes and different sparsity. Smoothing is needed: unsmoothed, only 26–41% of the cells from 60% epiboly on (median library) have an active identity program, against 56–93% smoothed (a check outside the pipeline); neighbourhoods found with or without the programs' genes give the same calls.

Top program and margin. The identity programs compete for each cell: the curated anatomy programs and the maternal and blastomere states that are supported at the stage (Section 8.1), i.e., 2–3 programs up to 50% epiboly, 7 at shield and 14–24 from 60% epiboly on. Unsupported programs are left out: a program at the level of random genes would win cells by chance. A cell's *top program* is its active program with the largest z, and the *margin* is how far it is ahead of the next competitor (z of the top minus the largest z of the others, floored at 0); a cell without an active program has no top program.

Is the top program reliable? We split every program's genes at random into two halves and call the top program from each half alone. From shield on, the two halves agree in 91% of the cells with a margin of at least 5, but only in 35% of the cells with a margin below 1 (Figure 17a). The low-margin disagreements are mostly between related programs: counting a program and its ancestors and descendants (`derives_from`) as the same, the halves agree in 86% of the cells with a margin below 1, and in 98% above 5. A small margin therefore mostly means "this program or its parent" (e.g., the margin or the mesoderm), not an arbitrary call. Each half has half of the genes, so the calls from all genes are more reliable than this. Up to 50% epiboly, with only 2–3 competitors, the halves agree in practically every cell.

Result. Up to 50% epiboly, 77–92% of the cells have no active identity program (Figure 17b): as in Section 8.1, these stages are dominated by programs that act on whole libraries. A top program is found in 29% of the cells at shield, 58% at 60% epiboly and 81–93% from 75% epiboly on, where about half of the cells have a margin of at least 3. Programs differ in how often they win the cells in which they are active (Figure S9). Distinct populations win almost all of them, often by large margins: the EVL and the PGCs, the endoderm, the non-neural ectoderm (97% at 75% epiboly) and the forebrain and eye (75–82% from 75% epiboly on). The broad parent programs rarely win (at bud, the margin and the mesoderm in 16% of their active cells), and lose almost only to their descendants (93–100%); the adaxial somite loses to the presomitic mesoderm and the mesoderm. The axial programs (dorsal organizer, notochord, prechordal plate) lose to each other and to the hatching gland, and at bud the prechordal plate also to the non-neural ectoderm and the forebrain, with which it shares genes. The active and top cells of every program on the stage UMAPs are in `results/figures/cell_programs_atlas.pdf`.

10.2 Cells excluded before population detection

Populations should describe cells on developmental trajectories. Four kinds of cells could get in the way: dying or stressed cells, cells that are not part of the embryo proper (the YSL), technical artefacts (doublets and broken cells), and cells whose identity is overlaid by a state. Only the first are removed (Table S9).

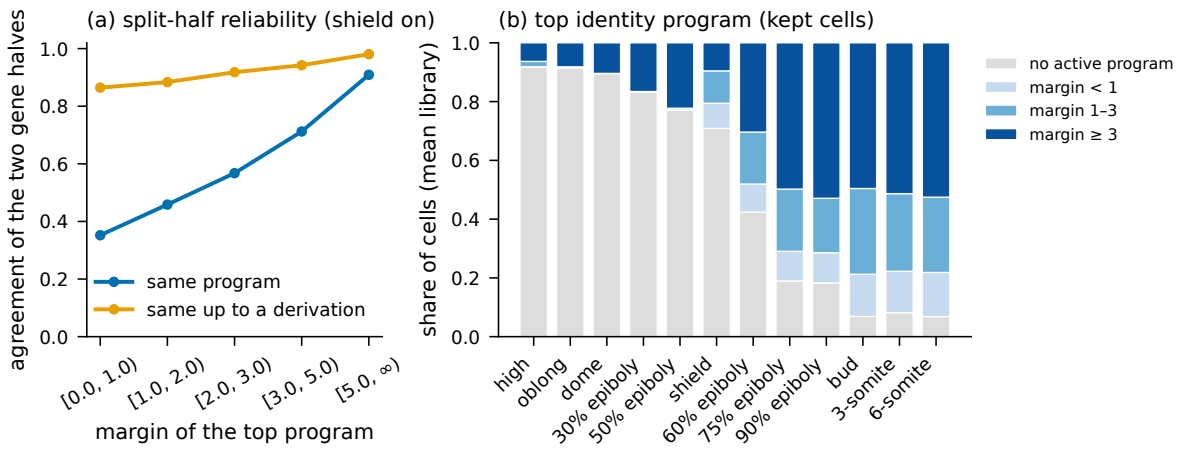


Figure 17: Top identity program per cell (`scripts/26_cell_programs.py`). (a) Agreement between the top programs called from two random halves of every program's genes, by the margin of the call from all genes, from shield on: the same program (blue), or the same up to a derivation (orange). (b) Share of the cells of a stage (mean over its libraries; stress cells excluded) without an active identity program, and with a top program by its margin.

Stress cells: removed. The p53 response, dissociation stress and heat shock compete with the identity programs where they are supported, and a cell is excluded if one of them is its top program with a margin of at least 2. This removes 460 cells (1.2%; 0–3.6% per library). 309 are p53 cells, from 50% to 90% epiboly (and 7 at high and oblong), with a median margin of 10.8: a distinct group of cells, not a p53 response on top of an identity. 151 are dissociation-stress cells, mostly at shield and 60% epiboly, with a median margin of 3.7. The p53 response is not supported from bud on, so no cells are removed there. The original study removed 618 "apoptotic-like" cells (Section 3.4), whose definition we do not know. Heat shock is never a top program.

Yolk and YSL: nothing to remove. The YSL control (Section 8.1) is active in 6% of the cells (3–9% per stage from dome on, more at later stages), but 92% of these also have an active identity program, and their top programs are mostly the non-neural ectoderm (44%), the endoderm (20%) and the EVL (14%). The control's genes that mark these cells (*tnfaip8l2b*, *sult2st1*, *sesn3*, *pleca*, *anxa1a*) are expressed in these tissues as well, whereas the yolk- and YSL-specific transcripts (*apoa1a*, *apoa1b*, *apoa2*, *apoa4b.1*, *apobb.1*, *afp4*, *itih2*, *tfa*) are nearly absent: 32 cells (0.08%) have three or more of them, mostly at bud and the somite stages. The deyolking removed the YSL, so there is no YSL population to remove; the YSL control measures non-neural and periderm genes rather than the YSL.

Heat shock, dissociation and cell cycle: genes, not cells. These states act on top of an identity. Heat shock differs mostly between libraries (the 50% epiboly DS4 pair, Section 5.3), dissociation stress is weak in most of the cells that show it, and the cell cycle runs in every cell type. Removing these cells would remove valid cells of every type, unevenly between libraries. Instead, their genes are left out of the neighbourhoods and embeddings used for the calls and the populations, so that they cannot pull cells of different types together.

Doublets and low-quality cells: no further filter. The GEO matrix already passed the original study's per-stage depth and complexity filters (Section 3.4). Within their libraries, 5 cells are more than 3 MADs below the median depth and 34 more than 3 MADs above. 279 cells (0.7%) have few detected genes for their depth, but up to shield 83% of them (of 144) have

the EVL as their top program: EVL cells, whose transcripts are dominated by a few keratins, not broken cells; later they are at most 1.1% of a library. Scrublet [210] flags 428 cells (1.1%) as doublets (score ≥ 0.25), but the scores decline smoothly without a separate group of cells, the flagged cells are only slightly deeper than their library (median 0.8 MADs above the median), and their top programs are distributed like those of all cells, without the combinations of distinct cell types that doublets of two types would show. This is about the rate expected after the upper depth limits of the original study, and we do not filter further.

10.3 Conclusions

- **Program calls.** Scored against its own permutation null and smoothed over neighbours found without its genes, every identity program can be compared with every other in every cell. From 75% epiboly on, 81–93% of the cells have a top program, and half of them by a margin of at least 3, at which the call is reproducible from half of the genes in 71–91% of the cells. Smaller margins are mostly ties between a program and its parent or sibling, which the population step has to resolve.
- **Exclusions.** Only the stress cells are removed (460 cells, 1.2%: p53 and dissociation stress). Yolk/YSL cells, doublets and low-quality cells are not a measurable problem in this dataset; heat shock, dissociation and the cell cycle are handled by leaving out their genes.
- **Next.** Populations: first the biological populations outside the Boolean model (PGCs, EVL), then the populations of the remaining cells.

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A Supplementary figures and tables

Table S1: Cell filtering per stage. Gene and UMI ranges are the per-stage thresholds of the original study (UMIs including mitochondrial RNA). Cells (GEO) is the number of cells in the GEO matrix; cells and embryos (Table S1) are taken from the supplementary Table S1.

Stage	Genes	UMIs	Cells (GEO)	Cells (Table S1)	Embryos (Table S1)
high	1,000–7,500	1,500–40,000	311	311	56
oblong	625–7,500	1,500–30,000	200	199	76
dome	800–3,800	2,000–20,000	1,174	1,147	28
30% epiboly	625–3,000	1,000–17,500	1,506	1,454	59
50% epiboly	600–4,000	1,500–25,000	5,864	5,695	91
shield	600–2,500	1,000–15,000	1,046	1,020	22
60% epiboly	600–3,500	1,500–22,500	4,201	4,099	79
75% epiboly	600–3,200	1,400–20,000	6,324	6,170	73
90% epiboly	500–3,500	1,000–20,000	5,626	5,474	66
bud	500–3,200	1,000–17,500	7,162	7,112	85
3-somite	500–3,000	1,000–12,500	1,631	1,602	19
6-somite	500–3,000	1,000–12,500	4,460	4,388	40
total			39,505	38,671	694

Table S2: The three tiers of heterogeneous genes, in the libraries of more than 1,000 cells (from dome on for the floors). Floor: minimum residual (median over libraries, and as a factor of the typical φ); the cutoff of a gene is the larger of the floor and the chance level at its mean. Sampling noise: genes selected in an independent simulated library in which no gene is heterogeneous. Well-expressed: genes of the tier with at least 1,000 UMIs; intrinsic noise: the estimated share of them that are merely noisier than typical (FDR of the empirical null).

Tier	Floor	Genes / library	Sampling noise	Well-expressed	Intrinsic noise
weak	0.49 (3.1×)	421–1,277	3–15 (median 8)	44–301	$\leq 50\%$
moderate	0.78 (6.0×)	258–920	1–9 (median 5)	20–183	$\leq 10\%$
strong	1.06 (11×)	157–667	0–5 (median 2)	11–122	$\leq 1\%$

Table S5: Signalling readouts and negative-control pathways (`data/curated/signalling_evidence.tsv`): role, type of evidence and its strength (strong: zebrafish, direct or loss/gain of the pathway; moderate: indirect zebrafish or other vertebrate evidence). Readouts mark cells that receive a signal; ligands and secreted antagonists mark cells that send or block one. Profile: the curated signalling profile of the gene (or why it is not used). Negative controls are pathways that should be inactive or uniform at 3–12 hpf, and dissociation-induced genes.

Pathway	Gene	Role	Evidence	Strength	Profile	References
readouts (receiving)						
BMP	*id1*	target	direct target (Smad-binding BMP response element in the Id1 promoter; induced by BMP in ES cells), mouse only	moderate	BMP	[74, 94]
BMP	*id3*	target	direct target (induced by BMP2/7 protein under cycloheximide; reduced in bmp7 mutants)	strong	BMP	[53, 74]
BMP	*bambia*	feedback	lost in swirl/bmp2b and expanded in chordino mutants; direct target (cycloheximide-resistant BMP induction)	strong	BMP	[193, 53]

Pathway	Gene	Role	Evidence	Strength	Profile	References
BMP	*szl*	feedback	reduced or absent in swirl/bmp2b and expanded in chordino mutants; direct target	strong	BMP	[212, 53]
BMP	*msx1b*	target	Msx expression changes with BMP level (BMP mutants, dominant-negative receptor); induced at intermediate BMP, so not monotonic	moderate	BMP	[191, 142]
BMP	*smad6a*	feedback	direct target in zebrafish (cycloheximide-resistant BMP induction); BMP-responsive element in the mouse Smad6 promoter	strong	BMP	[53, 78]
BMP	*smad6b*	feedback	direct target in zebrafish (cycloheximide-resistant BMP induction); BMP-responsive element in the mouse Smad6 promoter	strong	BMP	[53, 78]
BMP	*ved*	target	direct BMP target (cycloheximide); also downstream of Wnt3a/Wnt8 (lost in wnt3a/wnt8 morphants)	strong	not used: multi-readout and Wnt8 target	[53, 170, 169, 147]
BMP	*vox*	target	direct BMP target (cycloheximide); BMP-dependent and BMP-independent domains in BMP mutants; direct Wnt8/beta-catenin target	strong	not used: multi-readout and Wnt8 target	[53, 124, 148, 147, 77]
BMP	*vent*	target	direct BMP target (cycloheximide); mostly BMP-dependent in BMP mutants; direct Wnt8/beta-catenin target	strong	not used: multi-readout and Wnt8 target	[53, 124, 148, 147, 77]
Nodal	*lft1*	feedback	expression depends on Nodal signalling (antivin = lefty1); maintained by Cyclops; Lefty knockdown releases Nodal	strong	Nodal	[126, 35, 9, 186]
Nodal	*lft2*	feedback	Nodal-dependent transcription (Cyclops required for maintenance); Lefty knockdown releases Nodal	strong	Nodal	[35, 9]
Nodal	*ndr1*	auto-regulation	positive autoregulation through a Nodal response element (FoxH1/Smad) in the squint first intron (transgenic reporters)	strong	Nodal	[33, 35, 150]
Nodal	*ndr2*	auto-regulation	Nodal ligands induce their own transcription (positive autoregulatory loop), stated for both zebrafish Nodals	moderate	Nodal	[35, 150]
Wnt	*sp5l*	target	direct Wnt/beta-catenin target (microarray, induction under cycloheximide); required downstream of wnt8	strong	Wnt	[201, 188]
Wnt	*sp5a*	target	acts downstream of Wnt8 with sp5l (combined knockdown phenocopies loss of wnt8)	moderate	Wnt	[201]
Wnt	*axin2*	feedback	direct target (Tcf/Lef sites in the promoter and intron; rapid induction by Wnt), mouse	moderate	Wnt	[80]
Wnt	*nkd1*	feedback	expression mirrors Wnt/beta-catenin activity (ventrolateral margin, axial mesendoderm); Nkd1 inhibits canonical Wnt	moderate	not used: multi-readout and BMP target	[196, 53]
Wnt	*notum1a*	feedback	Wnt/beta-catenin-induced secreted inhibitor (self-regulatory loop; Wnt gain and loss in zebrafish)	strong	Wnt	[40]

Pathway	Gene	Role	Evidence	Strength	Profile	References
Wnt	*lef1*	target	direct Wnt target through a Wnt-responsive element in the LEF-1 promoter (human cells); in zebrafish Lef1 drives the TOPdGFP Wnt reporter	moderate	Wnt	[38, 24]
FGF	*spry2*	feedback	induced by FGF (gain and loss of FGF signalling), mouse and chick	moderate	FGF	[128]
FGF	*spry4*	feedback	FGF feedback-induced antagonist: lost in ace/fgf8 mutants and with SU5402, induced by FGF8	strong	FGF	[42]
FGF	*spry1*	feedback	induced by FGF (vertebrate Sprouty family), mouse and chick	moderate	FGF	[128]
FGF	*dusp6*	feedback	FGF synexpression; under FGF control after initiation by maternal beta-catenin on the dorsal side; FGF reporter transgene	strong	FGF	[194, 131]
FGF	*etv4*	target	lost in fgf8 mutants and with SU5402; induced by ectopic fgf8	strong	FGF	[155, 145]
FGF	*etv5b*	target	lost in fgf8 mutants and with SU5402; induced by ectopic fgf8	strong	FGF	[155, 145]
FGF	*il17rd*	feedback	FGF synexpression; positively regulated by FGF; feedback inhibitor of Ras/MAPK	strong	FGF	[192, 41]
RA	*cyp26a1*	feedback	induced by exogenous RA (zebrafish differential display; time- and dose-dependent up-regulation)	moderate	RA (blocking)	[203, 23, 100]
RA	*dhrs3a*	feedback	RA-dependent and strongly RA-inducible (microarray: opposite response to RA and RA antagonist; in situ)	strong	RA (receiving)	[37]
Notch	*her4.1*	target	activated by Notch1 intracellular domain (RNA injection); her4 regulatory elements report Notch activity (transgenic)	strong	Notch (receiving)	[181, 217]
Notch	*her4.2*	target	as HER4.1 (her4 cluster)	strong	Notch (receiving)	[181, 217]
Notch	*her4.3*	target	as HER4.1 (her4 cluster)	strong	Notch (receiving)	[181, 217]
Notch	*her4.4*	target	as HER4.1 (her4 cluster)	strong	Notch (receiving)	[181, 217]
Notch	*her15.1*	target	PSM expression regulated by Delta-Notch (mutants, morpholinos, inhibitor), in a complex manner; depends on her7	moderate	Notch (receiving)	[167]
Notch	*her15.2*	target	as HER15.1	moderate	Notch (receiving)	[167]
Notch	*her12*	target	PSM expression regulated by Delta-Notch (mutants, morpholinos, inhibitor), complex	moderate	Notch (receiving)	[167]
Notch	*nrarpa*	feedback	direct Notch target (Xenopus: Su(H)-dependent induction); zebrafish nrarpa also stabilises Lef1 (Wnt)	moderate	not used: multi-readout, also acts in Wnt/Lef1	[105, 79]
Notch	*nrarpb*	feedback	direct Notch target (Xenopus); direct BMP target in zebrafish (cycloheximide)	moderate	not used: multi-readout and BMP target	[105, 53]

Pathway	Gene	Role	Evidence	Strength	Profile	References
Notch	*hey1*	target	dynamic PSM expression impaired in deltaD (after-eight) mutants; ectopic deltaD induces hey1; promoter is a direct Notch target in mouse and human	strong	Notch (receiving)	[207, 120]
Hedgehog	*ptch1*	feedback	transcription depends on Hh signalling; responds to lower Hh levels than ptch2	strong	Hedgehog (receiving)	[115]
Hedgehog	*ptch2*	feedback	transcription follows shh activity; ectopic shh activates it; PKA-dependent	strong	Hedgehog (receiving)	[16, 115]
Hedgehog	*gli1*	target	Shh activates Gli1 transcription (Xenopus); in zebrafish Gli1 (detour) is an activator of Hh target genes	moderate	Hedgehog (receiving)	[108, 84]
Hedgehog	*hhip*	feedback	induced by ectopic Hh and lost in Hh mutants (mouse); zebrafish hhip is expressed in Hh-responding adaxial cells	moderate	Hedgehog (receiving)	[15, 138]
negative controls						
IFN	*isg15*	control	IFN-stimulated gene (induced by recombinant type I IFN and by virus via IFNR in larvae); also induced by morpholino injection	moderate	–	[114, 103, 104, 70]
IFN	*mxa*	control	IFN-stimulated gene (larvae, recombinant IFN and virus)	moderate	–	[114, 104, 70]
IFN	*rsad2*	control	IFN-stimulated gene (viperin; larvae, recombinant IFN and virus)	moderate	–	[114, 104, 70]
AhR	*cyp1a*	control	AHR target induced by TCDD and PCB126 in embryos (xenobiotic response elements)	strong	–	[4, 82]
AhR	*cyp1b1*	control	induced by PCB126 in embryos (xenobiotic response elements)	moderate	–	[82, 4]
AhR	*cyp1c1*	control	induced by PCB126 in embryos; high basal embryonic expression	moderate	–	[82]
AhR	*ahrrb*	control	induced by TCDD in ZF-L cells and embryos; AHR negative-feedback repressor	strong	–	[32]
GR	*fkbp5*	control	GR target: up-regulated by full-length GR mRNA at 6 hpf; GR-dependent and cortisol-induced in larvae	strong	–	[46, 47, 2]
GR	*klf9*	control	under-expressed in GR mutant larvae and induced by cortisol; feedforward regulator of fkbp5	strong	–	[46, 47, 2]
GR	*tsc22d3*	control	described as a GR target (gilz) with GR-dependent daily oscillation in larvae	moderate	–	[47, 2]
circadian	*per1b*	control	zygotic period1 transcription requires CLOCK (dominant-negative CLOCK blocks it)	moderate	–	[20]
circadian	*per2*	control	light-induced in all zebrafish cells through a D-box/E-box light-responsive module; light at 0-6 hpf transiently raises per2	moderate	–	[197, 218]
circadian	*cry1aa*	control	light-induced clock repressor (cry1a); induction correlates with clock resetting	moderate	–	[183]
circadian	*tefa*	control	light-inducible D-box factor that drives per2 light induction	moderate	–	[197]
dissoc.	*fosab*	control	immediate-early gene induced by enzymatic dissociation at 37 C (scRNA-seq), mouse	moderate	–	[195, 21]

Pathway	Gene	Role	Evidence	Strength	Profile	References
dissoc.	*junba*	control	immediate-early gene induced by dissociation (mouse)	moderate	–	[195, 21]
dissoc.	*jun*	control	immediate-early gene induced by dissociation (mouse)	moderate	–	[195, 21]
dissoc.	*atf3*	control	immediate-early gene induced by dissociation (mouse)	moderate	–	[195, 21]
dissoc.	*egr1*	control	immediate-early gene induced by dissociation (mouse)	moderate	–	[195, 21]
ligands (sending)						
Nodal	*ndr1*	ligand	in situ: maternal, then blastoderm margin at blastula; dorsal blastomeres at MBT and dorsal YSL, whole margin and YSL in early gastrula, dorsal forerunner cells at shield; sqt;cyc double mutants lack organizer, mesoderm and endoderm; misexpression (also only in the YSL) expands organizer fates; Squint/Cyclops required in the YSL for endoderm and head mesoderm	strong	Nodal	[150, 31, 36, 33]
Nodal	*ndr2*	ligand	in situ: blastoderm margin at blastula; dorsal mesendoderm and shield hypoblast at gastrula, prechordal plate until early somitogenesis, later left lateral plate and left diencephalon; cyc mutants lack floor plate and ventral brain, rescued non-autonomously by cyc RNA; redundant with sqt for organizer and mesendoderm	strong	Nodal	[150, 159, 36]
BMP	*bmp2b*	ligand	in situ: whole blastula, restricted ventrally before gastrulation (Fgf-dependent); ventral gastrula; swirl/bmp2b mutants dorsalized (loss of non-neural ectoderm, neural crest, ventral mesoderm), cell non-autonomous; overexpression ventralizes	strong	BMP	[92, 134, 135, 45, 163, 144]
BMP	*bmp4*	ligand	in situ: ventral gastrula and also the embryonic shield; null mutants affect late-gastrula specification of ventroposterior fates	strong	BMP	[135, 176]
BMP	*bmp7a*	ligand	snailhouse/bmp7 mutants strongly dorsalized, null identical to swirl/bmp2b; Bmp2b/Bmp7 synergize (likely heterodimers), transduced by Smad5 and antagonized by Chordino; Bmp genes expressed in the whole blastula, then restricted ventrally	strong	BMP	[22, 163, 45]
BMP	*admp*	ligand	in situ: dorsal only, from blastula in the organizer region (anterior neuroectoderm and axial mesoderm precursors), vanishing in an anterior-to-posterior wave during gastrulation; also tail organizer; knockdown enlarges the organizer and impairs ventral/posterior development, overexpression is Bmp-like; cooperates with Bmp2b/Bmp7	strong	BMP organizer (sending)	[113, 206]

Pathway	Gene	Role	Evidence	Strength	Profile	References
Wnt	*wnt8a*	ligand	in situ: maternal and zygotic; blastoderm margin at 50% epiboly (cleared from the dorsal midline in late gastrula), tailbud during segmentation; deficiency and morpholino: DV mesoderm and AP neural patterning defects; wnt3a/wnt8 double knockdown expands the organizer and removes the posterior body	strong	Wnt	[86, 111, 147, 169]
Wnt	*wnt3a*	ligand	in situ: blastoderm margin (overlapping wnt8) and later tailbud; wnt3a/wnt8 double knockdown anteriorizes the neuroectoderm, expands the organizer and blocks tail formation; sp5l in the tail depends on wnt3a and wnt8	strong	Wnt	[169, 188]
Wnt/PCP	*wnt5b*	ligand	maternal and zygotic at all gastrula stages; pipetail mutants: cell elongation and convergent extension defects in posterior regions of the gastrula (redundant with wnt11 anteriorly); Frizzled-2 acts downstream	strong	Wnt/PCP (sending)	[90]
Wnt/PCP	*wnt11*	ligand	in situ: blastoderm margin (presumptive mesoderm) from late blastula, induced by the yolk cell and FGF, independent of ntl; later somites and notochord; silberblick mutants: convergent extension defects and cyclopia, cell non-autonomous, rescued by a Dishevelled that does not signal through the canonical pathway	strong	Wnt/PCP (sending)	[121, 68]
FGF	*fgf8a*	ligand	in situ: dorsoventral gradient (dorsal high) at the gastrula margin, not in axial mesoderm; mesodermal precursors; MHB and somitogenesis domains; overexpression dorsalizes and represses ventral bmp2b/bmp4; acerebellar mutants lose the MHB organizer, with mild DV and mesoderm defects; with fgf24 required for posterior mesoderm	strong	FGF	[44, 151, 25, 45]
FGF	*fgf24*	ligand	in situ: co-expressed with fgf8 in mesodermal precursors during gastrulation; fgf24 knockdown in fgf8 mutants blocks most posterior mesoderm	strong	FGF	[25]
FGF	*fgf3*	ligand	in situ: dorsolateral margin from late blastula to early gastrula; overexpression expands the neural region and suppresses anterior neural fate, with little mesoderm-inducing activity	strong	FGF	[96]
FGF	*fgf4*	ligand	in situ: posterior notochord and Kupffer's vesicle region in early segmentation; knockdown randomizes visceral left-right patterning, removes lefty1 in the posterior notochord and impairs Kupffer's vesicle cilia	strong	FGF	[214]

Pathway	Gene	Role	Evidence	Strength	Profile	References
RA	*aldh1a2*	synthesis	in situ: involuting mesoderm during gastrulation, later paraxial mesoderm, branchial arches, eyes and fin buds; neckless and no-fin (raldh2) mutants: hindbrain/spinal cord AP defects and no pectoral fin, rescued by RA; RA from paraxial mesoderm acts non-autonomously on the neural tube	strong	RA (sending)	[52, 7, 23]
Notch	*dla*	ligand	in situ: large patches of contiguous cells in the neural plate (with deltaD) within which primary neurons are singled out; misexpression delivers lateral inhibition; itself repressed by lateral inhibition	strong	Notch (sending)	[60]
Notch	*dlb*	ligand	in situ: scattered cells within the deltaA/deltaD neural plate patches that become primary neurons; misexpression delivers lateral inhibition	strong	Notch (sending)	[60]
Notch	*dlc*	ligand	in situ: mesoderm from 50% epiboly, tailbud, stripes in the presomitic mesoderm, posterior half of each somite, notochord; also retina, ear and scattered epidermal cells; oscillates in the PSM; beamter/deltaC mutants and morphants disrupt the segmentation clock (her1 oscillation)	strong	Notch (sending)	[172, 73, 83]
Notch	*dld*	ligand	in situ: neural plate patches (with deltaA) and somite development; misexpression delivers lateral inhibition; after eight (aei)/deltaD mutants lose her1 oscillation in the PSM; deltaD itself does not oscillate	strong	Notch (sending)	[60, 72, 73]
Hedgehog	*shha*	ligand	in situ: notochord, floor plate (ventral midline of neural tube and brain, incl. ventral diencephalon) and fin mesoderm; ectopic expression consistent with floor plate induction and promotes proximal eye fates; sonic-you mutants: defects in somite patterning, lateral floor plate, fins and motor axons (medial floor plate still forms)	strong	Hedgehog (sending)	[98, 29, 162]
Hedgehog	*shhb*	ligand	in situ: ventral midline of neural tube and brain incl. floor of the diencephalon, not notochord; ectopic expression promotes proximal eye fates; lost with the ventral forebrain in cyclops mutants	strong	Hedgehog (sending)	[29]
secreted antagonists and degradation (blocking)						
Nodal	*lft1*	secreted antagonist	overexpression abolishes mesoderm induction, suppressed by Cyclops/Squint or Activin; expression depends on Nodal (feedback); midline expression overlapping cyc/sqt, later left diencephalon; lft1+lft2 knockdown: unchecked Nodal signalling, expanded mesendoderm and loss of ectoderm; lft1 alone: left-right defects	strong	Nodal	[186, 126, 9, 35]

Pathway	Gene	Role	Evidence	Strength	Profile	References
Nodal	*lft2*	secreted antagonist	midline expression overlapping cyc/sqt, later left lateral plate; lft2 knockdown alone has no obvious phenotype, lft1+lft2 knockdown expands mesendoderm through unchecked Squint signalling	strong	Nodal	[9, 35]
BMP	*chd*	secreted antagonist	in situ: organizer and transiently axial mesoderm, also paraxial mesoderm, ectoderm and brain regions; domains expand in dorsalized mutants by mid-gastrula; overexpression dorsalizes; chordino (dino) mutants ventralized	strong	BMP (blocking)	[127, 63, 92, 22]
BMP	*nog1*	secreted antagonist	noggin1 expressed in the fish organizer and dorsalizes on overexpression; all three zebrafish noggins antagonize ventralizing BMPs	moderate	BMP (blocking)	[43, 6]
BMP	*nog2*	secreted antagonist	overexpression antagonizes ventralizing BMPs; expression differs from noggin1/noggin3 (domain not stated in the abstract)	moderate	BMP (blocking)	[43]
BMP	*fsta*	secreted antagonist	in situ: from mid-gastrula in anterior paraxial mesoderm (head mesoderm and first five somites), transient in the posterior notochord anlage, later brain, eyes and somites; excluded from the organizer; domain regulated by Bmp2b and Chordino; overexpression dorsalizes like Xenopus follistatin	strong	BMP (blocking)	[6]
BMP	*szl*	secreted antagonist	in situ: ventral from late blastula through gastrula, posterior end during segmentation; ogon mutants/morphants ventralized; overexpression dorsalizes, needs Chordin; does not inhibit Wnt8-dependent transcription	strong	BMP	[212, 123]
Wnt	*dkk1b*	secreted antagonist	in situ: dorsal marginal blastoderm and dorsal YSL after MBT, entire margin at late blastula, embryonic shield during gastrulation, then anterior axial mesendoderm (prospective prechordal plate); maintenance reduced in boz, sqt and oep mutants; overexpression enlarges forebrain, eyes and axial mesendoderm	strong	Wnt (blocking)	[64]
Wnt	*frzb*	secreted antagonist	sfrp3 (frzb) cloned; expressed mainly in mesodermal and endodermal tissues (stages not given in the abstract)	moderate	Wnt (blocking)	[185]
Wnt	*sfrp1a*	secreted antagonist	in situ: anterior neural plate including the anterior neural border cells (with tlc), complementary to posterior wnt expression; later ventral forebrain, midbrain and hindbrain; sfrp1a loss of function upregulates canonical Wnt signalling (hematopoietic stem cells)	strong	Wnt (blocking)	[185, 76]

Pathway	Gene	Role	Evidence	Strength	Profile	References
RA	*cyp26a1*	degradation	in situ: presumptive anterior neural ectoderm and blastoderm margin from late blastula, circumblastoporal ring during gastrulation, tailbud during somitogenesis; complementary to hoxb1b; knockdown or giraffe mutation expands posterior (spinal cord) fates; encodes the RA 4-hydroxylase	strong	RA (blocking)	[100, 23, 30, 203, 122]

Table S6: Genes removed from the curated anatomy programs to make them disjoint (script 16; only genes that ranked within the cap of 30). Left: genes of a signalling profile or a cell-state program. Right: genes shared with another anatomy program that it neither derives from nor gives rise to (column derives_from of the annotation); the gene stays in the program with more ZFIN publications for it, or, if they have as many (marked $^{=}$), in the one where it ranks higher.

Program	In a signalling or state program	Shared with
EVL		tailbud: *sult6b1*$^{=}$; hatching gland: *klf17*, *pfn1*$^{=}$, *tagln2*$^{=}$; margin: *ved*, *aplnra*; non neural ectoderm: *gata3*; placodes: *foxi1*; intermediate mesoderm: *ftr83*$^{=}$; presomitic mesoderm: *gadd45ba*
PGC	*h1m* (maternal), *nanog* (blastomere)	
margin	*dkk1b* (Wnt (blocking)), *lft1* (Nodal), *nog1* (BMP (blocking)), *fgf8a* (FGF), *lft2* (Nodal)	forebrain eye: *noto*$^{=}$; EVL: *tagln2*, *grhl3*; neural crest: *foxd3*; PGC: *ca15b*
dorsal organizer	*chd* (BMP (blocking)), *dkk1b* (Wnt (blocking)), *frzb* (Wnt (blocking)), *admp* (BMP organizer (sending)), *id3* (BMP), *nog1* (BMP (blocking)), *id1* (BMP), *fgf3* (FGF), *bambia* (BMP), *shha* (Hedgehog (sending))	presomitic mesoderm: *gadd45ba*, *ripply1*; intermediate mesoderm: *osr1*, *lhx1a*; neural crest: *foxd3*; midbrain hindbrain boundary: *ism1*; forebrain eye: *arl4ab*$^{=}$; endoderm: *foxa3*; dorsal forerunner: *foxj1a*; hindbrain: *efnb2a*; tailbud: *foxd5*; lateral plate: *tbx1*
dorsal forerunner	*id1* (BMP), *sp5l* (Wnt), *fgf4* (FGF), *bmp2b* (BMP), *ndr1* (Nodal)	forebrain eye: *stm*; midbrain hindbrain boundary: *ism1*, *cnpy1*; endoderm: *sox32*, *sox17*; prechordal plate: *cdh1*
endoderm	*bmp7a* (BMP), *pou5f3* (blastomere)	EVL: *grhl3*; lateral plate: *gata6*, *tbx1*; mesoderm: *fscn1a*, *cxcl12b*; neural plate: *irx3a*; adaxial somite: *idh1*; placodes: *ptgs2a*$^{=}$; primary neurons: *neurod1*
mesoderm	*chd* (BMP (blocking)), *dkk1b* (Wnt (blocking)), *fosab* (dissociation), *szl* (BMP), *aldh1a2* (RA (sending)), *wnt8a* (Wnt), *msx1b* (BMP)	neural plate: *fzd8b*$^{=}$, *her6*; dorsal organizer: *gadd45ba*$^{=}$, *osr1*; endoderm: *gata5*; neural crest: *foxd3*; midbrain hindbrain boundary: *ism1*; forebrain eye: *six3b*; notochord: *lhx1a*
presomitic mesoderm	*nog1* (BMP (blocking)), *aldh1a2* (RA (sending)), *wnt8a* (Wnt), *fsta* (BMP (blocking))	neural plate: *her6*, *slc2a12*$^{=}$; tailbud: *hes6*$^{=}$, *foxd5*$^{=}$; lateral plate: *aplnrb*, *blf*$^{=}$; endoderm: *cxcl12b*$^{=}$; hindbrain: *her2*$^{=}$, *cyp26c1*
tailbud	*admp* (BMP organizer (sending)), *szl* (BMP), *wnt8a* (Wnt), *cyp26a1* (RA (blocking)), *shhb* (Hedgehog (sending)), *bmp4* (BMP), *bmp2b* (BMP), *bmp7a* (BMP)	dorsal forerunner: *sult6b1*$^{=}$, *foxj1a*; mesoderm: *eve1*; presomitic mesoderm: *msgn1*, *tbx6l*; neural plate: *nkx1.2la*; hatching gland: *cited4b*; hindbrain: *fam212ab*$^{=}$, *lfng*$^{=}$
adaxial somite	*aldh1a2* (RA (sending)), *hsp90aa1.2* (heat shock), *dlc* (Notch (sending))	forebrain eye: *arl4ab*$^{=}$; lateral plate: *aplnrb*; notochord: *cnn2*$^{=}$, *asph*$^{=}$; neural plate: *akap12b*; axial mesoderm: *ntd5*$^{=}$; tailbud: *foxd5*; hindbrain: *efemp2b*$^{=}$

Program	In a signalling or state program	Shared with
lateral plate	*wnt11* (Wnt/PCP (sending)), *aldh1a2* (RA (sending)), *atf3* (dissociation), *dhrs3a* (RA (receiving)), *her12* (Notch (receiving))	hatching gland: *klf17*; endoderm: *gata5*; intermediate mesoderm: *osr1*, *ftr82*$^{=}$; adaxial somite: *cxcl12b*; dorsal forerunner: *foxj1a*; tailbud: *znfl2a*
intermediate mesoderm	*wnt8a* (Wnt), *her12* (Notch (receiving))	dorsal forerunner: *foxj1a*; neural crest: *msx3*; placodes: *foxj1b*; midbrain hindbrain boundary: *emilin1a*; forebrain eye: *swap70b*; prechordal plate: *fzd8a*
prechordal plate	*dkk1b* (Wnt (blocking)), *frzb* (Wnt (blocking)), *id3* (BMP), *nog1* (BMP (blocking)), *lft2* (Nodal), *bambia* (BMP)	lateral plate: *klf17*; presomitic mesoderm: *aplnra*$^{=}$; neural plate: *fzd8b*$^{=}$; intermediate mesoderm: *osr1*; forebrain eye: *stm*, *six3b*; notochord: *ell2*$^{=}$, *calr3b*$^{=}$; adaxial somite: *idh1*, *prdm1a*; neural crest: *ovol1*$^{=}$; hindbrain: *dbx1a*$^{=}$; axial mesoderm: *tram1*$^{=}$; midbrain hindbrain boundary: *shisa2*$^{=}$, *otx1b*
hatching gland	*frzb* (Wnt (blocking)), *id3* (BMP), *lft2* (Nodal), *bambia* (BMP)	notochord: *xbp1*$^{=}$, *zgc:162964*$^{=}$; forebrain eye: *elovl6*$^{=}$, *si:ch211-153b23.5*$^{=}$; axial mesoderm: *tram1*$^{=}$; midbrain hindbrain boundary: *shisa2*$^{=}$
axial mesoderm	*chd* (BMP (blocking)), *dkk1b* (Wnt (blocking)), *frzb* (Wnt (blocking)), *admp* (BMP organizer (sending)), *nog1* (BMP (blocking)), *shha* (Hedgehog (sending))	hatching gland: *xbp1*; neural crest: *foxd3*, *fam212aa*$^{=}$; endoderm: *foxa2*, *gid8b*$^{=}$, *foxa1*; intermediate mesoderm: *lhx1a*; prechordal plate: *cdh1*; presomitic mesoderm: *btg2*
notochord	*admp* (BMP organizer (sending)), *id3* (BMP), *nog1* (BMP (blocking)), *shha* (Hedgehog (sending))	endoderm: *foxa2*; adaxial somite: *cdkn1ca*$^{=}$, *ntd5*$^{=}$; midbrain hindbrain boundary: *ism1*; intermediate mesoderm: *lhx1a*; hatching gland: *ell2*$^{=}$, *calr3b*$^{=}$, *rrbp1b*$^{=}$, *stk10*$^{=}$, *hspa5*$^{=}$, *tram1*$^{=}$, *cited4b*$^{=}$; dorsal forerunner: *rassf7b*; neural plate: *akap12b*, *tdgf1*; neural crest: *fam212aa*
neural plate	*wnt11* (Wnt/PCP (sending)), *sox3* (blastomere), *msx1b* (BMP), *id1* (BMP), *fgf3* (FGF), *cyp26a1* (RA (blocking)), *sox19a* (blastomere)	dorsal organizer: *noto*; non neural ectoderm: *tfap2a*; dorsal forerunner: *stm*$^{=}$; adaxial somite: *arl4ab*$^{=}$; presomitic mesoderm: *mdka*; notochord: *klf6a*$^{=}$
forebrain eye	*lft1* (Nodal), *lft2* (Nodal)	dorsal organizer: *noto*; prechordal plate: *elovl6*; dorsal forerunner: *cxcr4a*; hindbrain: *efnb2a*$^{=}$, *dbx1a*$^{=}$, *rnd1l*$^{=}$; midbrain hindbrain boundary: *mdkb*$^{=}$, *shisa2*$^{=}$, *otx1b*$^{=}$, *otx1a*$^{=}$; margin: *arl4ab*; presomitic mesoderm: *fgf17*
midbrain hindbrain boundary	*dkk1b* (Wnt (blocking)), *fgf8a* (FGF)	hindbrain: *irx7*$^{=}$; forebrain eye: *otx2*; placodes: *irx1a*$^{=}$
hindbrain	*dkk1b* (Wnt (blocking)), *fgf3* (FGF)	mesoderm: *cxcl12a*; presomitic mesoderm: *efemp2b*; dorsal forerunner: *si:ch211-152c2.3*$^{=}$; forebrain eye: *sox2*; midbrain hindbrain boundary: *sox21a*; adaxial somite: *uncx4.1*
primary neurons	*dlb* (Notch (sending))	non neural ectoderm: *gata3*, *tfap2c*; neural crest: *tfap2a*; adaxial somite: *cdkn1ca*; midbrain hindbrain boundary: *prdm12b*; placodes: *scrt2*, *six4b*, *insm1b*, *eya1*
non neural ectoderm		EVL: *grhl3*; neural plate: *dlx3b*; lateral plate: *gata2a*; prechordal plate: *cdh1*
neural crest	*msx1b* (BMP)	margin: *ved*; placodes: *foxi1*; hatching gland: *cited4b*; primary neurons: *zbtb10*$^{=}$; forebrain eye: *zic2b*; axial mesoderm: *cnn3a*
placodes	*sox3* (blastomere), *sp5a* (Wnt), *fgf24* (FGF)	primary neurons: *gata3*, *pnocb*$^{=}$, *neurod1*, *insm1a*$^{=}$; neural crest: *tfap2a*; midbrain hindbrain boundary: *prdm12b*

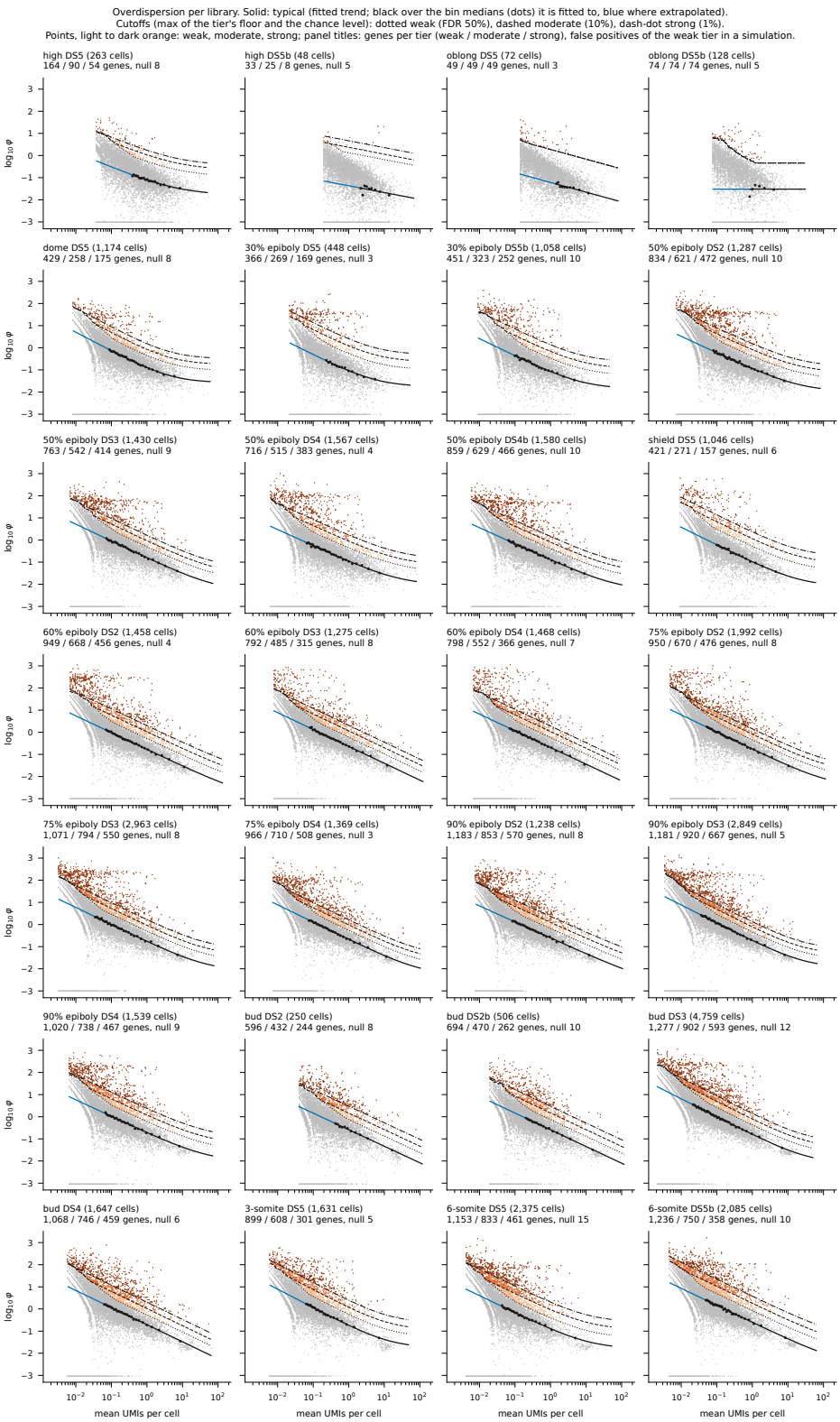


Figure S1: Overdispersion against the mean in every library, as in Figure 5a: estimates, the fitted typical value (blue where extrapolated) and the cutoffs of the three tiers (`scripts/01b_heterogeneous_genes.py`).

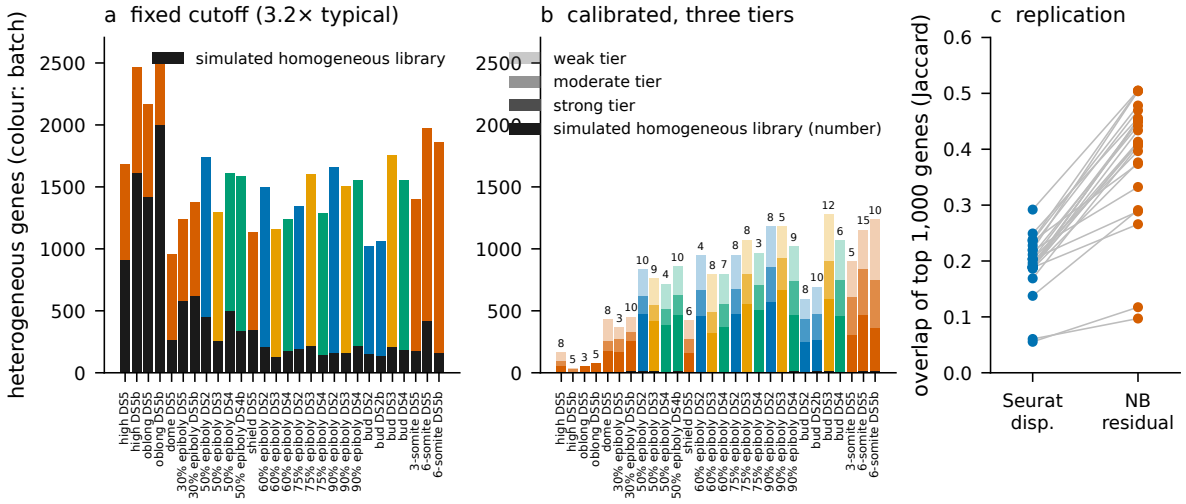


Figure S2: Heterogeneous genes per library. (a) With a fixed cutoff (residual at least 0.5), in the data (coloured by sequencing batch) and in a simulated library with the same depths and means in which no gene is heterogeneous (black, in front). (b) The same for the three tiers (light to full colour: weak, moderate, strong); the simulation is independent of the ones used for the calibration, and its weak-tier genes (at most 15, too few to be visible as bars) are given as numbers. (c) Overlap (Jaccard index) of the top 1,000 genes of two libraries of the same stage, ranked by Seurat's normalized dispersion or by the NB residual; one line per pair of libraries.

Table S3: Heterogeneous genes vs. HVG methods, ranges over the stages from dome on. Top 2,000: genes also heterogeneous, and genes with fewer than 10 UMIs in every library. ZFIN, top k: share of the ZFIN genes with a restricted (Restr.) or whole-organism (Whole) pattern at the stage among the top k genes (k = number of heterogeneous genes). Jaccard: overlap of the top 1,000 genes of two libraries of a stage (median pair per stage, from 30% epiboly on).

	Top 2,000		ZFIN, top k		
Method	Het.	Unmeas.	Restr. %	Whole %	Jaccard
Seurat (v1/v2)	339–917	816–1074	16–40	1–2	0.14–0.29
Seurat (v1/v2), ≥ 3 cells	371–1027	510–849	19–41	1–3	0.16–0.34
Seurat v3 (vst)	420–1296	379–703	26–55	1–3	0.29–0.49
Seurat v3, per library	420–1255	282–755	26–54	1–3	–
Pearson residuals	412–1217	448–792	25–54	1–4	0.24–0.46
Pearson res., per library	412–1203	1–623	25–56	1–5	–
NB residual (01b)	421–1369	0–0	29–59	1–4	0.29–0.50
heterogeneous (01b)	–	–	32–61	1–4	–

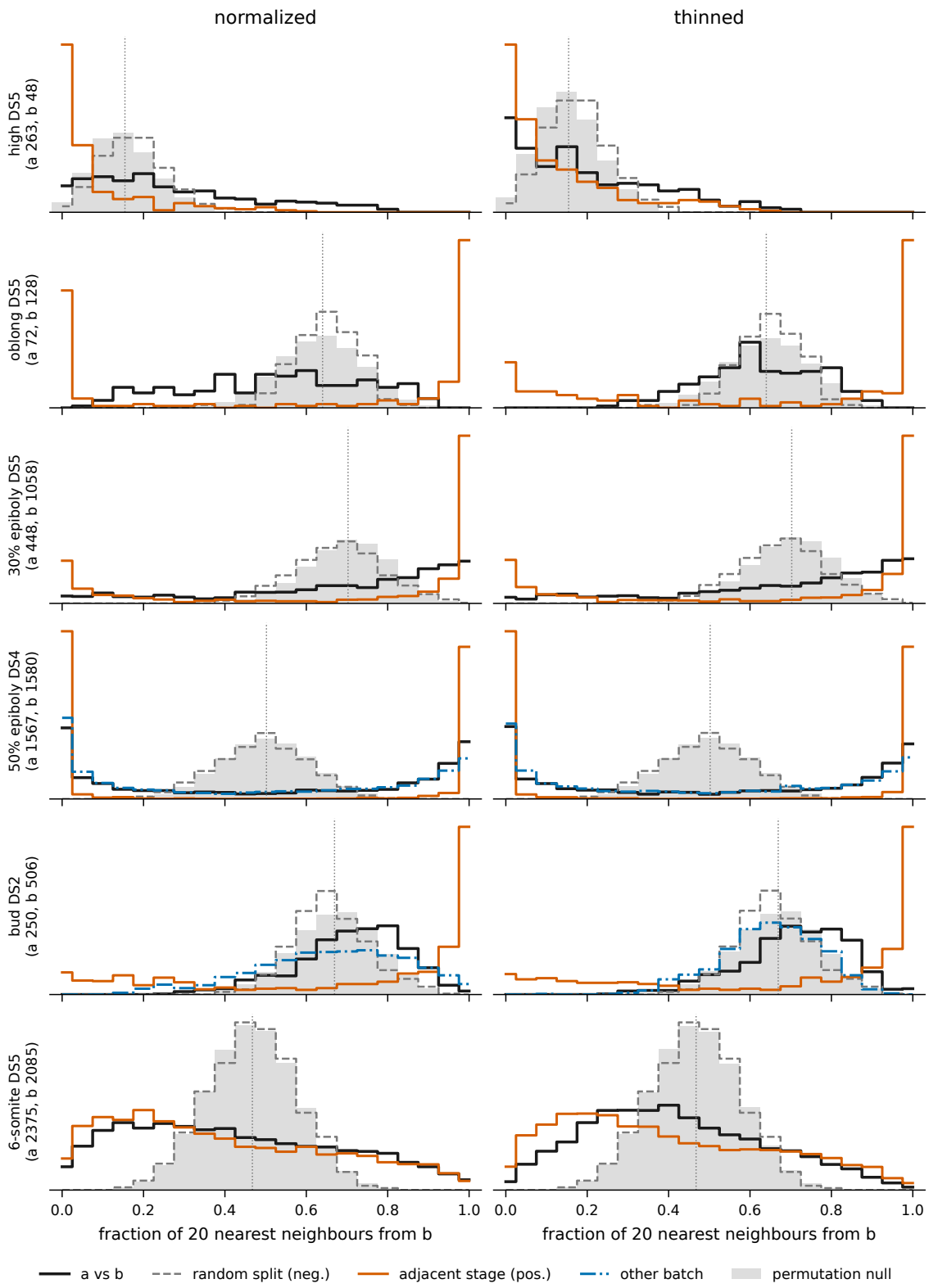


Figure S3: Mixing of the a/b libraries: per-cell distributions of the neighbourhood composition for every comparison of Figure 7 (`scripts/02_replicate_mixing.py`).

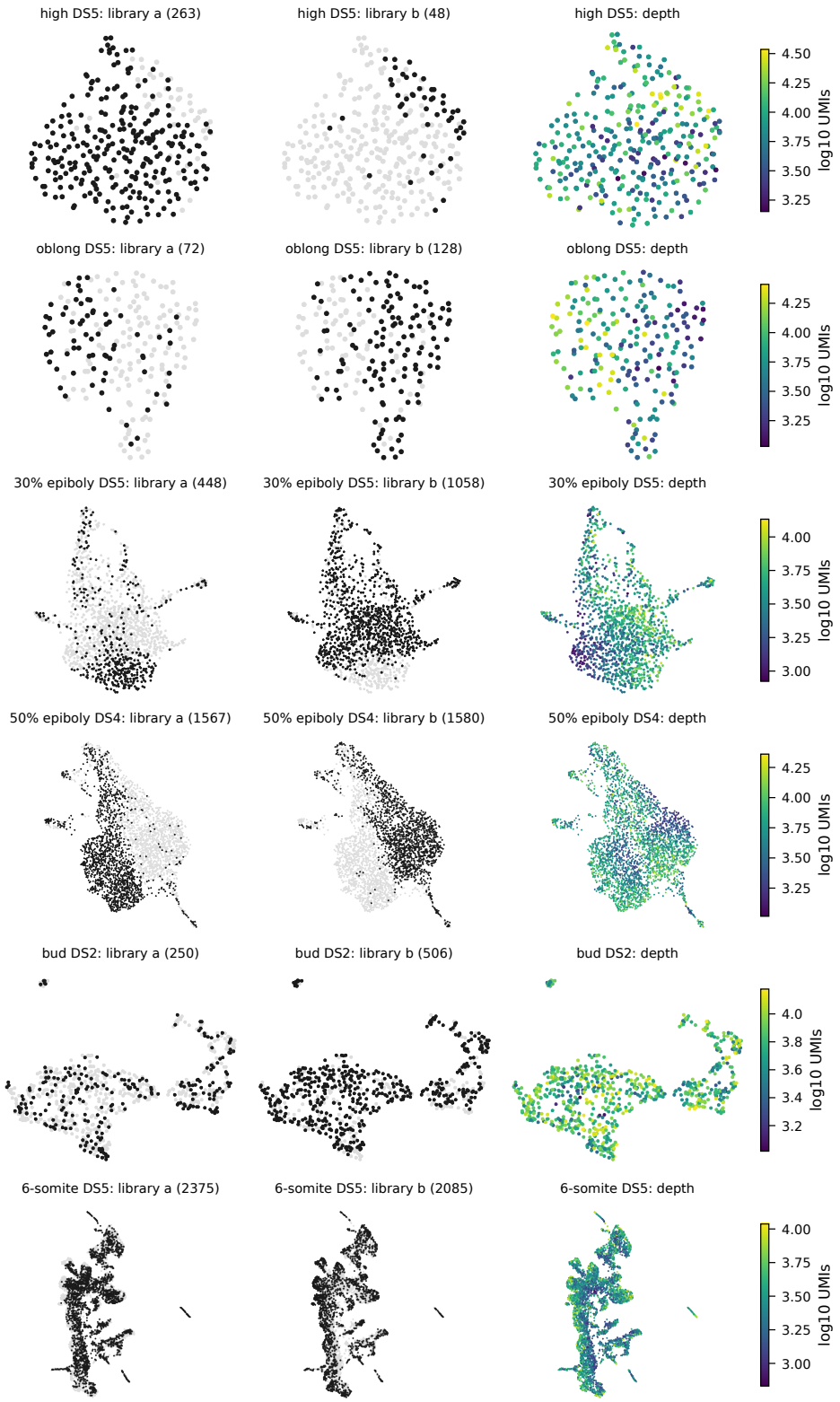


Figure S4: UMAPs of the six a/b pairs (library-size normalization only), coloured by library (`scripts/02_replicate_mixing.py`).

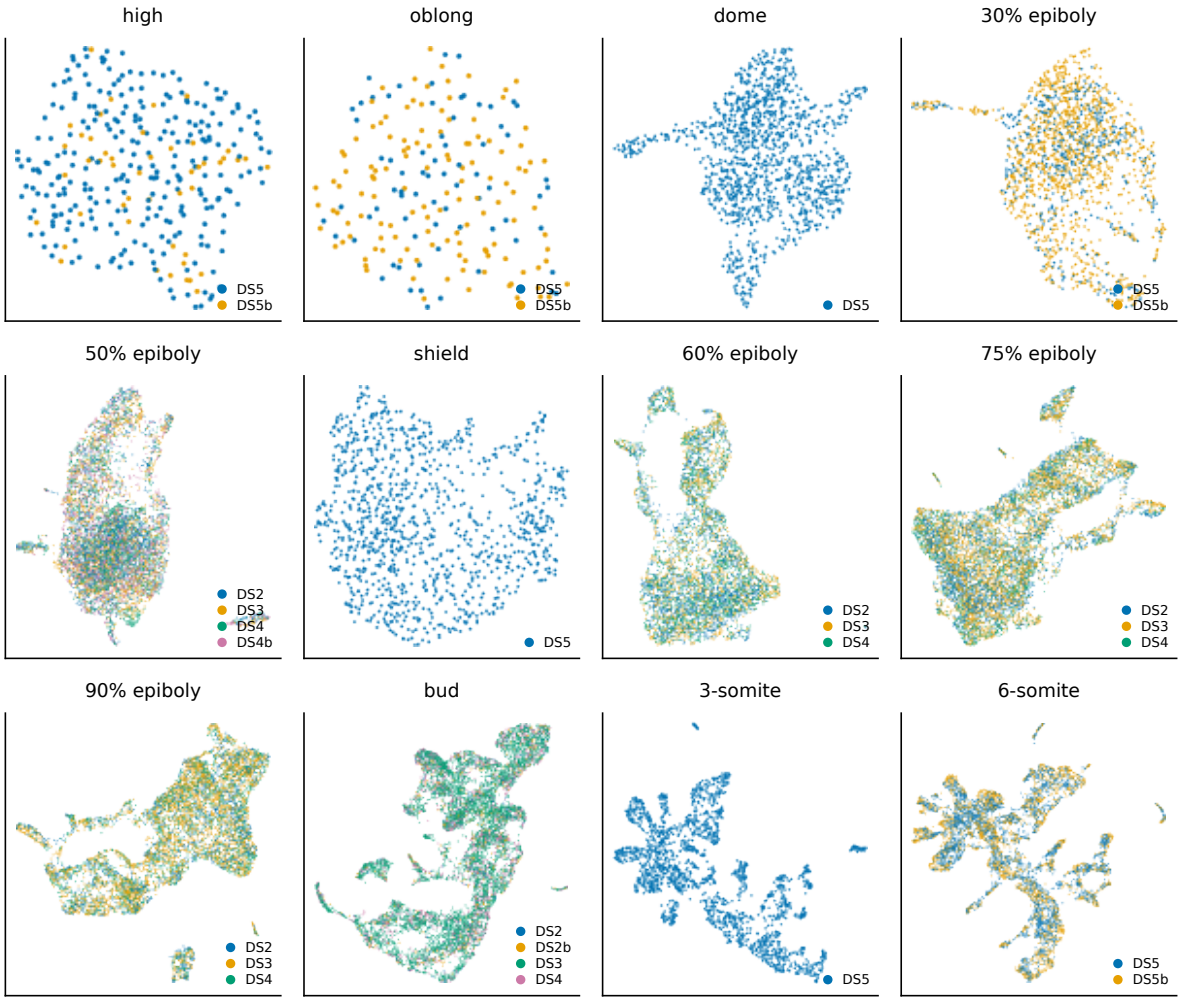


Figure S5: UMAPs of the aligned libraries of every stage; colour: library (sequencing batch, *b* for the second library of a batch).

Table S4: Selected genes supporting the explanations of the a/b differences. Mean expression is in UMIs per 10,000 UMIs; the fold change is b relative to a. Detection is the fraction of cells with at least one UMI, after thinning the deeper library to the median depth of the other.

Pair	Gene	Reading	Mean/10k (a/b)		$\log_2$FC	Detected (a/b)	
high DS5	*gbgt1l4*	on/off	0.1	2.3	+3.5	6%	71%
	si:dkey-261j4.3	on/off	0.9	0.1	−2.7	43%	4%
	nanos3	germ plasm	2.3	9.1	+1.9	50%	44%
	dnd1	germ plasm	2.5	6.2	+1.2	43%	21%
	tbx16	early zygotic	2.5	0.5	−2.2	64%	15%
oblong DS5	*aldh1l1*	near-absent in b	0.7	0.1	−2.3	19%	2%
	chia.6	near-absent in b	0.7	0.0	−2.7	21%	1%
	krt8	EVL	0.0	1.1	+3.0	4%	12%
	krt18	EVL	0.5	3.3	+2.5	7%	21%
	cldne	EVL	0.3	2.8	+3.0	4%	18%
50% epiboly DS4	*hsp90aa1.2*	heat shock	8.4	0.4	−4.0	81%	14%
	hsp70.3	heat shock	1.4	0.1	−3.1	34%	3%
	serpinh1b	heat shock	4.4	0.5	−2.9	57%	14%
	si:dkey-108k21.10	on/off	1.8	0.1	−3.2	49%	5%
bud DS2	*myod1*	muscle (adaxial)	0.7	0.1	−2.2	4%	1%
	myl10	muscle (adaxial)	1.7	0.4	−2.0	12%	8%
	actc1a	muscle (adaxial)	0.7	0.2	−1.6	13%	5%
	sox19b	early neuroectoderm	0.2	0.5	+1.0	10%	21%

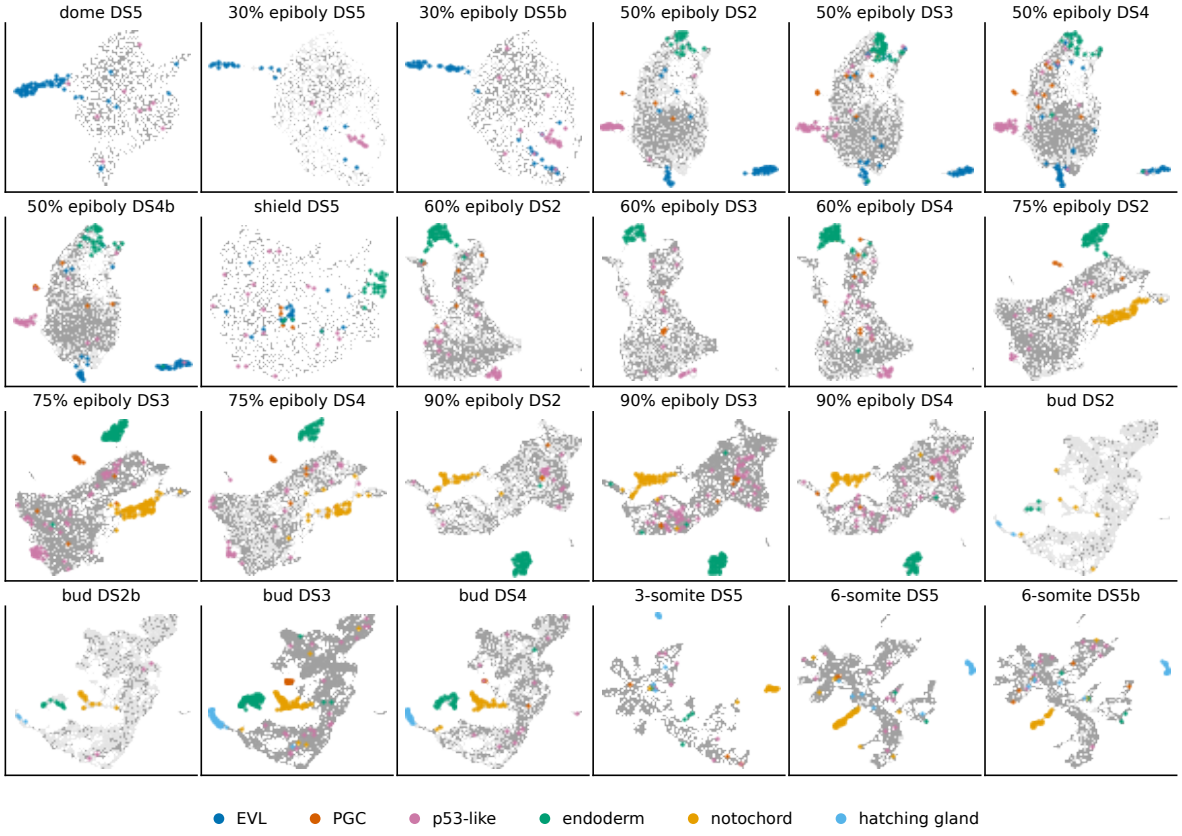


Figure S6: Marker-called populations (check 4) in every library from dome on, on the aligned UMAP of the stage (Figure S5). Dark grey: the cells of the library; light grey: the other libraries of the stage. A population is shown only at the stages where its markers are unambiguous; the marker genes are excluded from the features.

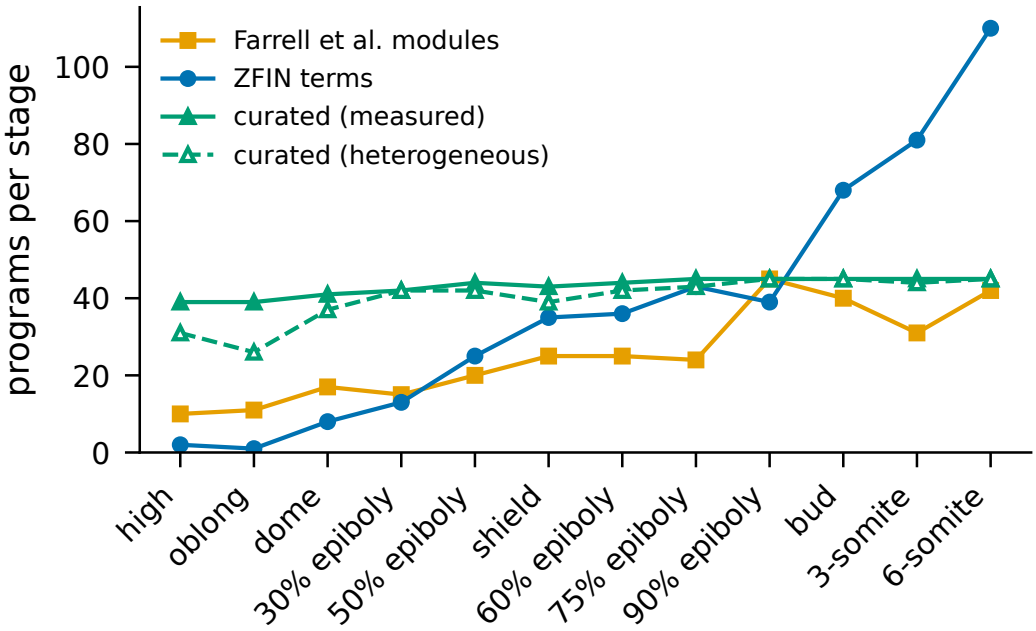


Figure S7: Programs per stage of the Farrell et al. modules, the ZFIN terms (all years) and the curated set (Section 7.1). Curated (measured): programs with at least one gene measured at the stage (10 or more UMIs); curated (heterogeneous): programs with at least one gene heterogeneous at the stage (the filled markers of Figure 10).

Table S7: Curated programs: source (ZFIN anatomy terms grouped by `data/curated/zfin_anatomy_programs.tsv`, or the signalling and state tables), the stages at which ZFIN records it (anatomy programs; all curated programs are defined at every stage), the program of Table 1 it corresponds to and the programs it is presumed to derive from (the arrows of Figure 10), where and under which signals it arises, its role for the model and interpretation, and its first genes (by mean heterogeneity over the stages, then records). Annotation: `data/curated/curated_programs.tsv`.

Program	ZFIN stages	Source	Table 1 / from	Location, signals	Role and interpretation	First genes
early states						
maternal		literature table	Maternal	whole embryo; degraded after MBT	Library-level timing control. Maternal transcripts; acts on whole libraries [204, 205, 69].	*h1m, cth1, buc, gdf3, mos, cldnd*
blastomere		literature table	Blastomeres; from maternal	whole blastoderm after MBT	The undifferentiated state from which the first decisions start. Pluripotency genes (pou5f3, nanog, soxB1); lost from 60% epiboly in most cells [110, 66, 211, 139].	*pou5f3, nanog, sox19b, lin28a, sox3, sox19a*
blastoderm and margin						
EVL	dome – 6-somite	ZFIN: EVL; periderm; superficial blastomere	EVL; from blastomere	outermost cell layer of the blastoderm	First lineage split after MBT; a distinct population at every stage. Enveloping layer, an epithelium covering the embryo; becomes the periderm; a clear switch-like program [157, 97, 156, 171].	*krt4, krt18, cldne, cldnb, lye, krt8*
PGC	30% epiboly – 6-somite	ZFIN: primordial germ cell; germ line cell	Primordial germ cells; from maternal	scattered clusters; specified by inherited germ plasm	A population outside the somatic decisions; a positive control for a rare population. Primordial germ cells; few cells per embryo [102, 200, 65].	*nanos3, ddx4, dnd1, ca15b, tdrd7a, gra*
margin	dome – bud	ZFIN: margin; germ ring; marginal blastomere	Marginal mesendoderm; from blastomere	vegetal edge of the blastoderm (germ ring); Nodal from the YSL and the margin itself	The mesendoderm before internalization; the hypoblast/epiblast decision around 6 hpf starts here. Marginal cells that internalize to form mesoderm and endoderm; split by fate along the margin [154, 36, 62].	*gsc, ved, ta, aplnra, mespab, eve1*
dorsal organizer	30% epiboly – 75% epiboly	ZFIN: shield; presumptive shield; presumptive dorsal mesoderm	Dorsal organizer; from margin	dorsal margin; maternal beta-catenin targets, then Nodal	The dorsal-ventral decision: the organizer protects the dorsal side from BMP. The shield (Spemann organizer); gives rise to prechordal plate and notochord [213, 170, 175, 124].	*gsc, noto, insb, xbp1, pitx2, twist2*
dorsal forerunner	50% epiboly – 6-somite	ZFIN: forerunner cell group; Kupffer's vesicle	Dorsal forerunner cells; from EVL	dorsal margin, ahead of the shield	A small population at the focus stages (shield, 60% epiboly). Dorsal forerunner cells, which form the Kupffer's vesicle (left-right organ); shares genes with EVL and endoderm [125].	*sult6b1, cxcr4a, rassf7b, vgll4l, foxj1a, si:ch73-234b20.5*
endoderm and mesoderm						

Program	ZFIN stages	Source	Table 1 / from	Location, signals	Role and interpretation	First genes
endoderm	30% epiboly – 6-somite	ZFIN: endoderm; presumptive endoderm; endodermal cell	Endoderm; from margin	margin, closest to the edge; highest Nodal	The endoderm/mesoderm decision of the hypoblast. Endoderm (sox32/sox17 cells); gut and its derivatives [1, 88, 87, 129].	*gata5, foxa2, ackr3b, sox32, mixl1, sox17*
mesoderm	30% epiboly – 6-somite	ZFIN: hypoblast; mesoderm; presumptive mesoderm	Marginal mesendoderm (internalized); from margin	hypoblast; Nodal, FGF, Wnt and BMP along the margin	The internalized side of the hypoblast/epiblast decision. Mesoderm before it splits into paraxial, lateral and intermediate parts; broad, largely unsupported [55, 75, 164].	*ved, mespab, tbx16, eve1, her1, mespaa*
presomitic mesoderm	30% epiboly – 6-somite	ZFIN: segmental plate; presumptive segmental plate; paraxial mesoderm; presumptive paraxial mesoderm	Paraxial mesoderm; from mesoderm	lateral margin, then segmental plate; Wnt, FGF, Notch oscillator	Paraxial mesoderm, the source of somites. Tbx6/tbx16 paraxial mesoderm; segmental plate with the her/Notch clock from bud on [75, 161, 137, 73].	*aplnra, gadd45ba, mespab, tbx16, msgn1, ripply1*
tailbud	bud – 6-somite	ZFIN: tail bud	Tailbud; from margin	ventral margin converging at the vegetal pole; Wnt, FGF, BMP	Posterior progenitors (neuromesodermal) at the end of gastrulation. Tailbud with cdx and ntl/tbxta genes; feeds the posterior body [122, 81, 169].	*vox, hes6, p4ha2, foxd5, znfl2a, nfkbiaa*
adaxial somite	bud – 6-somite	ZFIN: somite; adaxial cell; myotome; fast muscle cell	Paraxial mesoderm (adaxial cells, somites); from presomitic mesoderm	next to the notochord; Hedgehog from the axis	First muscle and somite fates. Adaxial cells (slow muscle precursors) and forming somites [202, 17, 168, 208].	*ved, ripply1, mespaa, gadd45bb, cdkn1ca, rdh10a*
lateral plate	bud – 6-somite	ZFIN: lateral plate mesoderm; anterior lateral plate mesoderm; posterior lateral plate mesoderm; lateral mesoderm (+2 more)	Lateral plate mesoderm; from mesoderm	ventral and lateral margin; BMP	Lateral plate decisions (blood, vessels, heart, kidney) begin at bud. Lateral plate mesoderm with haemato-vascular precursors [50, 12, 132].	*gata6, aplnrb, lmo2, blf, gata2a, tbx1*
intermediate mesoderm	bud – 6-somite	ZFIN: pronephric mesoderm; presumptive pronephric mesoderm; intermediate mesoderm	Lateral plate mesoderm (kidney precursors); from mesoderm	between paraxial and lateral plate mesoderm	Kidney precursors, a lateral fate. Intermediate (pronephric) mesoderm [132, 141].	*ftr83, osr1, ftr82, acy3.2, lhx1a, twist1a*
heart	6-somite	ZFIN: heart primordium	Lateral plate mesoderm (heart precursors); from lateral plate	anterior lateral plate mesoderm	Heart precursors at the end of the range (6-somite). Heart field (nkx2.5/2.7, hand2) [109, 216, 136].	*nkx2.7, tmem88a, tbx20, nkx2.5, rbfox1l*
axial						
prechordal plate	50% epiboly – 6-somite	ZFIN: prechordal plate; anterior axial hypoblast	Prechordal plate; from dorsal organizer	first internalized dorsal cells; highest Nodal	The anterior axial fate, which patterns the forebrain. Prechordal plate (gsc, frzb); its anterior part becomes the hatching gland [57, 187, 165].	*gsc, tagln2, pitx2, elovl6, icn, ctslb*

Program	ZFIN stages	Source	Table 1 / from	Location, signals	Role and interpretation	First genes
hatching gland	bud – 6-somite	ZFIN: polster; hatching gland	Prechordal plate (anterior part); from prechordal plate	anterior prechordal plate (polster)	A terminal fate of the prechordal plate. Hatching gland (he1a, ctsl): a clear switch-like program from bud [199].	*gsc, klf17, pfn1, tagln2, pitx2, cldn7b*
axial mesoderm	50% epiboly – 6-somite	ZFIN: axial chorda mesoderm; axial mesoderm; axial hypoblast	Notochord (axial mesoderm); from dorsal organizer	dorsal midline; Nodal, then low Wnt	The posterior axial fate before the notochord differentiates. Axial chorda mesoderm (noto/flh, foxa2) during gastrulation [177, 182, 164].	*noto, ta, serpinh1b, tob1a, cnn2, perp*
notochord	bud – 6-somite	ZFIN: notochord; notochord posterior region	Notochord; from axial mesoderm	dorsal midline, under the neural plate	Differentiated axial structure; source of Hedgehog. Notochord (col2a1, col8a1, ntd5) from bud on [182, 215, 48].	*ta, xbp1, zgc:162964, serpinh1b, cnn2, waslb*
ectoderm						
neural plate	50% epiboly – 6-somite	ZFIN: neural plate; presumptive neural plate; anterior presumptive neural plate; neuroectoderm (+4 more)	Neural ectoderm; from blastomere	dorsal ectoderm; low BMP; recorded in ZFIN from 50% epiboly	The neural/non-neural decision of the epiblast, then the neural plate before and during A-P subdivision. Presumptive neural plate, neuroectoderm, neural plate, keel and rod; broad, mostly unsupported (covers all neural regions) [19, 152, 56].	*fzd8b, her3, her6, hspb1, irx7, rx3*
forebrain eye	75% epiboly – 6-somite	ZFIN: anterior neural plate; anterior neural keel; anterior neural rod; forebrain neural plate (+8 more)	Neural ectoderm (anterior); from neural plate	anterior neural plate; Wnt off	A-P subdivision of the neural plate: anterior. Forebrain and eye field (six3, rx, hesx1/anf, lhx2) [14, 166, 85, 3].	*fzd8b, stm, six3b, rx3, hesx1, zic2a*
midbrain hindbrain boundary	75% epiboly – 6-somite	ZFIN: midbrain hindbrain boundary neural plate; midbrain hindbrain boundary neural keel; midbrain hindbrain boundary neural rod; presumptive midbrain hindbrain boundary (+4 more)	Neural ectoderm (midbrain-hindbrain boundary); from neural plate	midbrain-hindbrain boundary; FGF8 organizer	A-P subdivision: the isthmic organizer. Midbrain-hindbrain boundary (eng, pax2/5/8, gbx2) [28, 141, 89, 151].	*her3, ism1, her5, mdkb, her11, shisa2*
hindbrain	75% epiboly – 6-somite	ZFIN: hindbrain neural plate; hindbrain neural keel; hindbrain neural rod; presumptive hindbrain (+8 more)	Neural ectoderm (hindbrain); from neural plate	posterior neural plate; FGF, RA	A-P subdivision: hindbrain rhombomeres. Hindbrain (krox20/egr2b, hox genes) [140, 130, 143].	*irx7, her2, cyp26c1, efnb2a, dbx1a, hoxb1b*

Program	ZFIN stages	Source	Table 1 / from	Location, signals	Role and interpretation	First genes
primary neurons	bud – 6-somite	ZFIN: primary neuron; primary neuron hindbrain; primary motor neuron; Rohon-Beard neuron (+6 more)	(extends Table 1: neurogenesis in the neural ectoderm); from neural plate	proneural clusters; Notch lateral inhibition	The first neuronal differentiation. Primary neurons (motor, Rohon-Beard, interneurons) from bud on [181].	*pnocb, pnx, elavl3, neurod4, smox, isl1*
non neural ectoderm	75% epiboly – 6-somite	ZFIN: non neural ectoderm; integument	Non-neural ectoderm; from blastomere	ventral epiblast; high BMP	The non-neural side of the ectoderm decision. Non-neural ectoderm, future epidermis (tp63, gata3); its early genes (dlx3b, foxi1) are pooled into the placodes [5, 133, 149].	*gata3, foxi1, tfap2a, tp63, foxi3a, tfap2c*
neural crest	3-somite – 6-somite	ZFIN: neural crest; neural crest cell; cranial neural crest; cranial neural crest cell	Neural plate border; from neural plate, non neural ectoderm	neural plate border; intermediate BMP, Wnt, FGF	The border decision: neural crest. Neural crest (sox10, tfap2a, foxd3) from 3-somite [26, 93, 116, 71].	*tfap2a, foxd3, ovol1, cdh6, fam212aa, fam46ba*
placodes	75% epiboly – 6-somite	ZFIN: preplacodal ectoderm; otic placode; trigeminal placode; neurogenic placode (+2 more)	Neural plate border; from neural plate, non neural ectoderm	neural plate border, anterior; preplacodal ectoderm	The border decision: placodes. Preplacodal ectoderm and placodes (six1, eya1, dlx3b, foxi1) [173, 174, 158, 8].	*foxi1, dlx3b, cldna, ptgs2a, tp63, pax8*
signalling profiles						
Nodal		readouts, ligands and antagonists (signalling table)		margin, then dorsal mesendoderm and axial mesoderm	General (received and sent in the same cells). Nodal readouts, ligands (ndr1, ndr2; autoregulated) and lefty (feedback, secreted) [150, 36, 186, 35].	*lft1, lft2, ndr1, ndr2*
BMP		readouts, ligands and antagonists (signalling table)		ventral gastrula, then tailbud	General (received and sent in the same cells). BMP readouts and the ventral ligands (bmp2b, bmp4, bmp7a), which are BMP targets themselves; szl (feedback, secreted) [92, 134, 22].	*id1, id3, bambia, szl, msx1b, bmp2b*
BMP (blocking)		secreted antagonists, degradation (signalling table)		dorsal organizer (chd, nog1), paraxial mesoderm (fsta)	Where the signal is blocked (secreted antagonists). BMP antagonists made by cells with low BMP; shape the dorsal-ventral gradient [63, 43, 212].	*chd, nog1, nog2, fsta*
BMP organizer (sending)		ligands, synthesis (signalling table)		dorsal organizer, then axial mesoderm	Where the signal is sent (ligands). The organizer's BMP ligand admp, which goes with low BMP readouts [113, 206].	*admp*
Wnt		readouts, ligands and antagonists (signalling table)		margin, then tailbud	General (received and sent in the same cells). Wnt/beta-catenin readouts and the canonical ligands (wnt8a, wnt3a) [86, 111, 169].	*sp5l, sp5a, notum1a, lef1, wnt8a, wnt3a*

Program	ZFIN stages	Source	Table 1 / from	Location, signals	Role and interpretation	First genes
Wnt (blocking)		secreted antagonists, degradation (signalling table)		organizer and prechordal plate, then anterior neural plate	Where the signal is blocked (secreted antagonists). Wnt antagonists (dkk1b, frzb, sfrp1a); protect the anterior from Wnt [64, 185].	*dkk1b, frzb, sfrp1a*
Wnt/PCP (sending)		ligands, synthesis (signalling table)		margin and mesoderm	Where the signal is sent (ligands). Non-canonical Wnt ligands (wnt5b, wnt11) of convergent extension; no readouts [68, 90].	*wnt5b, wnt11*
FGF		readouts, ligands and antagonists (signalling table)		margin, then tailbud and midbrain-hindbrain boundary	General (received and sent in the same cells). FGF readouts and ligands (fgf8a, fgf24, fgf3, fgf4) [44, 151, 25].	*etv4, il17rd, fgf8a, fgf24, fgf3, fgf4*
Notch (receiving)		readouts (signalling table)		neural plate, presomitic mesoderm	Where the signal is received (readouts). Her/hey targets of Notch.	*her4.1, her4.2, her4.3, her15.1, her15.2, her12*
Notch (sending)		ligands, synthesis (signalling table)		neural plate (primary neurons), presomitic mesoderm	Where the signal is sent (ligands). Delta ligands (dla, dlb, dlc, dld): lateral inhibition and the segmentation clock [60, 73].	*dla, dlb, dlc, dld*
RA (receiving)		readouts (signalling table)		central nervous system (somite stages)	Where the signal is received (readouts). Dhrs3a (feedback); a weak readout.	*dhrs3a*
RA (sending)		ligands, synthesis (signalling table)		involuting and paraxial mesoderm	Where the signal is sent (synthesis). RA synthesis (aldh1a2) [7, 52].	*aldh1a2*
RA (blocking)		secreted antagonists, degradation (signalling table)		anterior neural plate, margin, then tailbud	Where the signal is blocked (degradation). RA degradation (cyp26a1); RA-inducible, but its pattern follows anterior identity [100, 23].	*cyp26a1*
Hedgehog (sending)		ligands, synthesis (signalling table)		axial mesoderm, notochord, floor plate	Where the signal is sent (ligands). Hedgehog ligands (shha, shhb); the readouts are not heterogeneous [98, 29].	*shha, shhb*
cell states						
cell cycle s		literature table		all proliferating cells	A known non-identity axis, to be separated from identity. S phase genes [189].	*pcna, mcm2, mcm5, mcm6, cdc6, slbp*
cell cycle g2m		literature table		all proliferating cells	A known non-identity axis, to be separated from identity. G2/M phase genes [189].	*cdk1, ube2c, top2a, nusap1, cenpf, tpx2*
heat shock		literature table		handling of embryos and cells	Technical control (stress during handling). Heat shock response (hsp70, hsp90) [99, 112, 198].	*hsp70.3, hsp90aa1.2, bag3, dnajb1a, hsph1*
p53 response		literature table		stressed or damaged cells	Technical and biological stress control. P53 targets (mdm2, cdkn1a, ccng1) [39, 106, 153].	*mdm2, cdkn1a, phlda3, tp53inp1, gadd45aa, rrm2b*

Program	ZFIN stages	Source	Table 1 / from	Location, signals	Role and interpretation	First genes
controls						
whole organism		ZFIN: whole organism only		whole embryo	Negative control: uniform genes should not vary between cells. Genes ZFIN records only in the whole organism from high to shield.	*yes1, hdac1, ikzf5, fyna, stk11, dgcr8*
YSL	high – 6-somite	ZFIN: yolk syncytial layer; external yolk syncytial layer; internal yolk syncytial layer (genes of curated programs left out)	YSL	syncytium between the blastoderm and the yolk; source of Nodal signals	Negative control: removed with the yolk before dissociation. No YSL cells are expected; genes of the curated programs and other controls are left out, so that only YSL-specific genes remain [154].	*apoeb, tnfaip8l2b, tfa, sesn3, pleca, rgs4*
IFN		readouts (signalling table)		whole embryo	Negative control: should not vary between cells. Type I interferon response; no immune system yet.	*isg15, mxa*
AhR		readouts (signalling table)		whole embryo	Negative control: should not vary between cells. Aryl hydrocarbon receptor (xenobiotic) response; embryo-wide.	*cyp1a, cyp1b1*
GR		readouts (signalling table)		whole embryo	Negative control: should not vary between cells. Glucocorticoid receptor response; embryo-wide.	*fkbp5, klf9, tsc22d3*
circadian		readouts (signalling table)		whole embryo	Negative control: should not vary between cells. Circadian clock; synchronised across the embryo.	*per1b, per2, cry1aa, tefa*
dissociation		readouts (signalling table)		whole embryo	Technical control: stressed cells. Immediate early genes induced by tissue dissociation; flags stressed cells.	*fosab, junba, jun, atf3, egr1*

Table S8: Phenotype transitions (`scripts/21_signal_coupling.py`): for every presumed derivation (parent → program), the signalling profiles that are higher or lower in the program's active cells than in the parent's other active cells, beyond random genes, at two or more of the first three stages at which the program is supported (Stages), and never on the other side there. Expected: where and under which signals the program arises, from the annotation (Table S7). Profiles: general unless marked (sending, blocking, receiving).

Transition	Stages	Expected	Higher	Lower
maternal → blastomere	60% epiboly – 90% epiboly	whole blastoderm after MBT	–	–
blastomere → EVL	high – dome	outermost cell layer of the blastoderm	–	–
maternal → PGC	high – dome	scattered clusters; specified by inherited germ plasm	–	–
blastomere → margin	high – 50% epiboly	vegetal edge of the blastoderm (germ ring); Nodal from the YSL and the margin itself	Nodal, FGF, Wnt/PCP (sending), RA (sending)	–
margin → dorsal organizer	shield – 75% epiboly	dorsal margin; maternal beta-catenin targets, then Nodal	BMP organizer (sending), Hedgehog (sending), BMP (blocking), Wnt (blocking)	Wnt, Wnt/PCP (sending), Notch (sending)
EVL → dorsal forerunner	not supported	dorsal margin, ahead of the shield	–	–
margin → endoderm	shield – 75% epiboly	margin, closest to the edge; highest Nodal	Wnt (blocking)	RA (blocking), Wnt/PCP (sending), RA (sending), Notch (sending)
margin → mesoderm	shield – 75% epiboly	hypoblast; Nodal, FGF, Wnt and BMP along the margin	RA (sending)	Hedgehog (sending), BMP (blocking)
mesoderm → presomitic mesoderm	shield – 75% epiboly	lateral margin, then segmental plate; Wnt, FGF, Notch oscillator	Nodal, RA (sending), Wnt (blocking)	BMP, RA (blocking)
margin → tailbud	bud – 6-somite	ventral margin converging at the vegetal pole; Wnt, FGF, BMP	BMP, Wnt, FGF, RA (blocking), Notch (receiving), Wnt/PCP (sending)	RA (sending)
presomitic mesoderm → adaxial somite	60% epiboly – 90% epiboly	next to the notochord; Hedgehog from the axis	Wnt (blocking)	RA (blocking)
mesoderm → lateral plate	bud – 6-somite	ventral and lateral margin; BMP	–	RA (blocking), Notch (receiving), Notch (sending)
mesoderm → intermediate mesoderm	3-somite – 6-somite	between paraxial and lateral plate mesoderm	–	FGF, RA (blocking), Notch (receiving), Notch (sending)
lateral plate → heart	bud – 6-somite	anterior lateral plate mesoderm	–	Wnt
dorsal organizer → prechordal plate	60% epiboly – 90% epiboly	first internalized dorsal cells; highest Nodal	–	Wnt, FGF, BMP organizer (sending), Wnt/PCP (sending), RA (sending)
prechordal plate → hatching gland	60% epiboly – 90% epiboly	anterior prechordal plate (polster)	–	Notch (sending)
dorsal organizer → axial mesoderm	60% epiboly – 90% epiboly	dorsal midline; Nodal, then low Wnt	–	–
axial mesoderm → notochord	60% epiboly – 90% epiboly	dorsal midline, under the neural plate	Nodal	Wnt, Notch (sending)
blastomere → neural plate	not supported	dorsal ectoderm; low BMP; recorded in ZFIN from 50% epiboly	–	–
neural plate → forebrain eye	75% epiboly – bud	anterior neural plate; Wnt off	RA (blocking)	Wnt

Transition	Stages	Expected	Higher	Lower
neural plate → midbrain hindbrain boundary	75% epiboly – 3-somite	midbrain-hindbrain boundary; FGF8 organizer	Wnt	–
neural plate → hindbrain	bud – 6-somite	posterior neural plate; FGF, RA	–	RA (blocking)
neural plate → primary neurons	3-somite – 6-somite	proneural clusters; Notch lateral inhibition	–	–
blastomere → non neural ectoderm	oblong – 75% epiboly	ventral epiblast; high BMP	BMP	Wnt, FGF, BMP (blocking)
neural plate → neural crest	3-somite – 6-somite	neural plate border; intermediate BMP, Wnt, FGF	BMP, Wnt	–
non neural ectoderm → neural crest	3-somite – 6-somite	neural plate border; intermediate BMP, Wnt, FGF	Wnt	–
neural plate → placodes	90% epiboly – 3-somite	neural plate border, anterior; preplacodal ectoderm	BMP	FGF, RA (blocking), BMP (blocking), Wnt (blocking)
non neural ectoderm → placodes	90% epiboly – 3-somite	neural plate border, anterior; preplacodal ectoderm	–	Nodal, Wnt, FGF

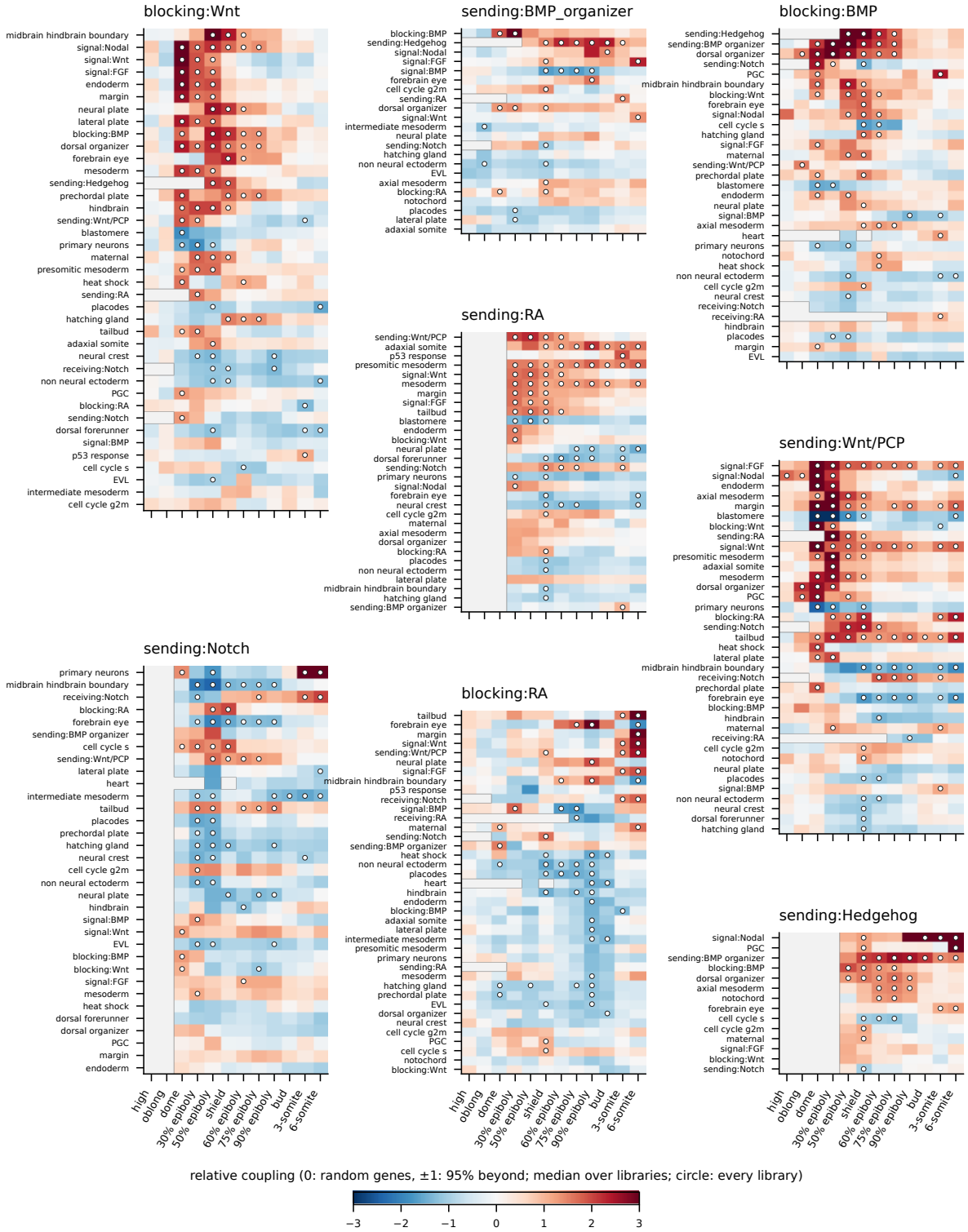


Figure S8: Relative coupling of the sending and blocking signalling profiles with the other curated programs, as in Figure 16.

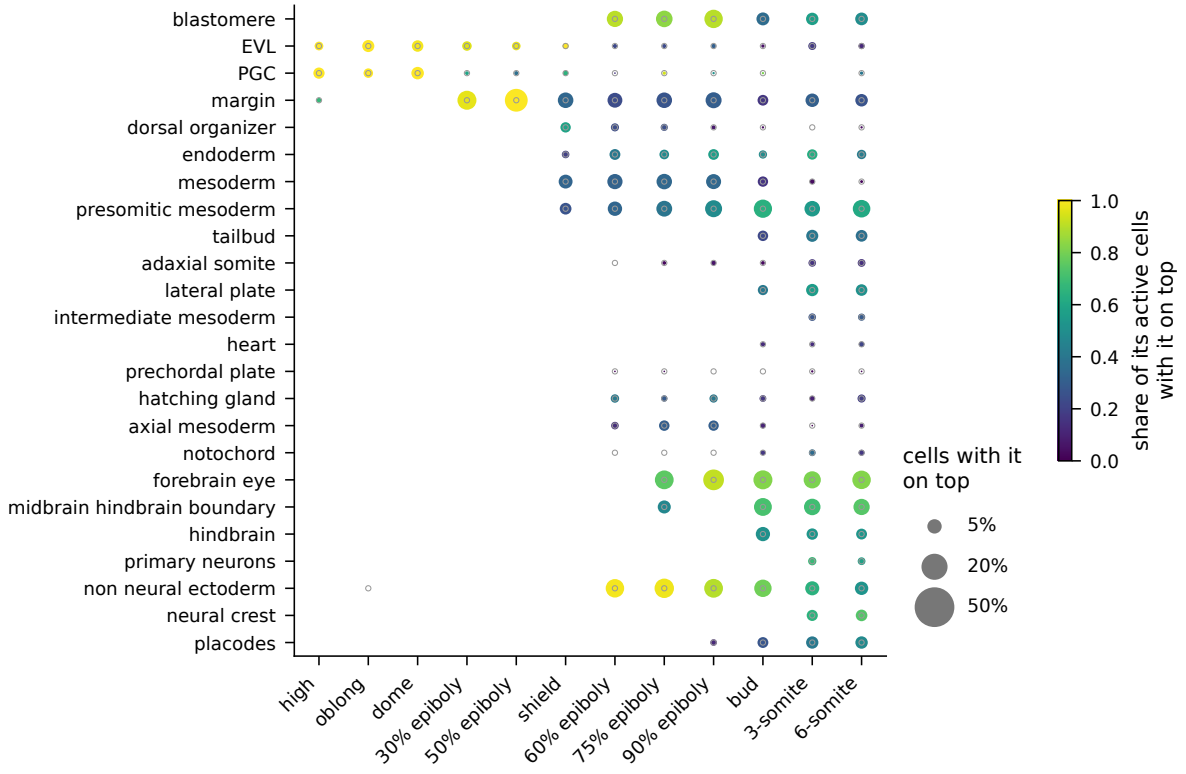


Figure S9: Identity programs as top programs (`scripts/26_cell_programs.py`; stress cells excluded; median over the libraries of a stage). Dot area: share of the cells with the program on top; colour: share of the cells in which the program is active that have it on top. Small grey rings: the program is supported but has (almost) no top cells. Only supported programs compete.

Table S9: Cells per library (`scripts/26_cell_programs.py`): excluded as stress cells (top program p53 response, dissociation stress or heat shock with a margin of at least 2); with the YSL control active (in brackets: of these, with an identity program active as well); with at least 3 yolk/YSL-specific transcripts detected; scrublet doublet score of at least 0.25; depth and complexity more than 3 MADs below the library's median (in brackets: of these, with the EVL on top). Only the stress cells are excluded.

Stage	Library	Cells	p53	Dissoc.	Heat sh.	YSL ctrl.	Yolk	Doublet	Low depth	Low compl.
high	DS5	263	5	2	0	1 (0)	0	0	0	5 (0)
high	DS5b	48	0	0	0	0 (0)	0	0	0	0 (0)
oblong	DS5	72	1	0	0	0 (0)	0	0	0	0 (0)
oblong	DS5b	128	1	0	0	0 (0)	0	0	0	4 (1)
dome	DS5	1,174	0	0	0	37 (37)	0	8	0	16 (13)
30% epiboly	DS5	448	0	0	0	21 (12)	0	4	0	9 (9)
30% epiboly	DS5b	1,058	0	0	0	38 (24)	0	6	0	20 (16)
50% epiboly	DS2	1,287	31	0	0	65 (42)	0	23	0	31 (25)
50% epiboly	DS3	1,430	29	0	0	66 (38)	0	14	0	23 (21)
50% epiboly	DS4	1,567	38	0	0	56 (23)	0	29	0	7 (7)
50% epiboly	DS4b	1,580	19	0	0	61 (46)	0	20	0	25 (25)
shield	DS5	1,046	0	29	0	12 (4)	0	7	0	4 (3)
60% epiboly	DS2	1,458	25	28	0	62 (48)	0	8	0	16 (3)
60% epiboly	DS3	1,275	6	34	0	29 (24)	1	13	0	5 (0)
60% epiboly	DS4	1,468	16	25	0	35 (28)	0	21	1	11 (1)
75% epiboly	DS2	1,992	8	6	0	61 (59)	0	16	0	9 (1)
75% epiboly	DS3	2,963	52	16	0	134 (128)	0	60	0	8 (2)
75% epiboly	DS4	1,369	17	5	0	45 (42)	0	22	0	4 (0)
90% epiboly	DS2	1,238	11	0	0	85 (81)	0	10	0	6 (0)
90% epiboly	DS3	2,849	32	1	0	240 (236)	0	36	0	14 (6)
90% epiboly	DS4	1,539	18	5	0	98 (93)	0	9	0	2 (0)
bud	DS2	250	0	0	0	21 (21)	1	0	2	1 (0)
bud	DS2b	506	0	0	0	54 (54)	0	2	2	6 (0)
bud	DS3	4,759	0	0	0	439 (438)	5	57	0	15 (1)
bud	DS4	1,647	0	0	0	142 (141)	3	15	0	13 (0)
3-somite	DS5	1,631	0	0	0	150 (150)	11	17	0	3 (0)
6-somite	DS5	2,375	0	0	0	185 (185)	1	13	0	11 (0)
6-somite	DS5b	2,085	0	0	0	228 (228)	10	18	0	11 (0)